Spectroscopie de photo-tension de la conductance différentielle négative dans des matériaux de moiré à base de graphène

Paul JAÏS, ICFO

Supervisé par Dr. Koppens

Encadré par R.Bertini et J.Barrier

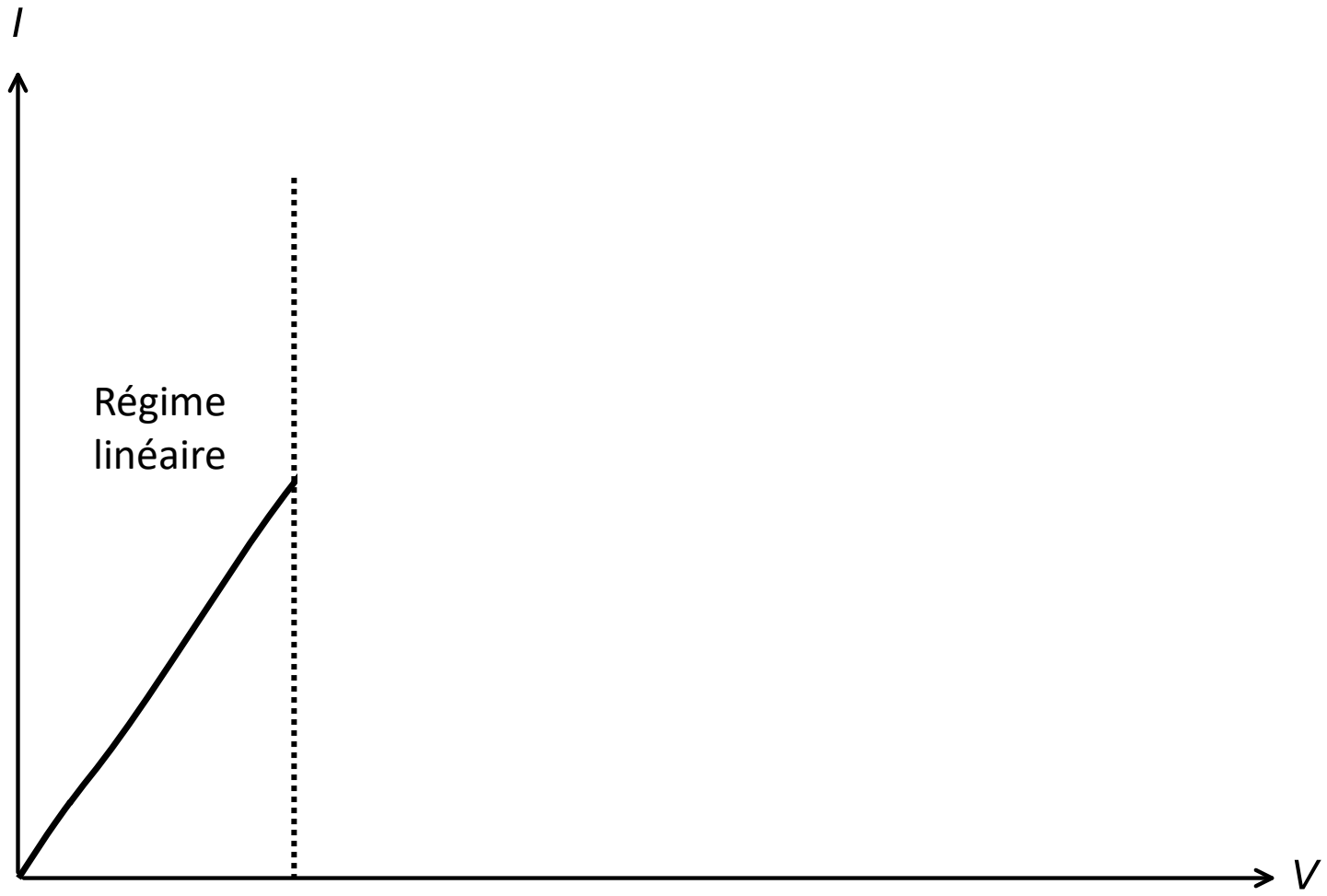
école normale supérieure paris—saclay

11/04/2026

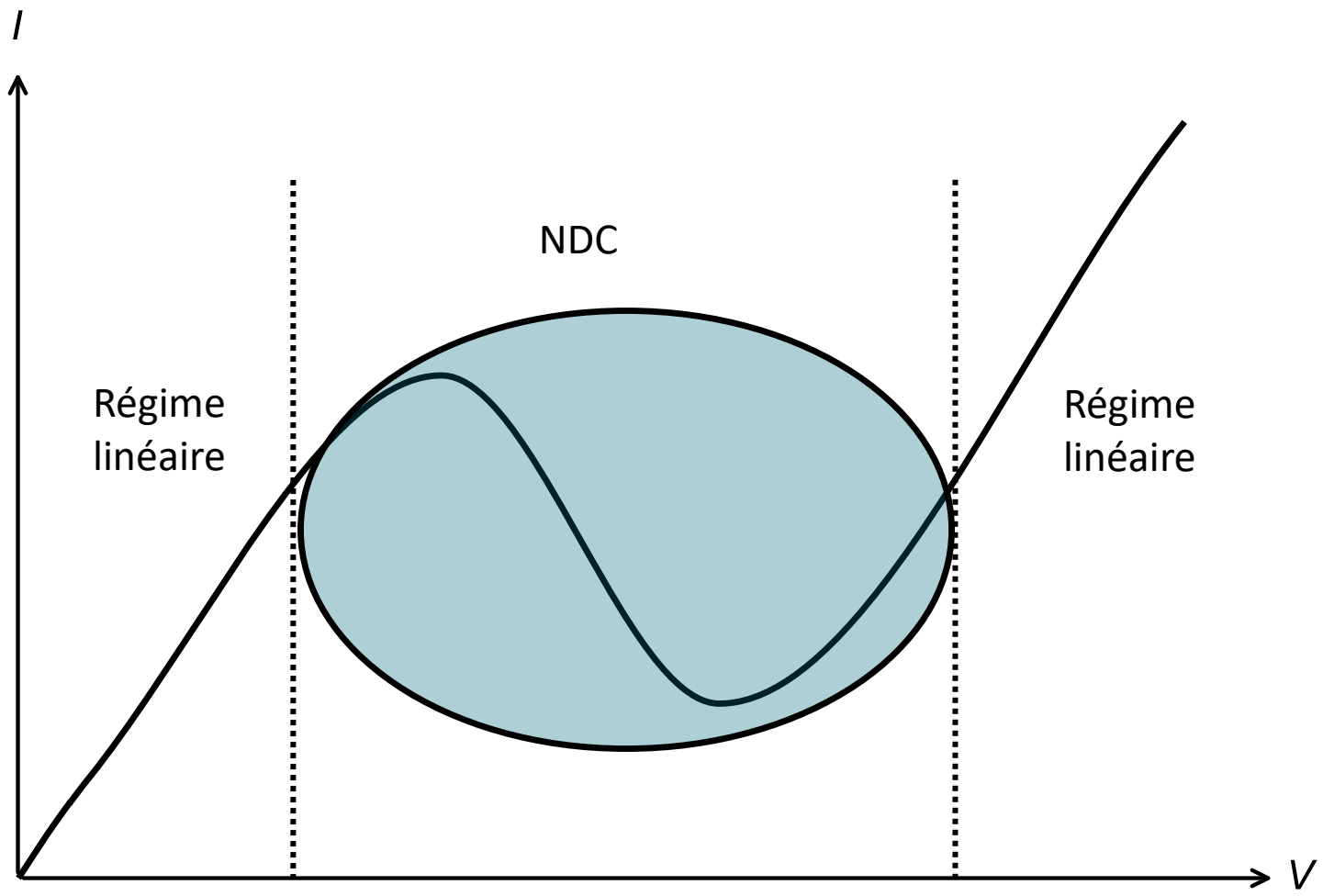
ICFO

NDC : Plus de tension pour moins de courant

- Source une tension
- Mesure un courant



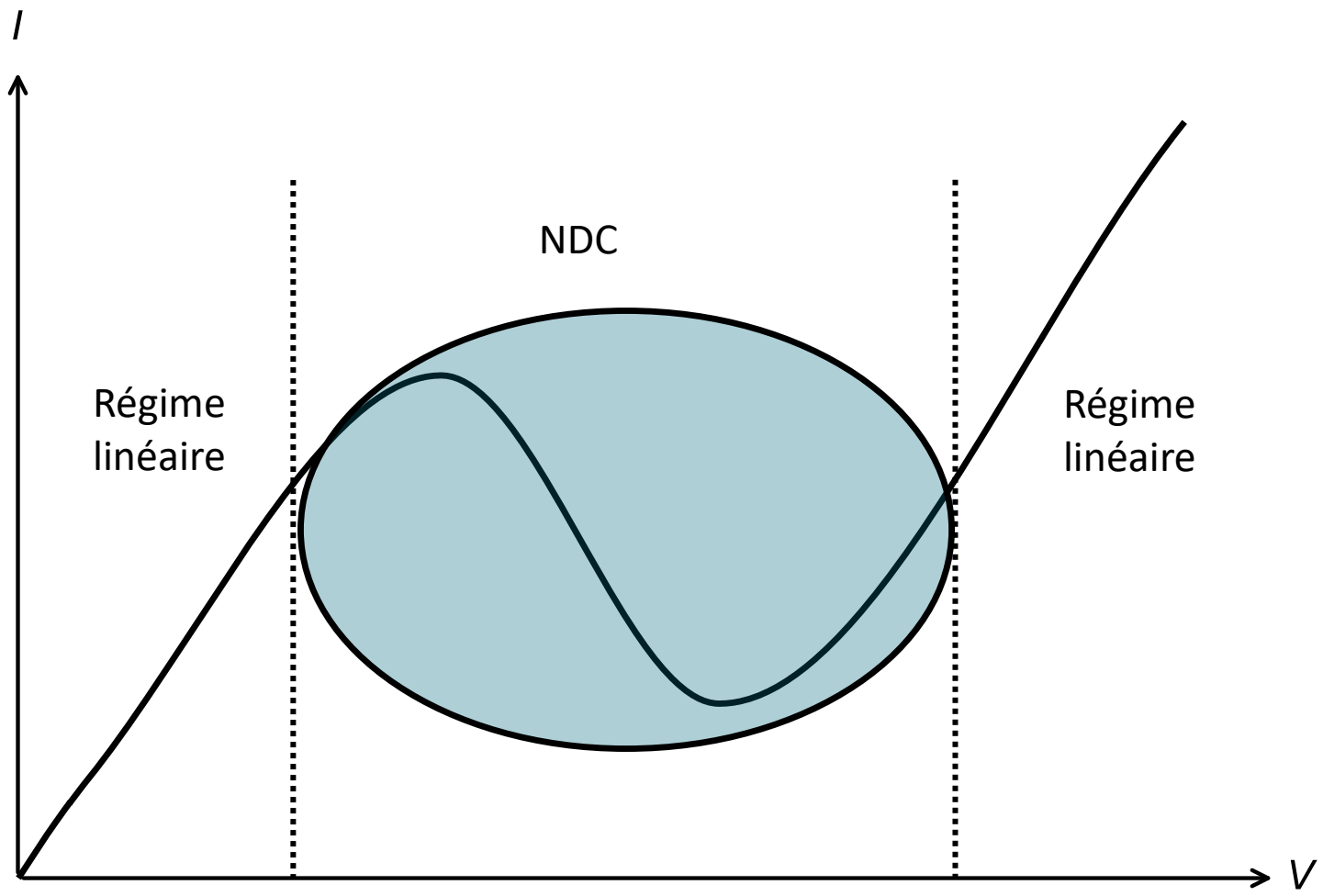
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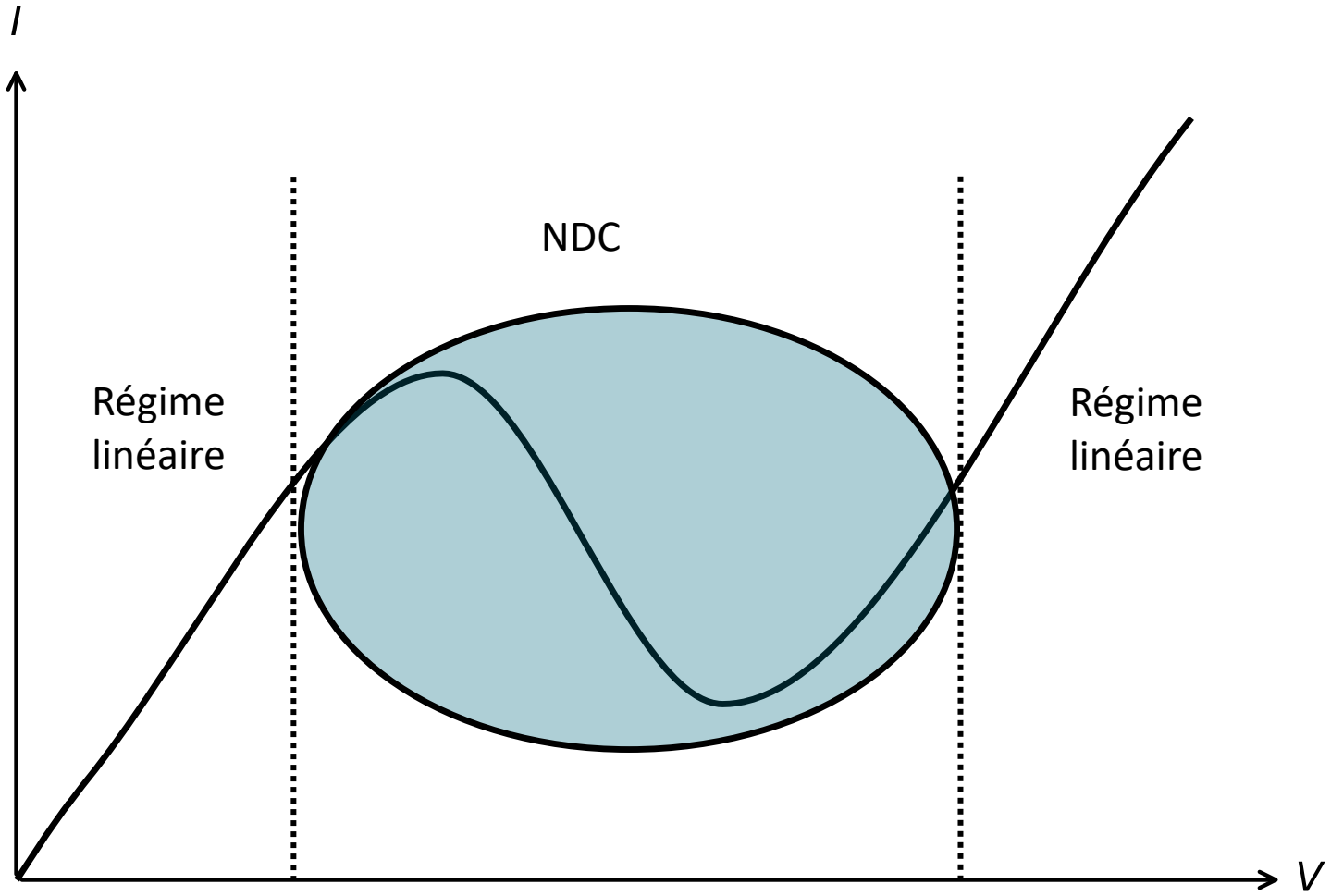


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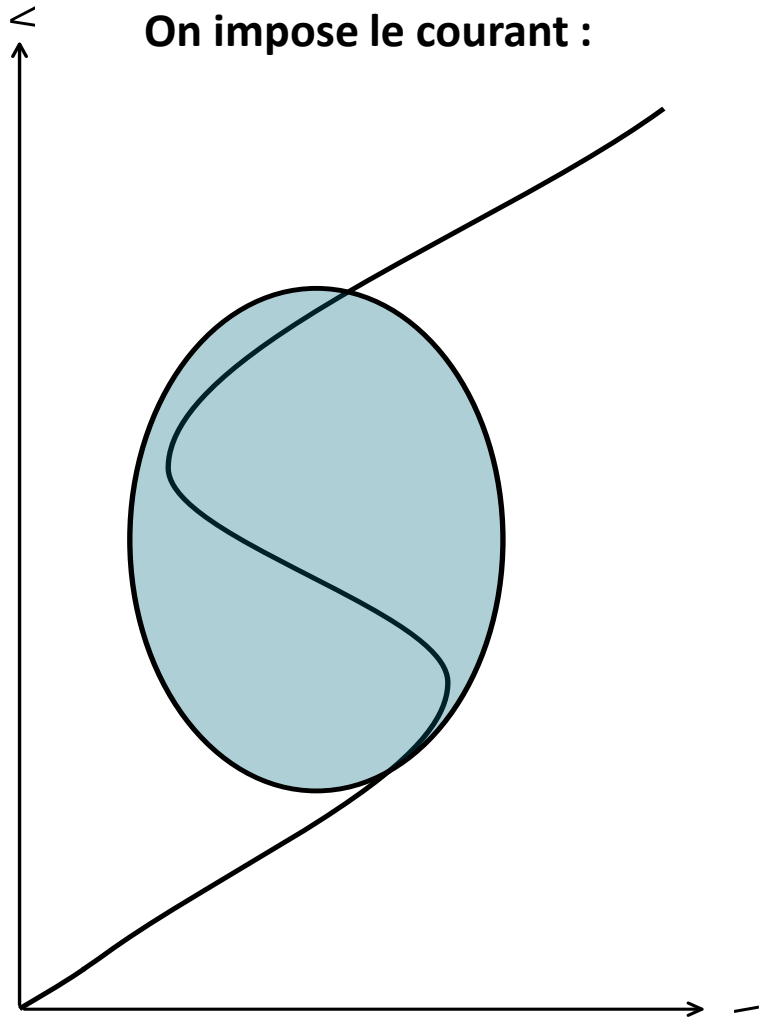
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- **Pourquoi c'est intéressant ?**

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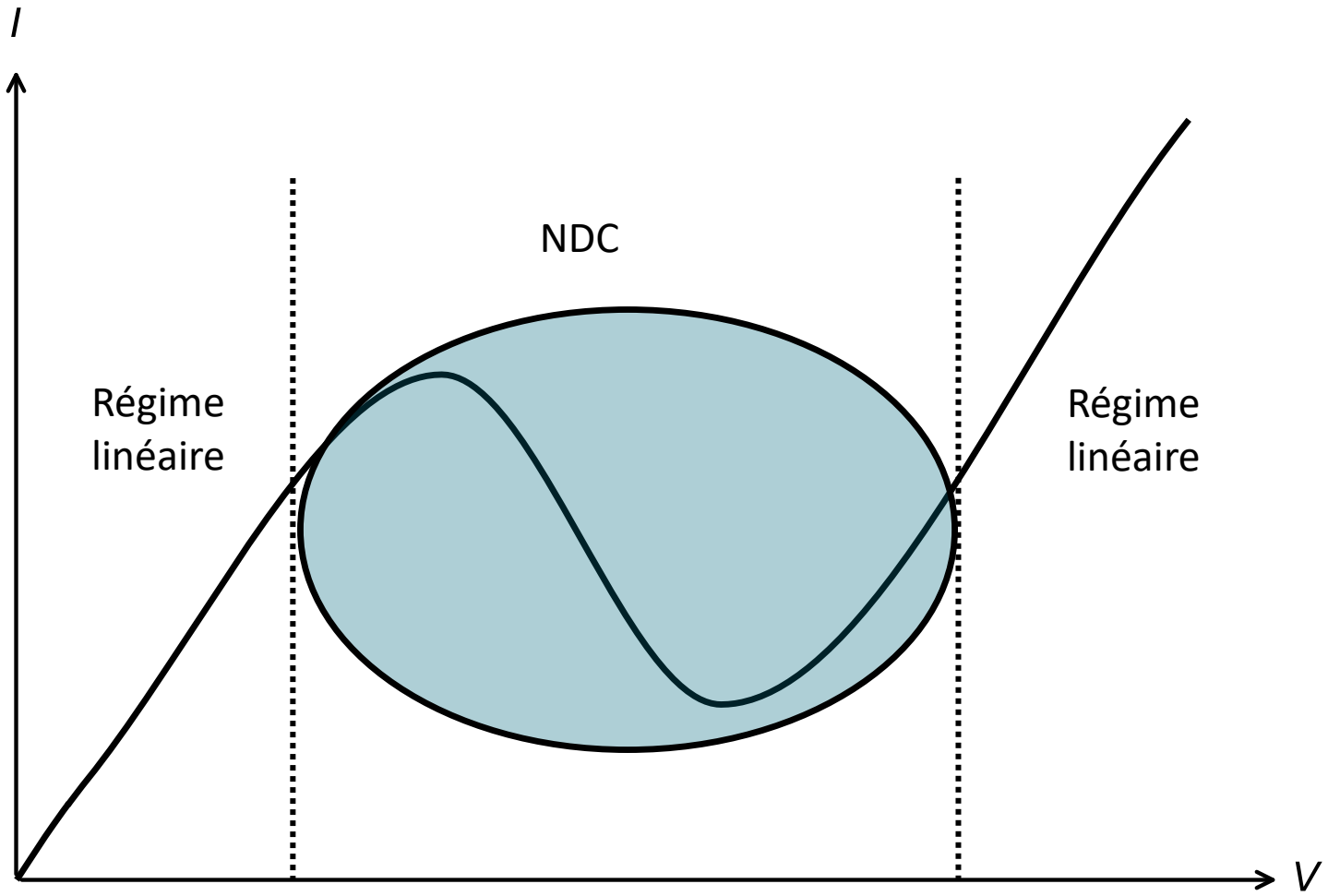


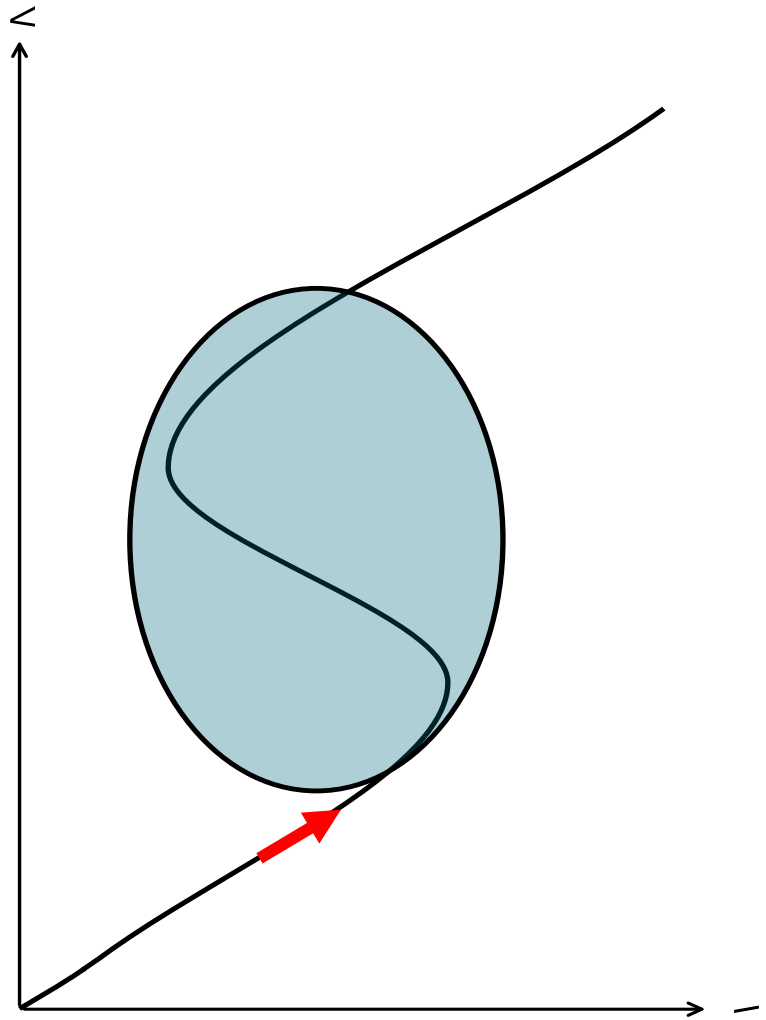
On impose le courant :



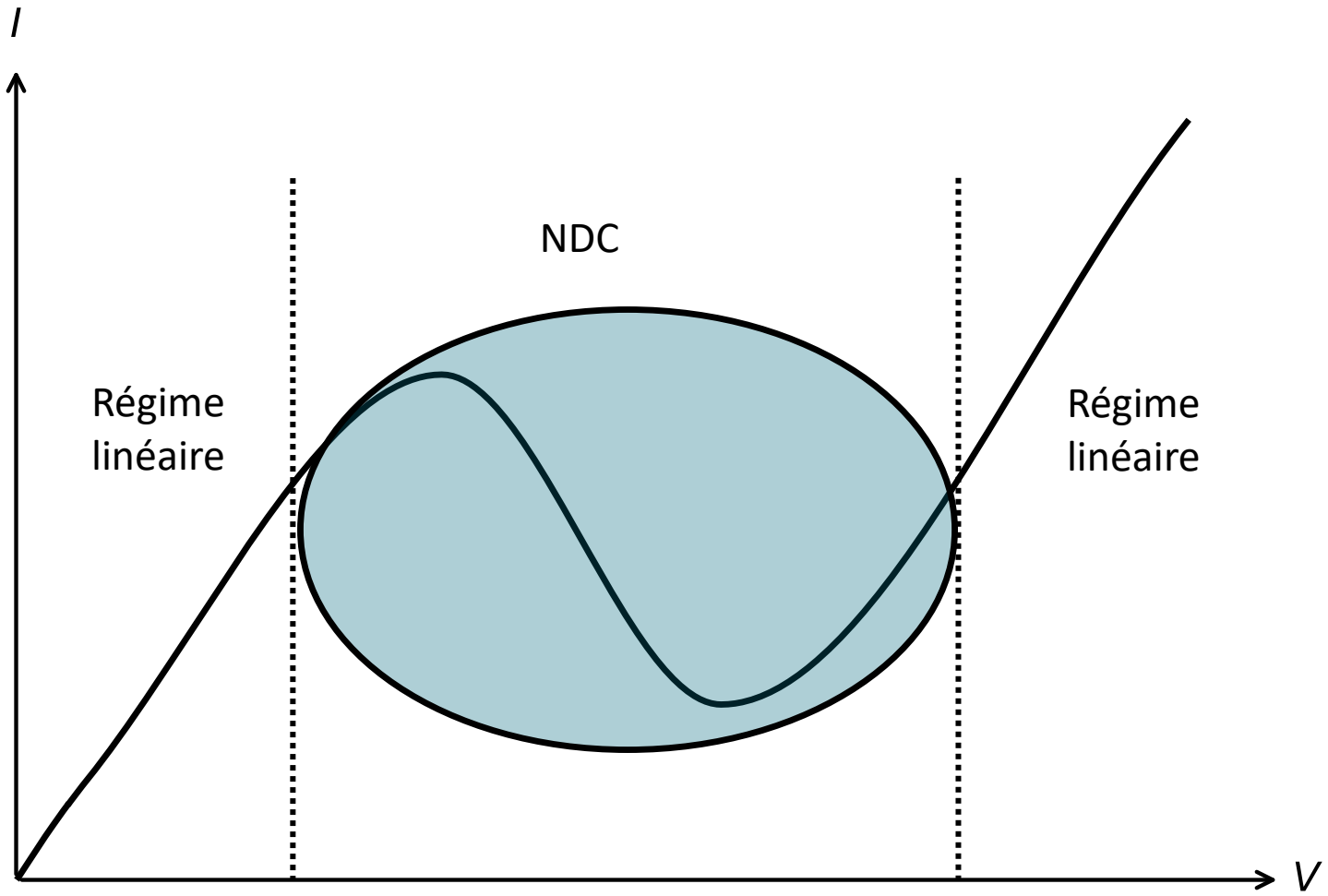
- On passe d'une courbe en « N » à une courbe en « S »

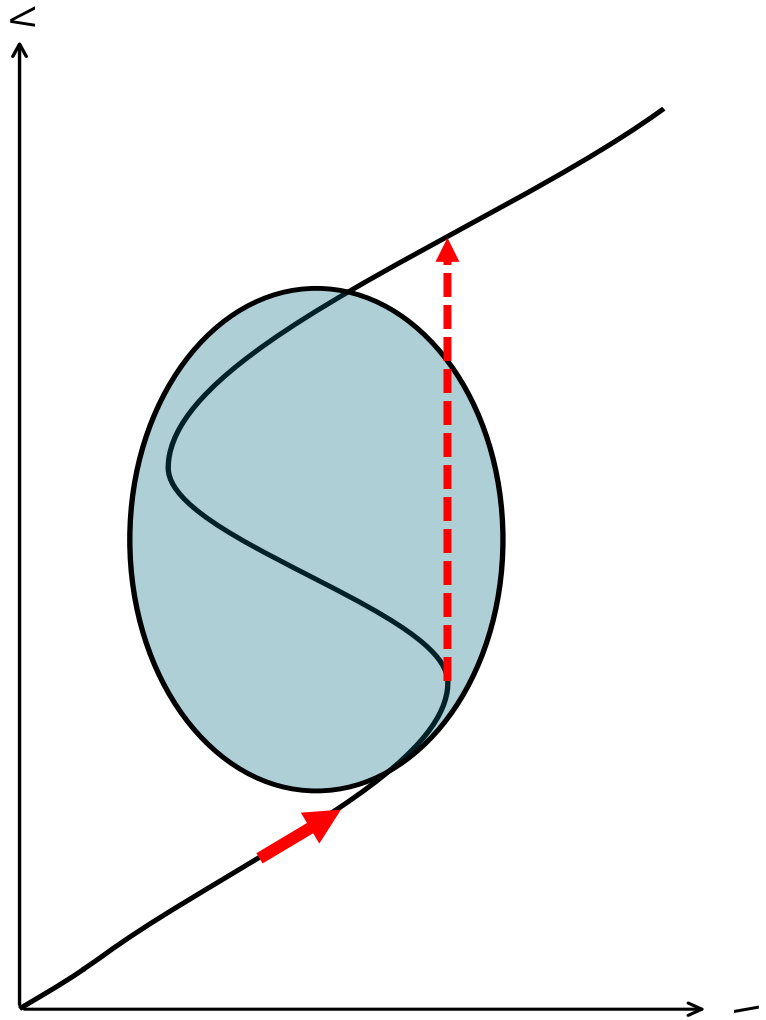
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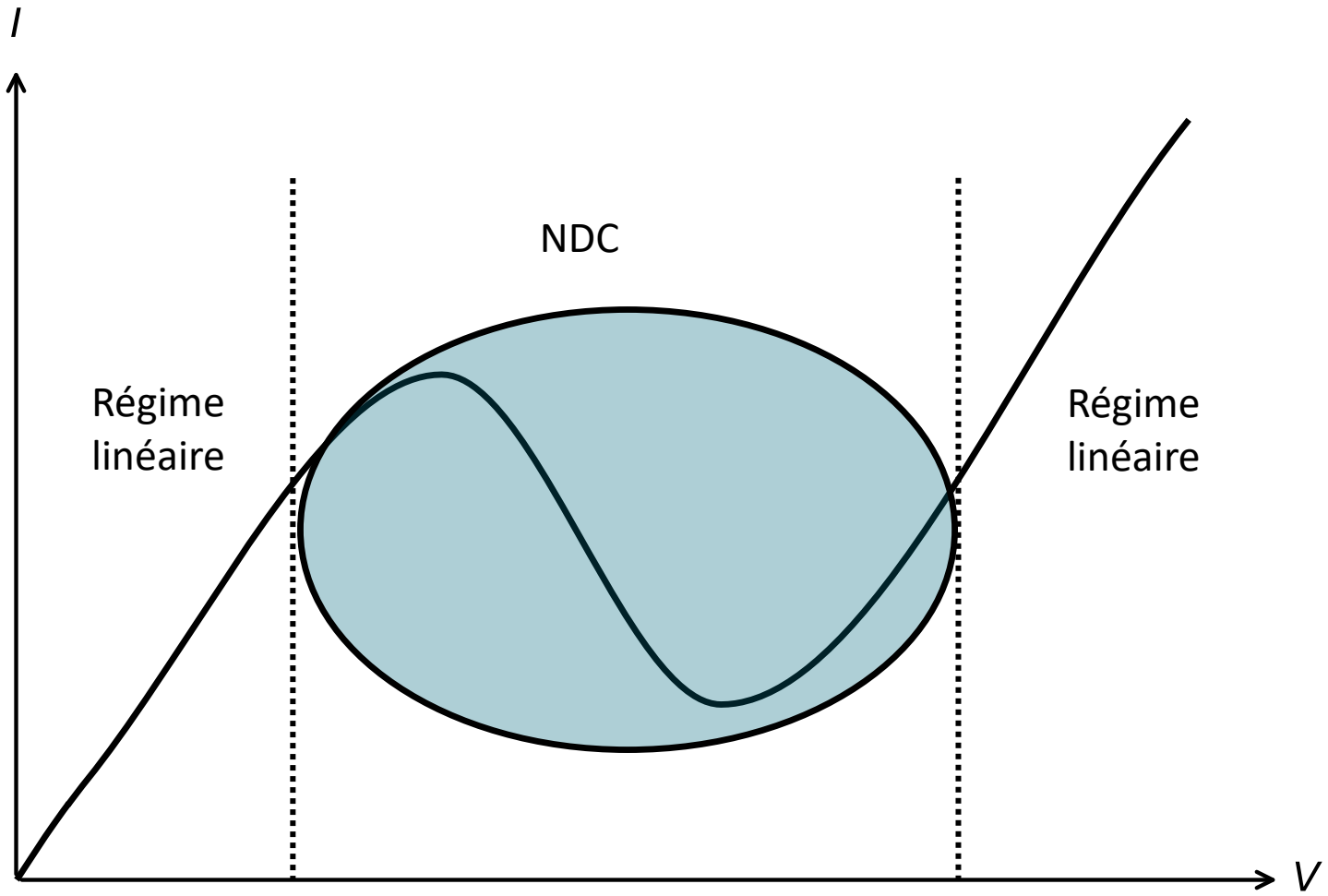


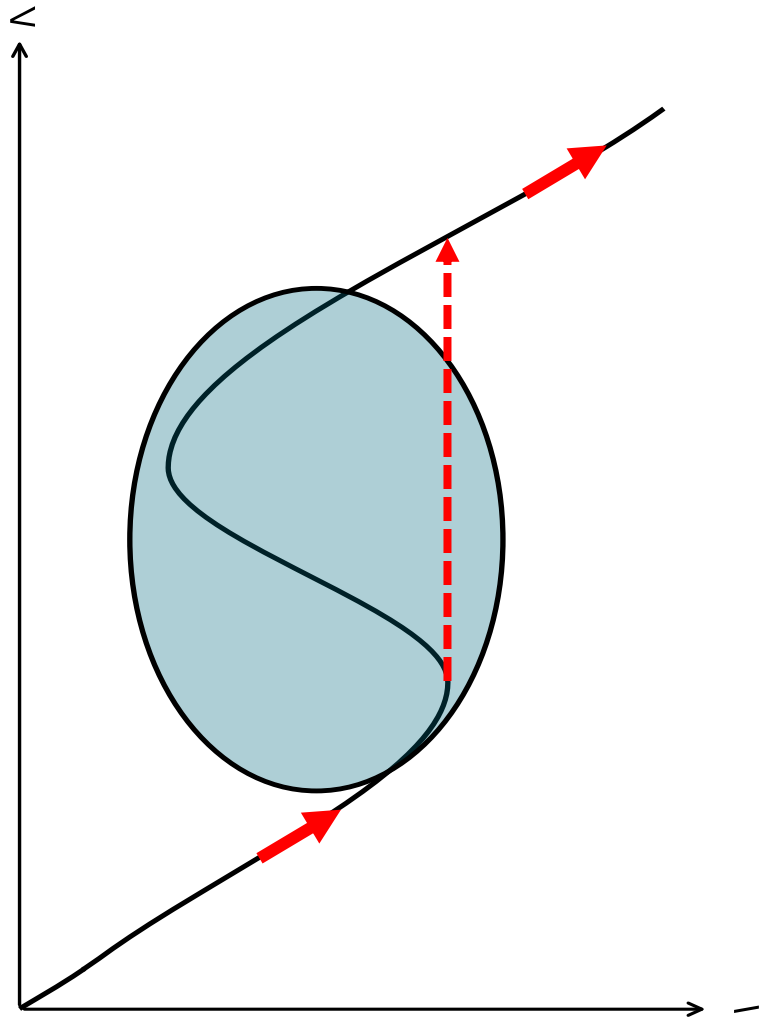
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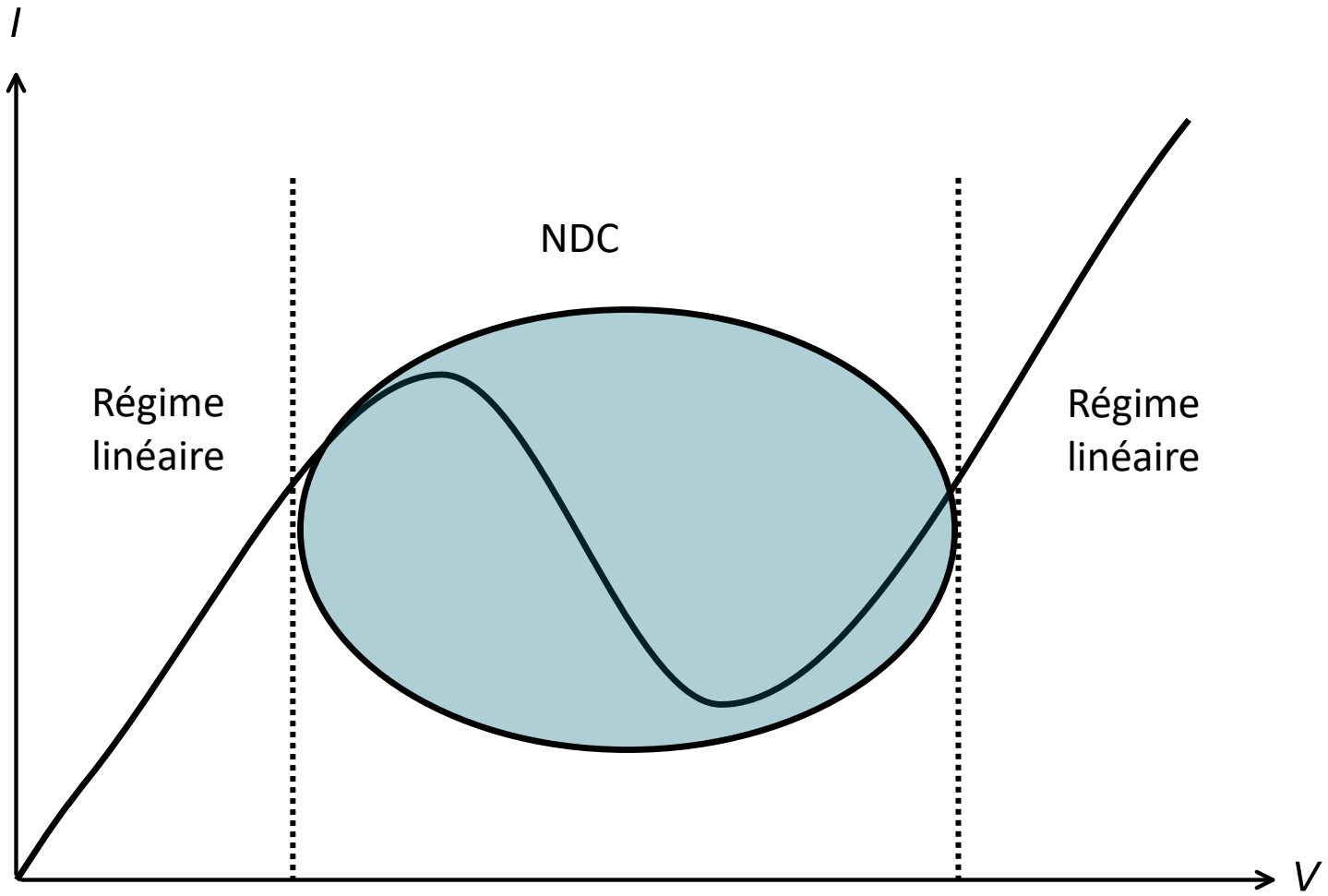


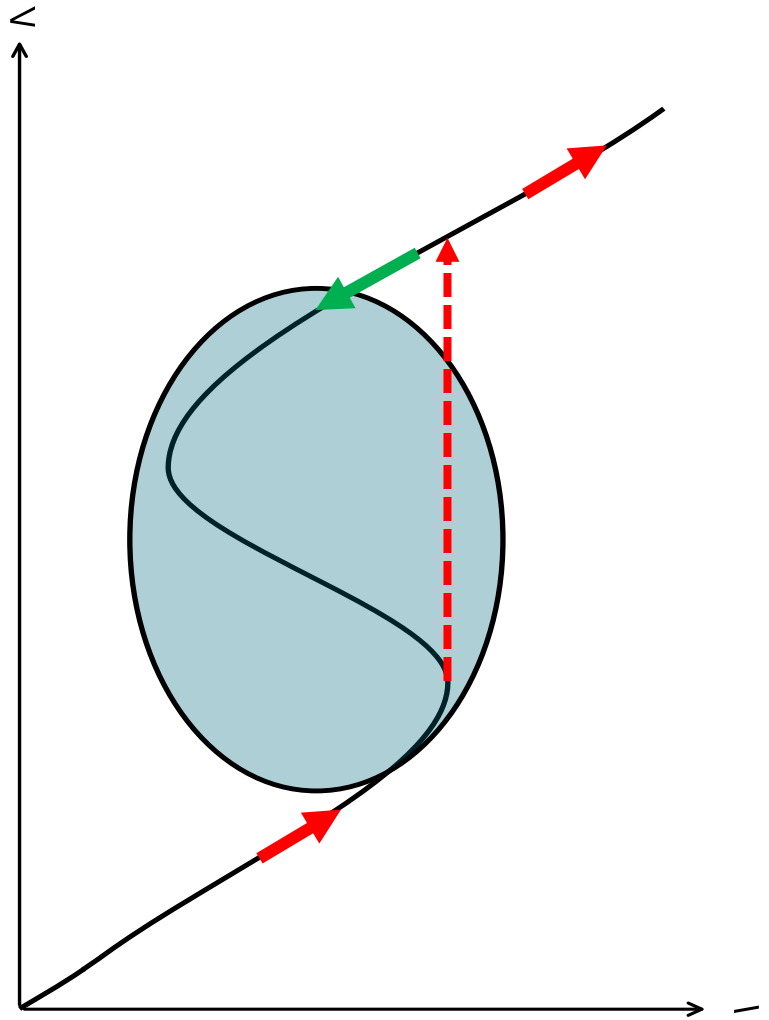
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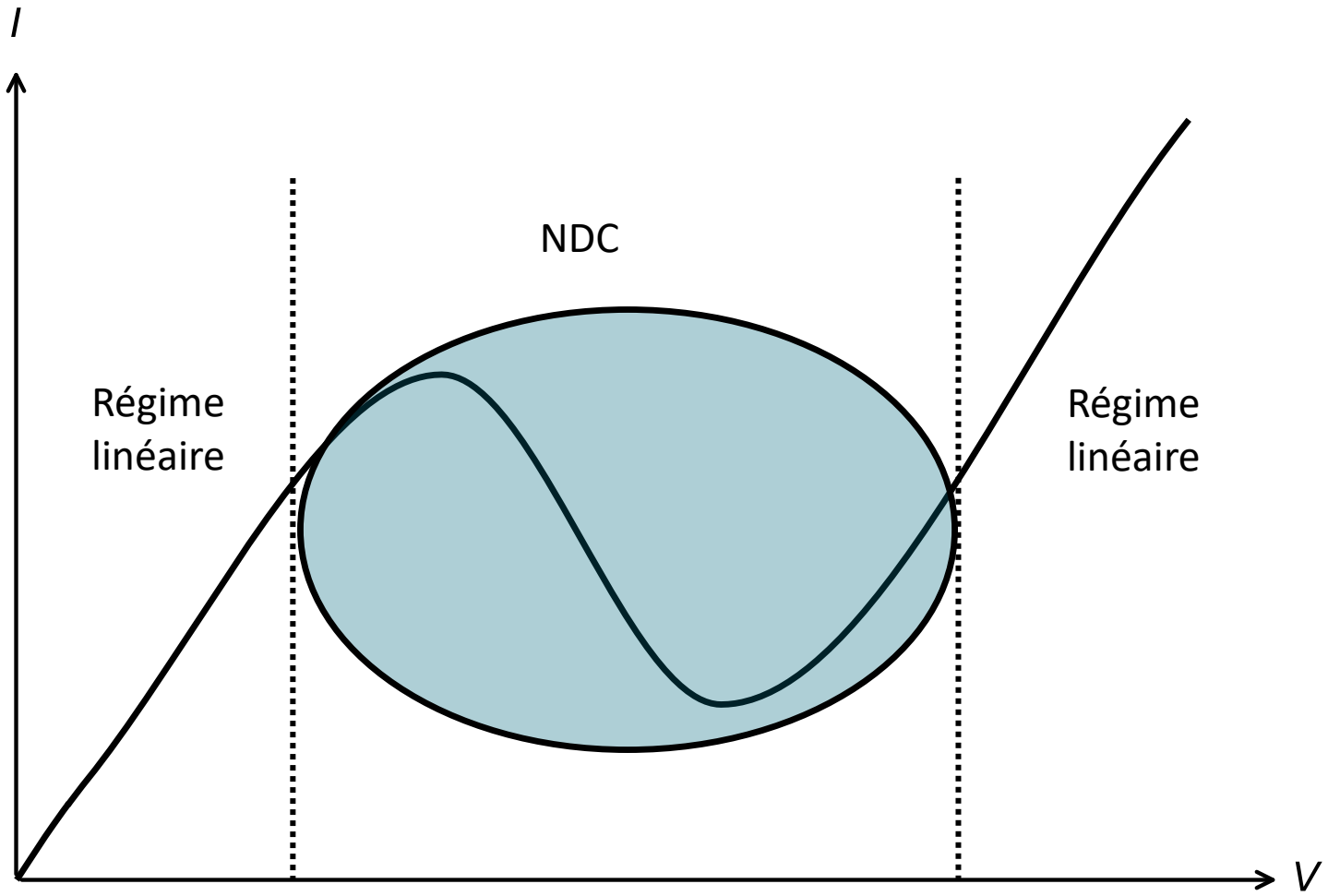


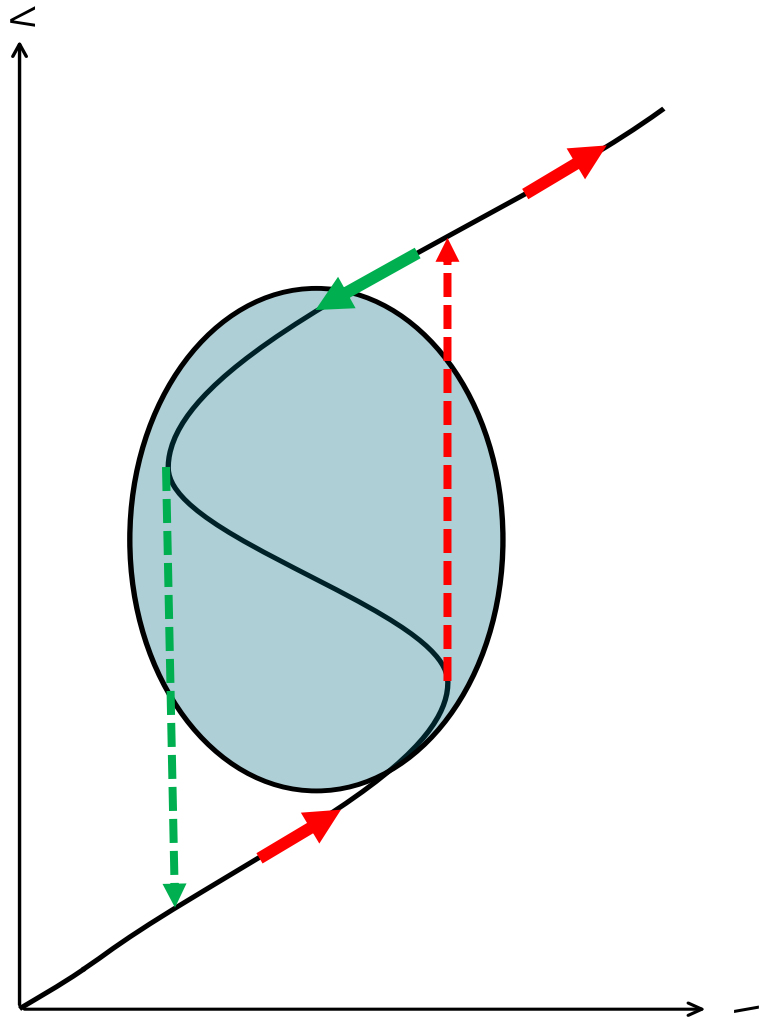
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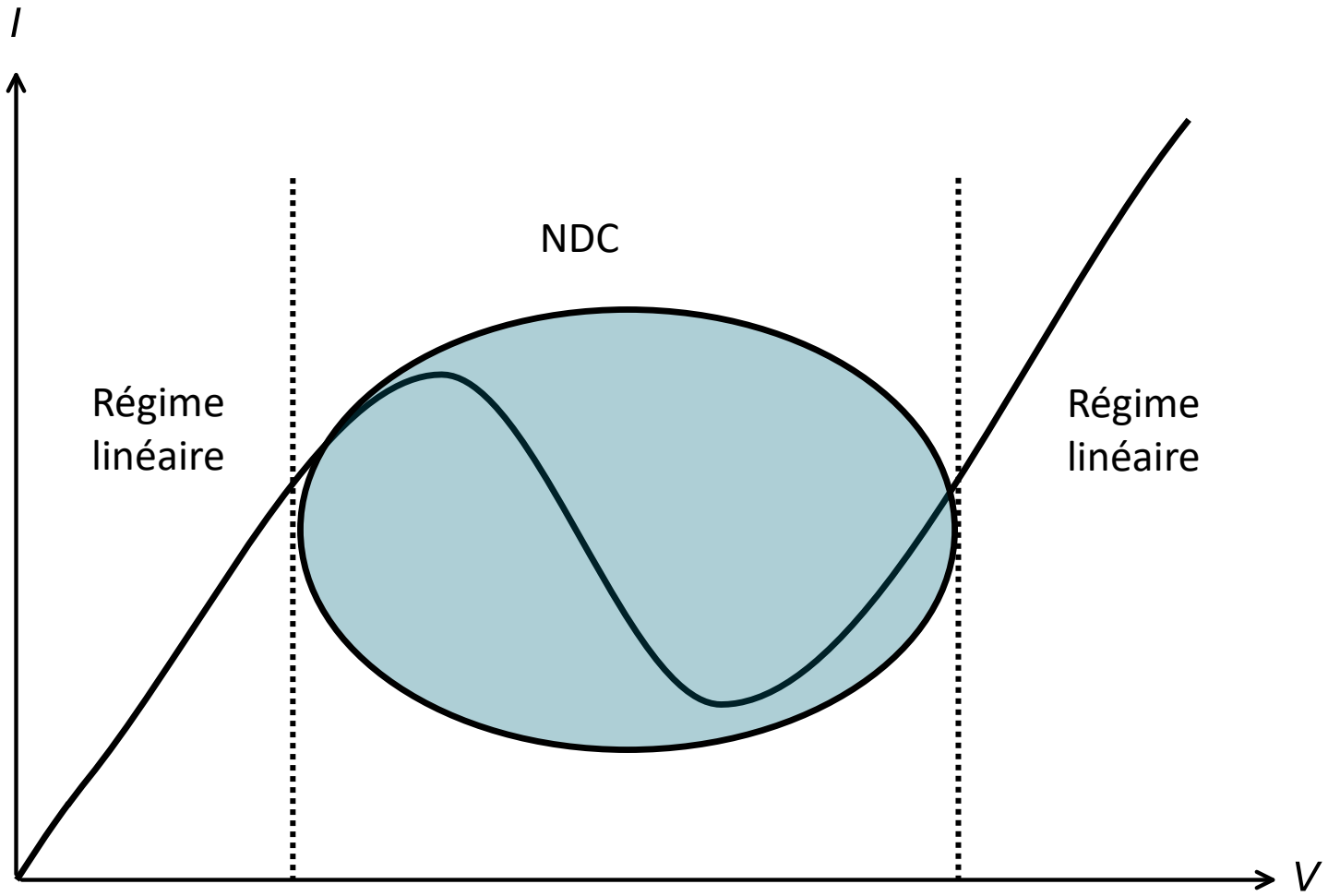


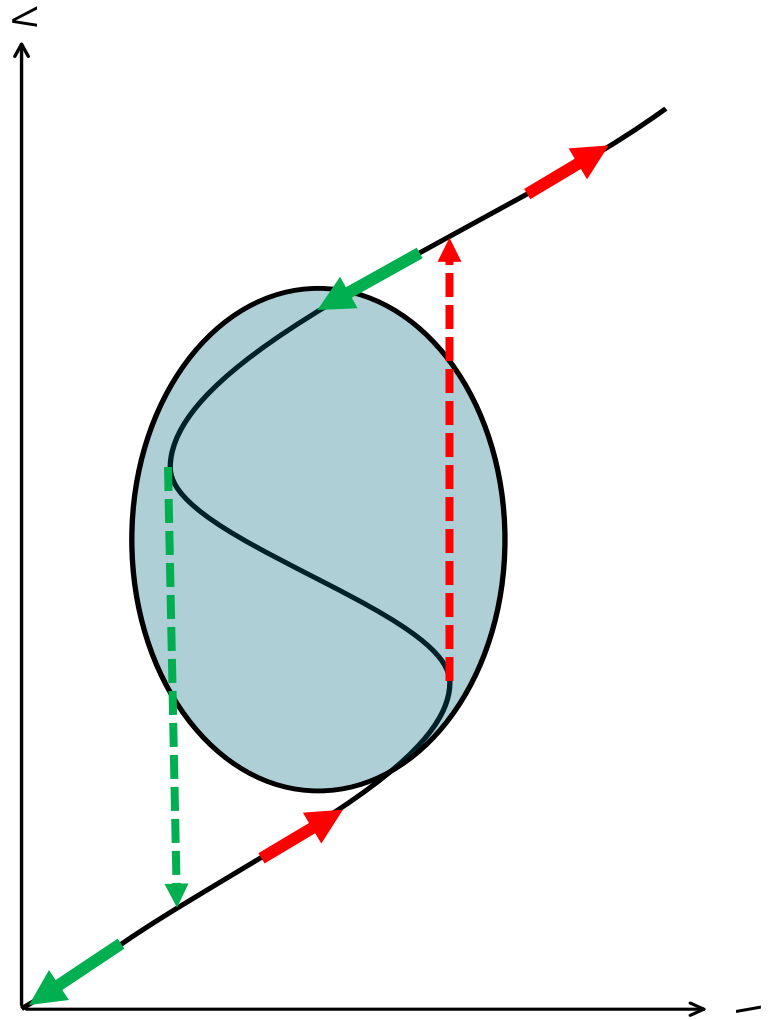
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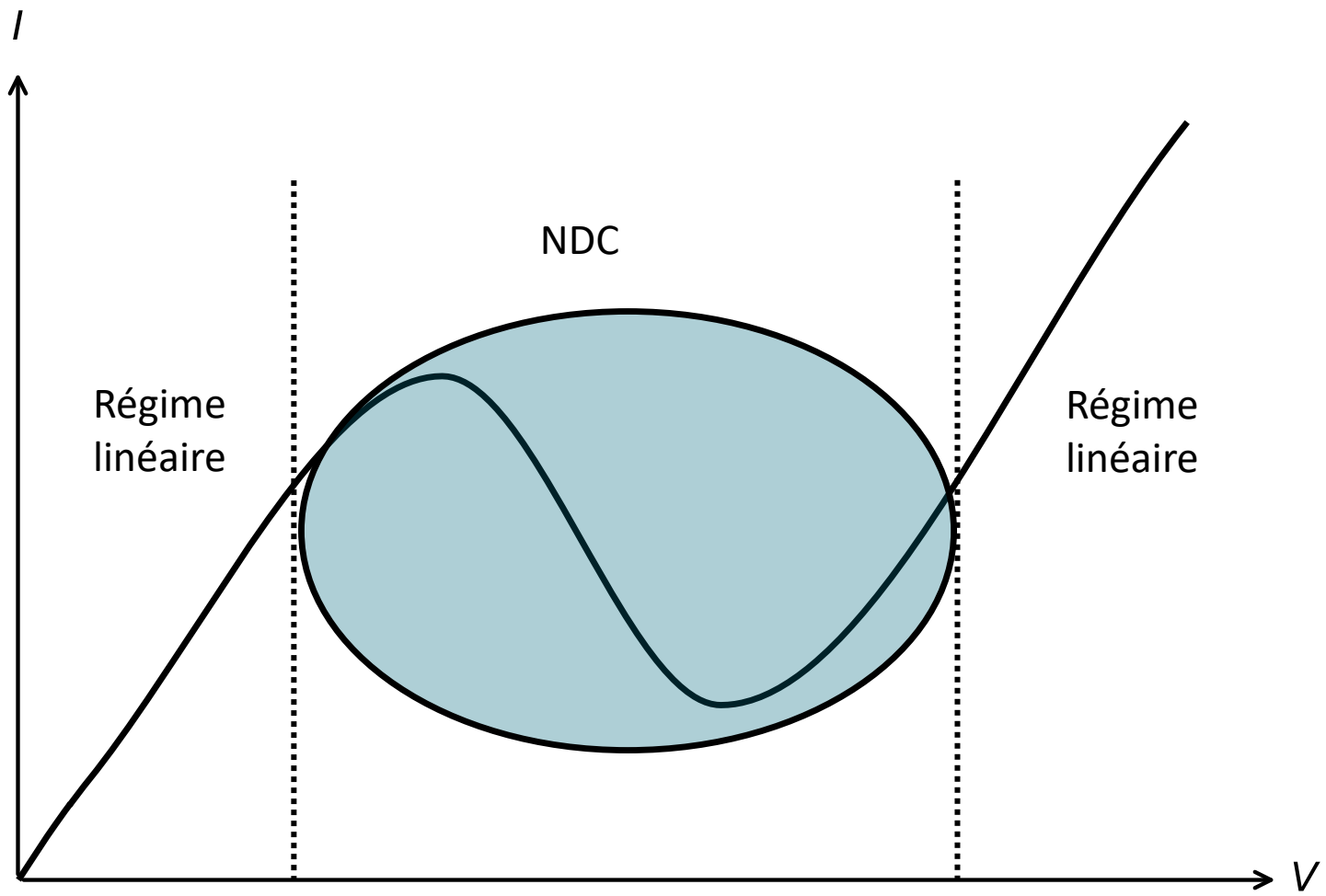


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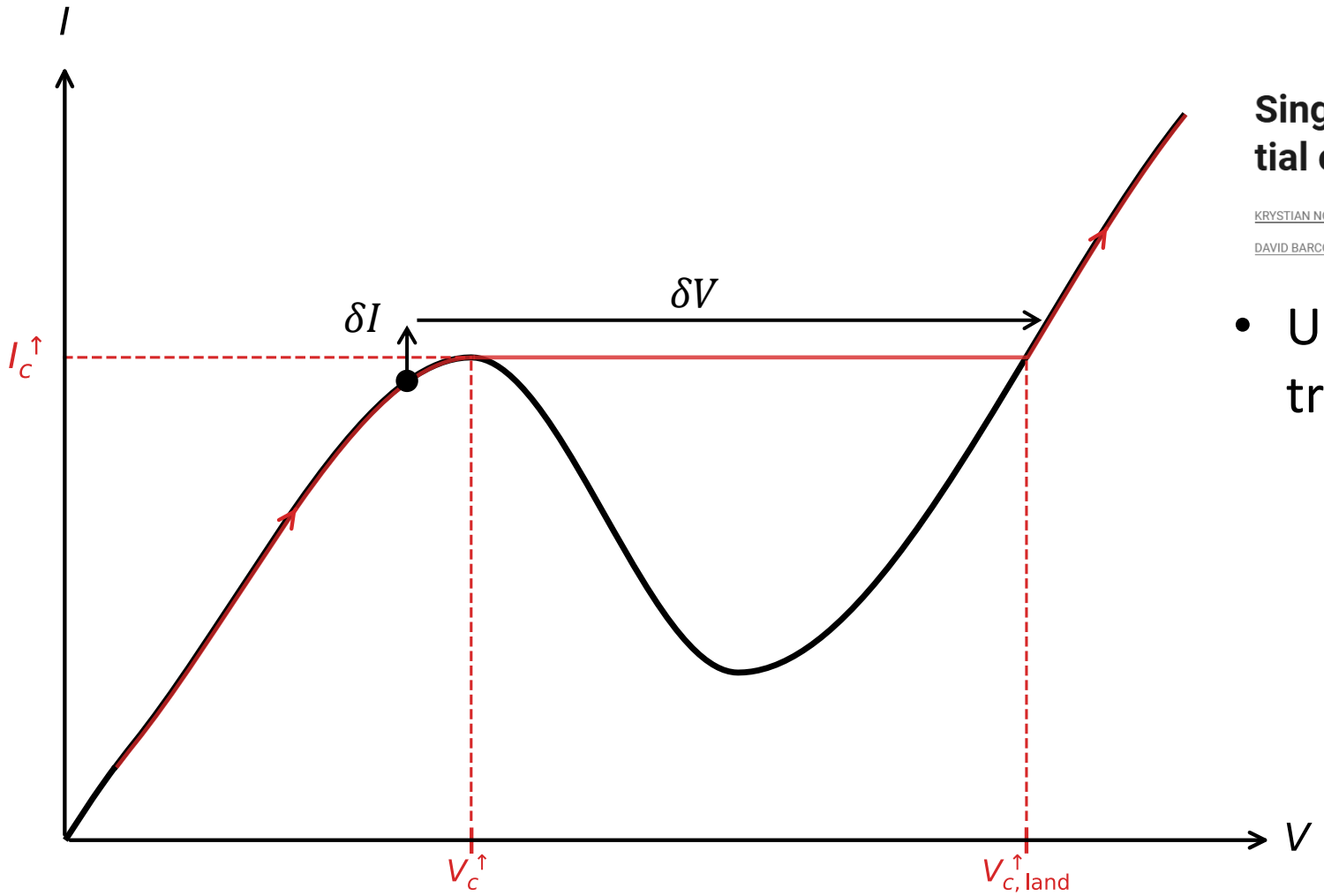


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- **Pourquoi c'est intéressant ?**
  - Bistabilité
  - Hystérésis

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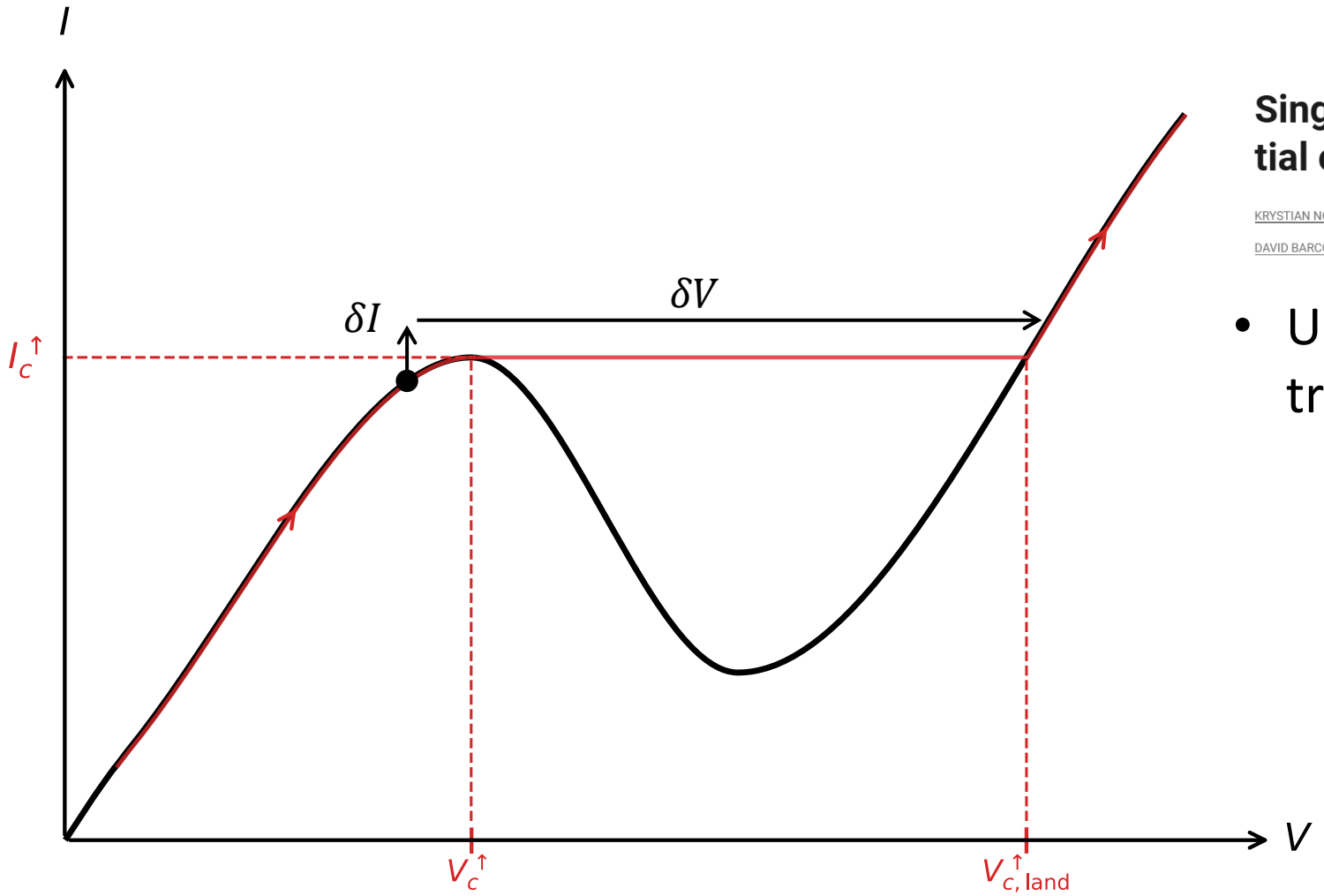


Single-photon detection enabled by negative differential conductivity in moiré superlattices

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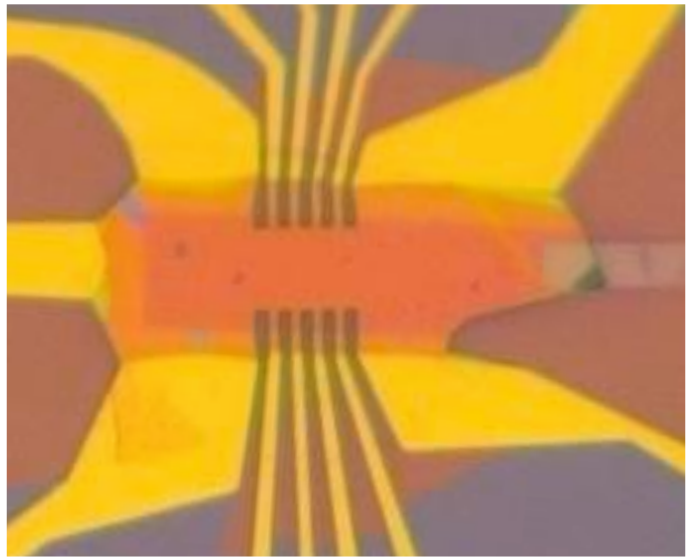
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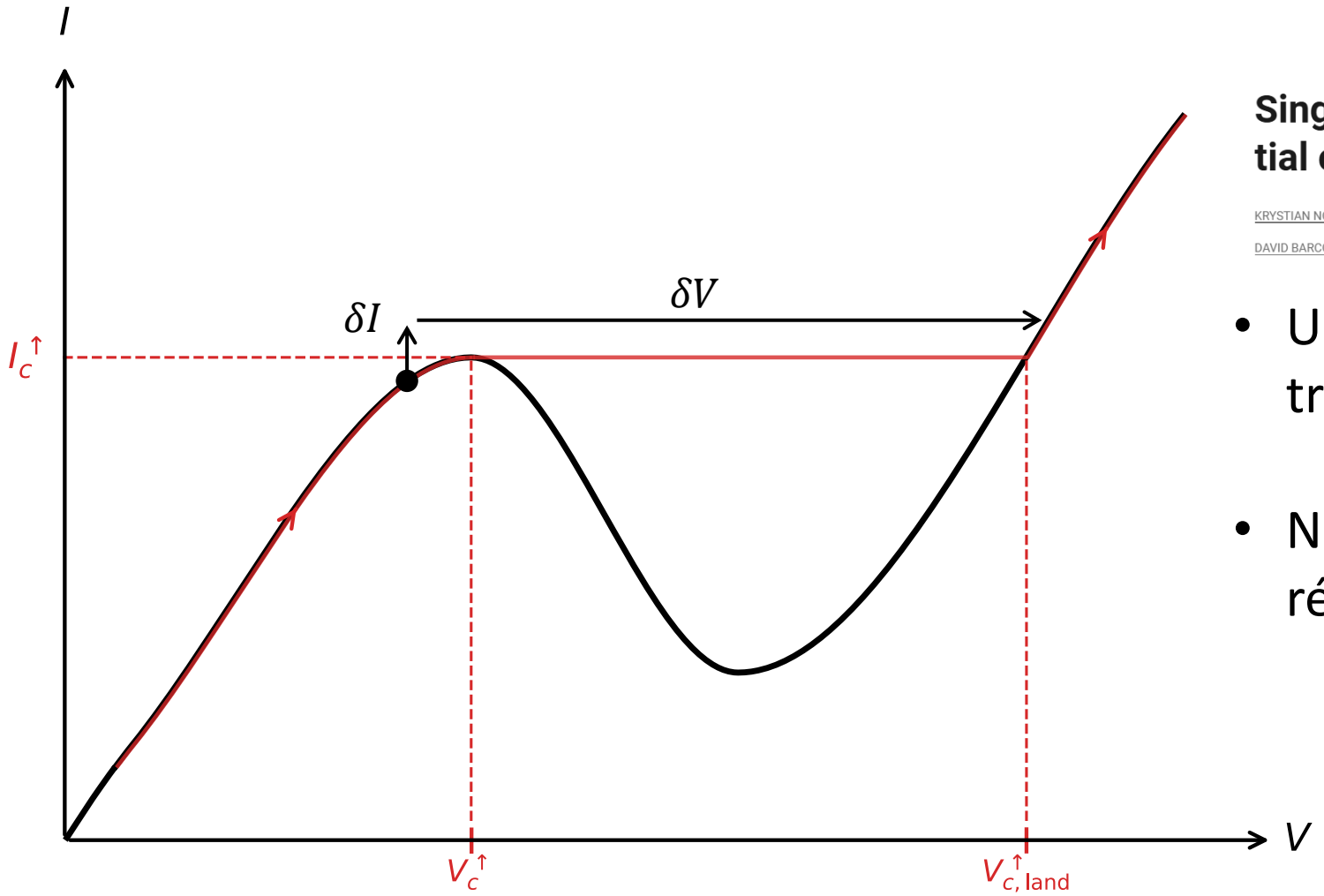
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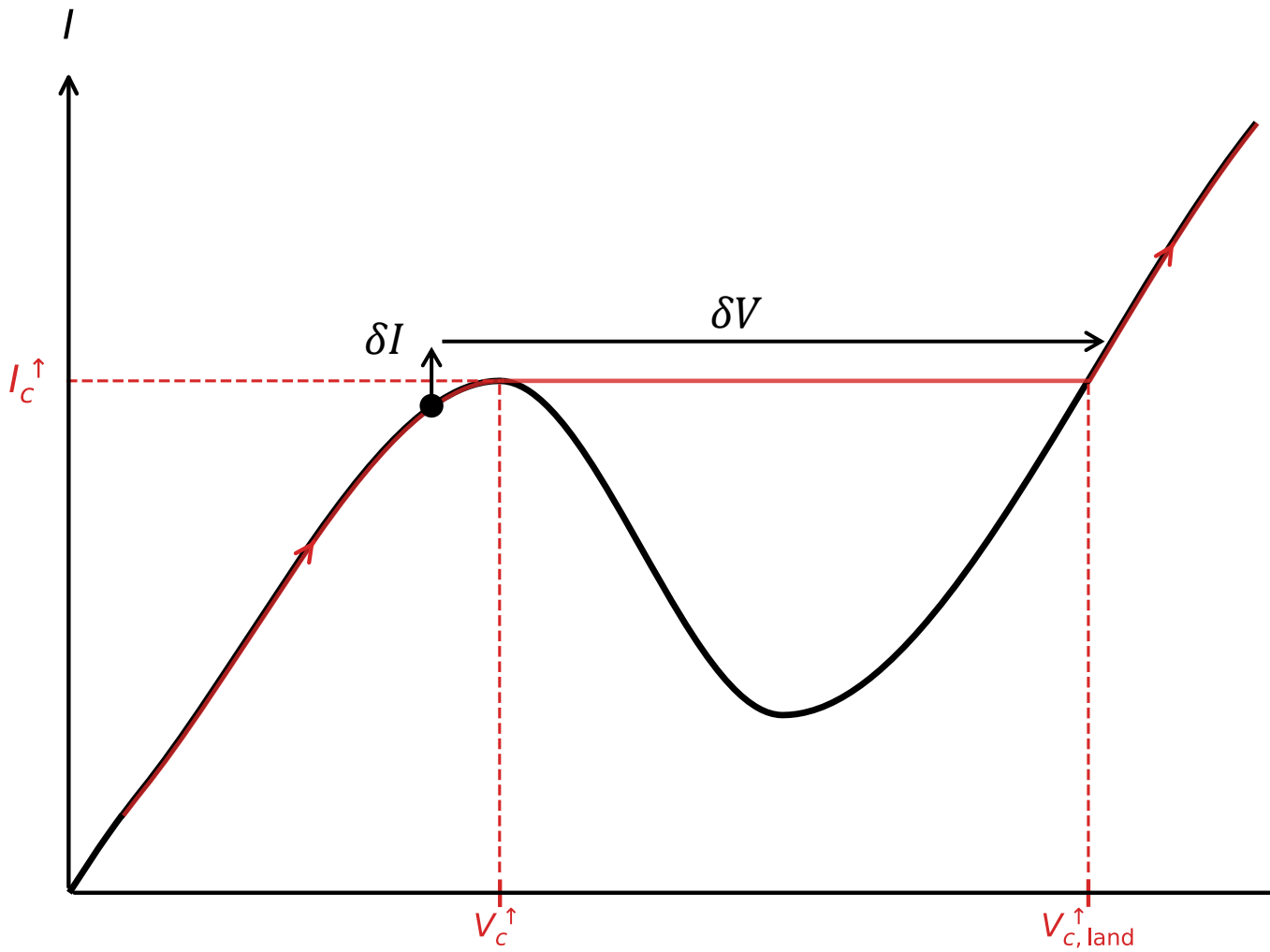


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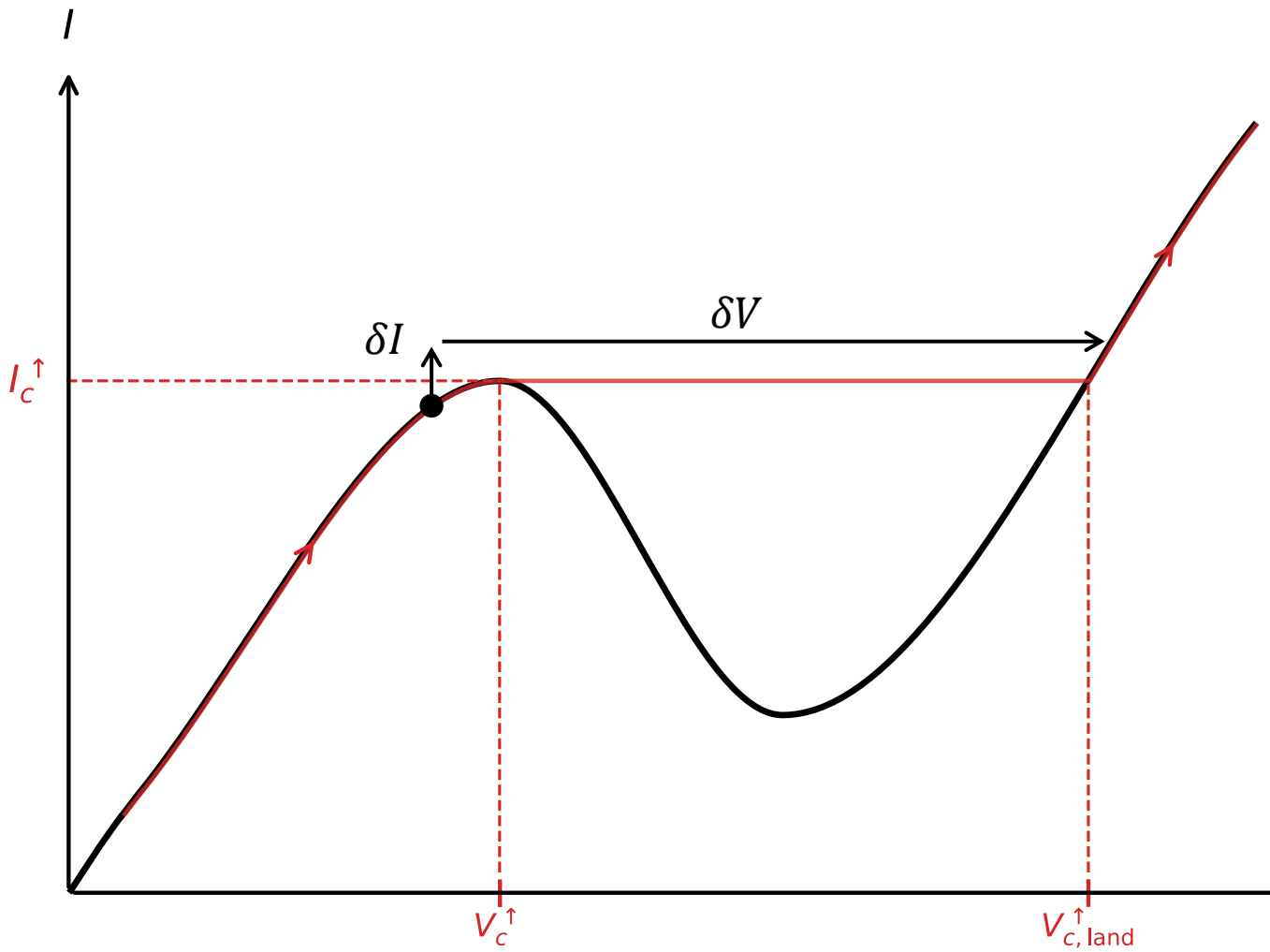


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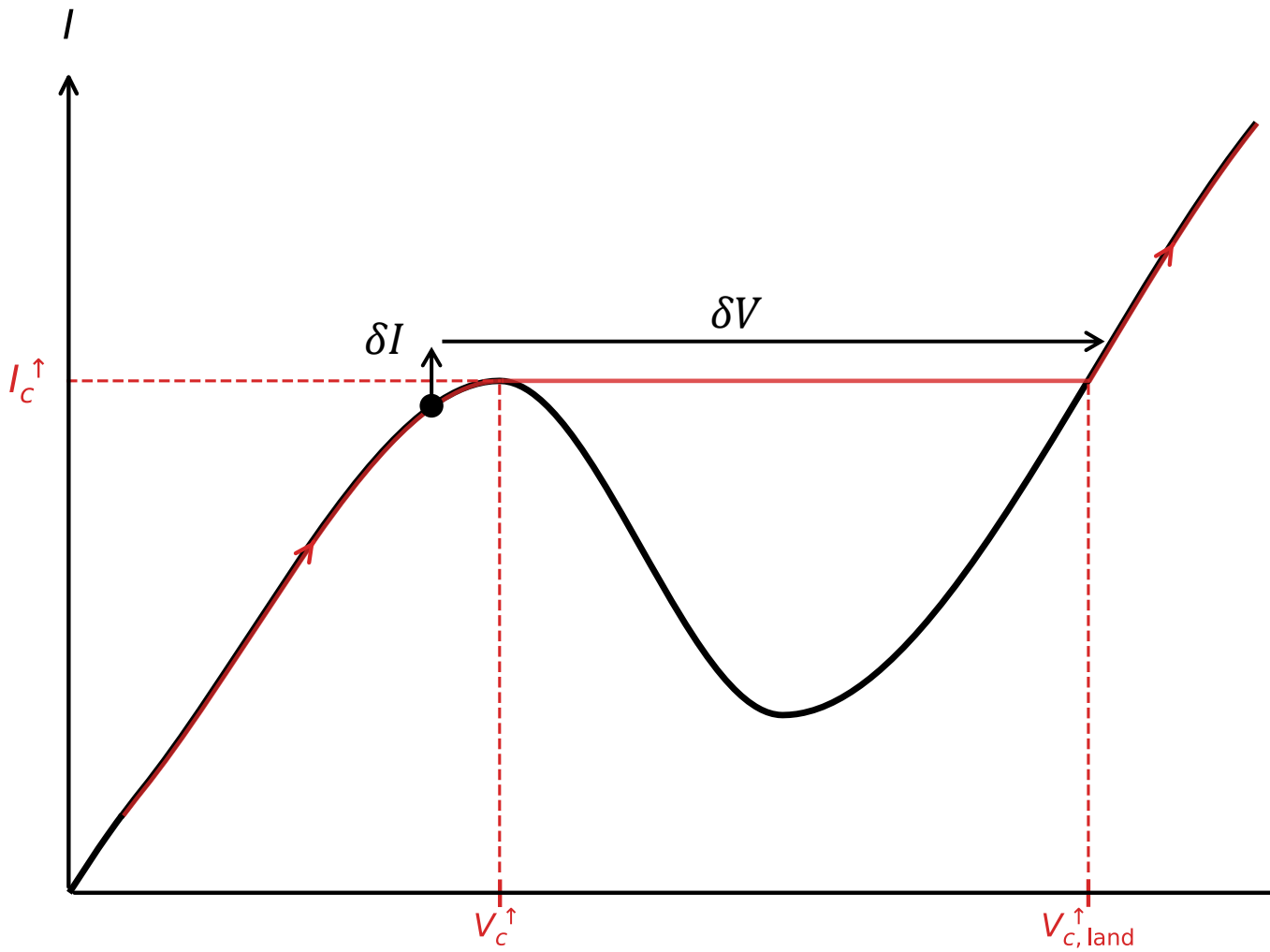


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Quel est le comportement spatial ?

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  - Locale ?
  - Affecte par les impuretés ?
  - Affecte par la géométrie ?

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- A quoi ressemble la NDC spatialement ?
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  - Locale ?
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  - Affecte par la géométrie ?

- C’est essentiel à caractériser pour comprendre et renforcer le phénomène de NDC dans notre échantillon

Plan :

- Introduction
  - théorie de la NDC en transport
  - Pourquoi les matériaux de moiré

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  - Mesure en fonction du biais (courant et tension)
  - Mesure en fonction de la densité de porteurs de charge

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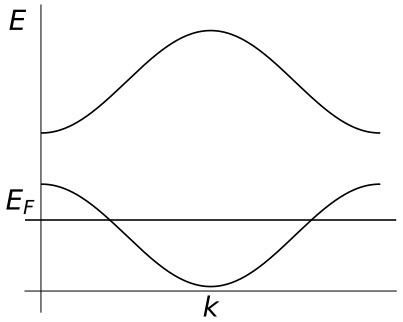
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- Mesures locales de photo-tension
  - Le principe
  - L’importance du self-gating

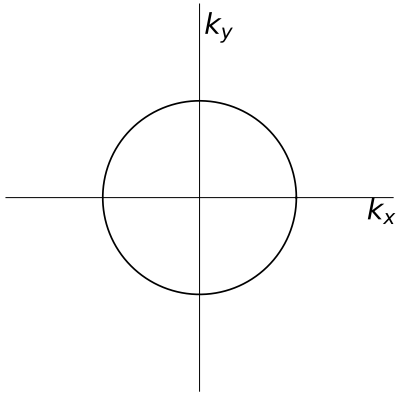
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- Simulations

Introduction

La NDC dans les criticités de transport hors équilibre



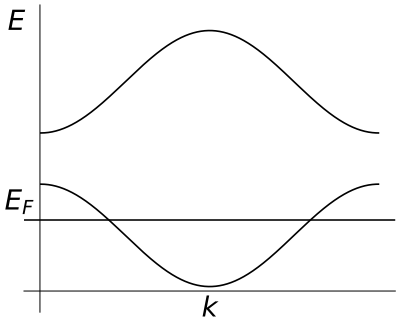


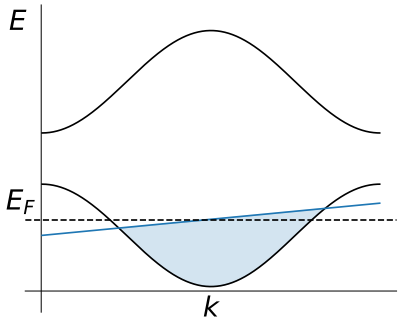
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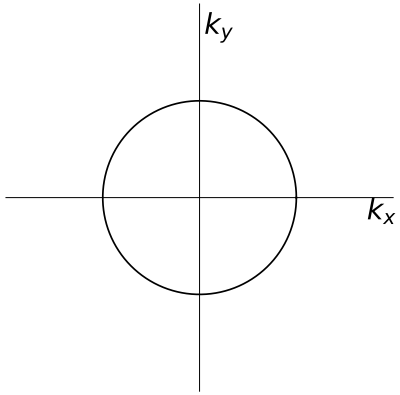
- Equilibre

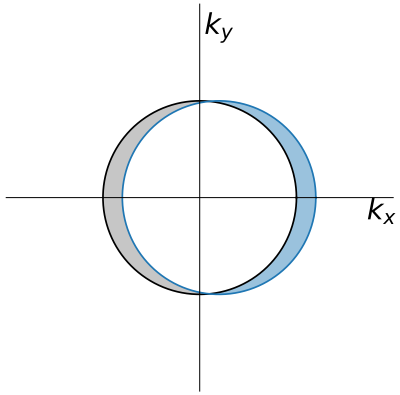


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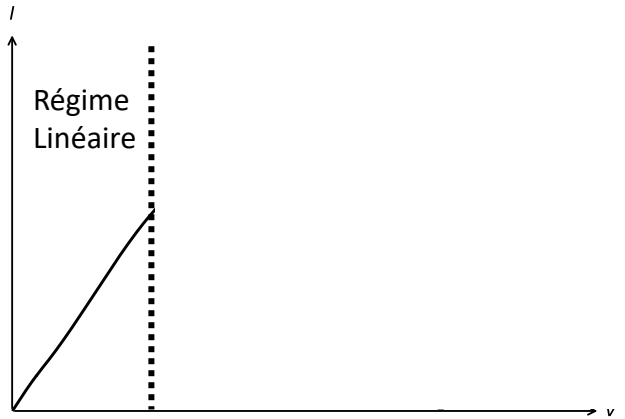




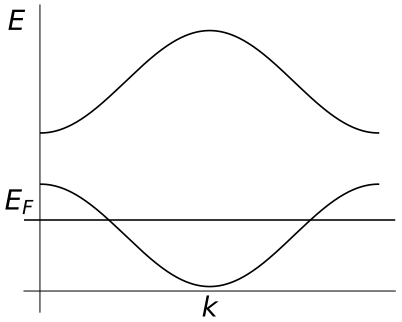
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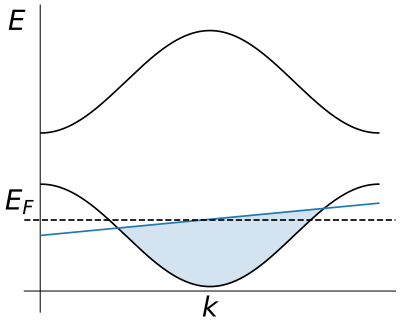
$J = nev$

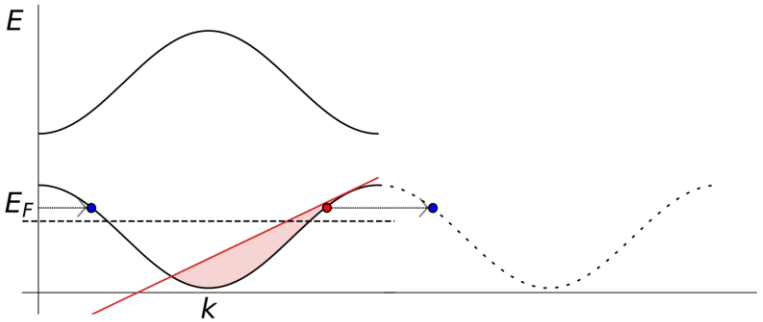
- Equilibre
- Régime linéaire



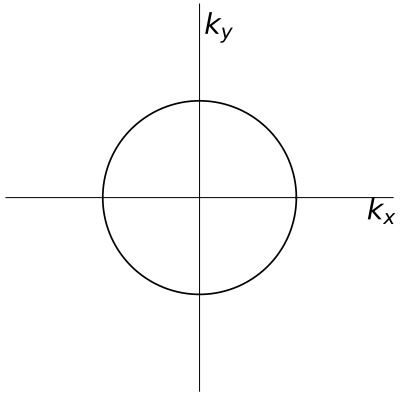
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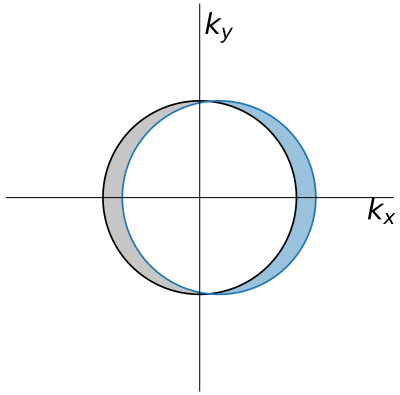


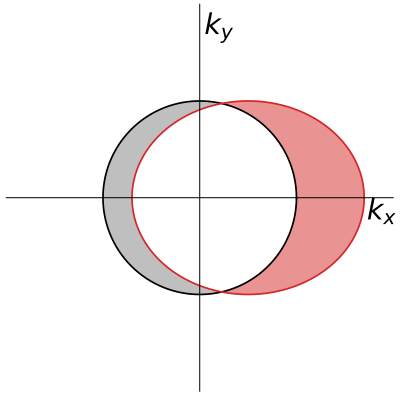




- Equilibre
- Régime linéaire
- Saturation et NDC







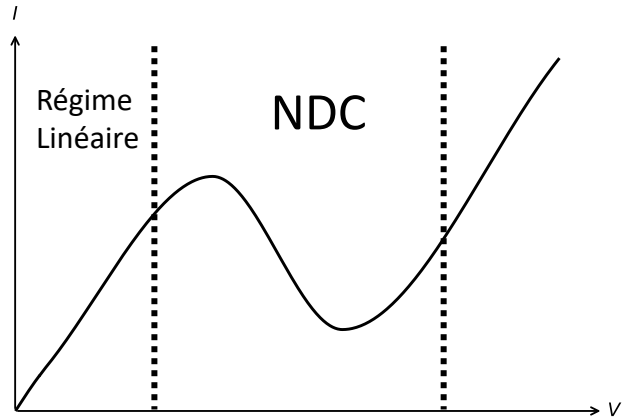
J = 0

$J = nev$

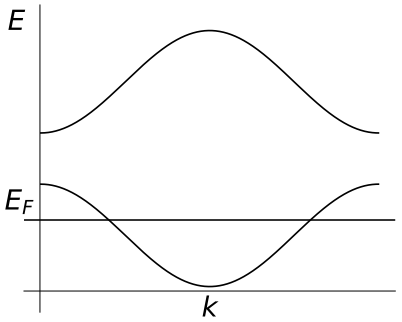
$J_C = nev_F$

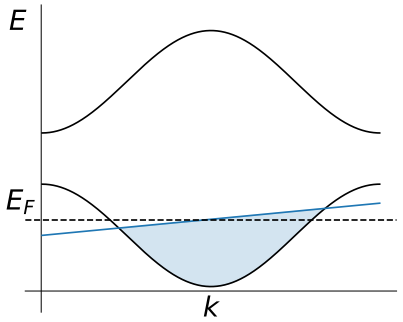
Out-of-equilibrium criticalities in graphene superlattices

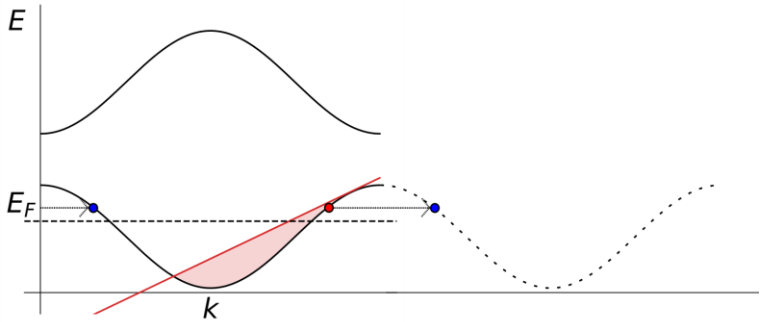
ALEXEY I. BERDYUGIN, NA XIN, HAOYANG GAO, SERGEY SLIZOVSKIY, ZHIYU DONG, SHUBHADEEP BHATTACHARJEE, P. KUMARAVADIVEL, SHUIGANG XU, L. A. PONOMARENKO, [...], AND A. K. GEIM +12 authors Authors Info & Affiliations



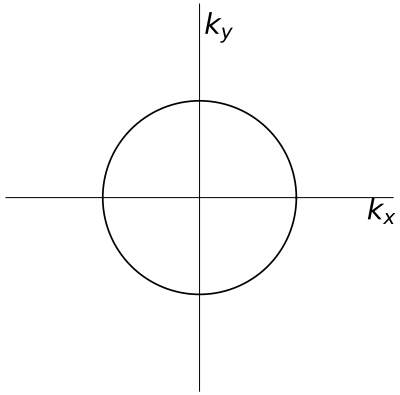
La NDC dans les criticités de transport hors équilibre

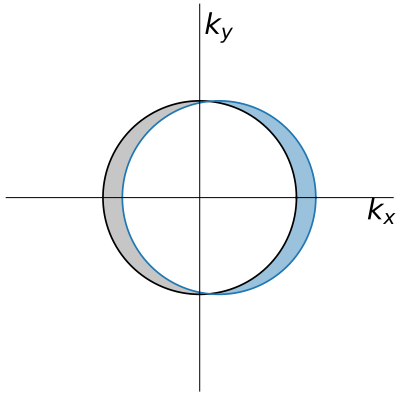


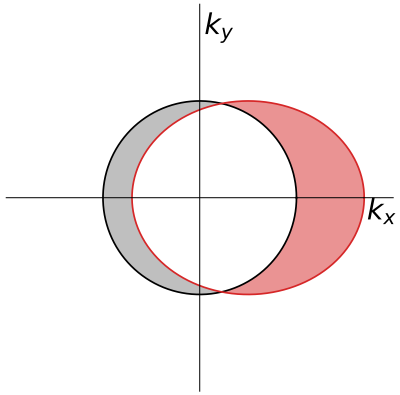




- Equilibre
- Régime linéaire
- Saturation et NDC







J = 0

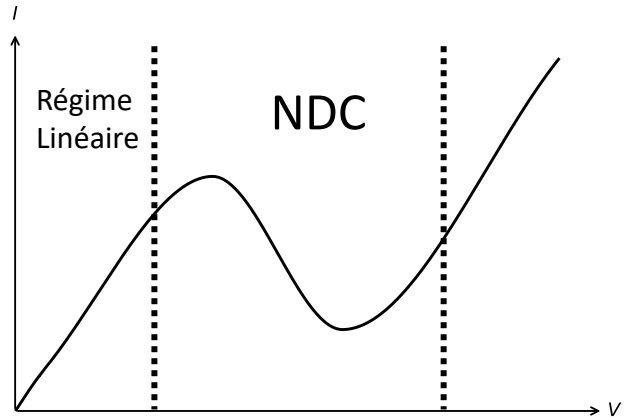
$J = nev$

$J_C = nev_F$

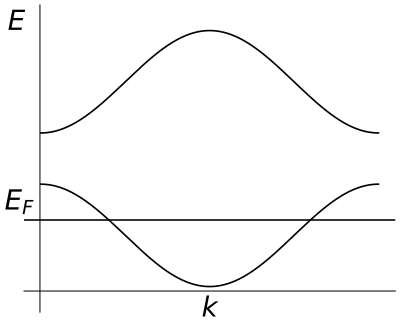
- Pourquoi ne le voit on pas régulièrement ?

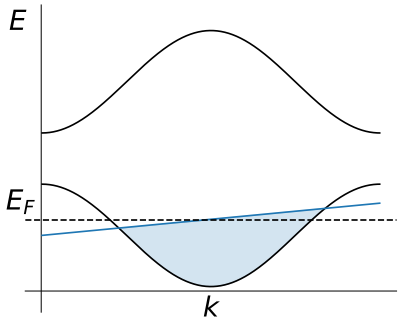
Out-of-equilibrium criticalities in graphene superlattices

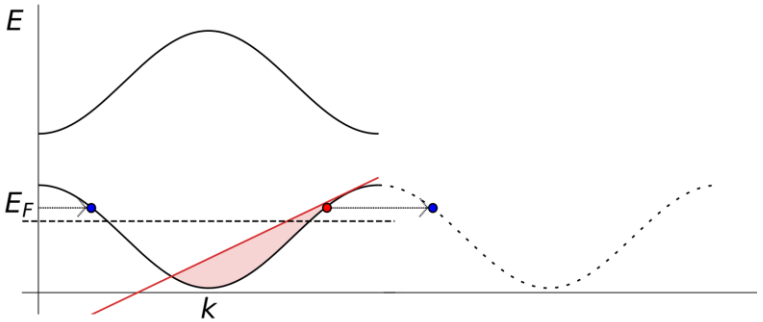
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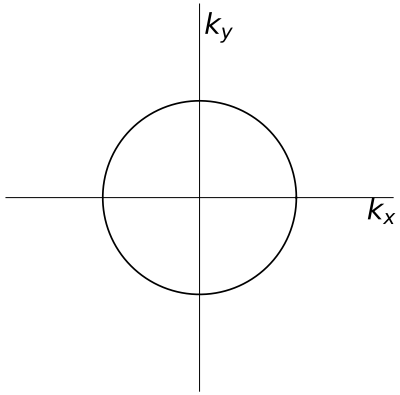
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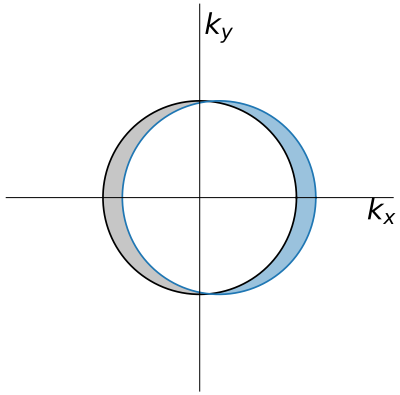


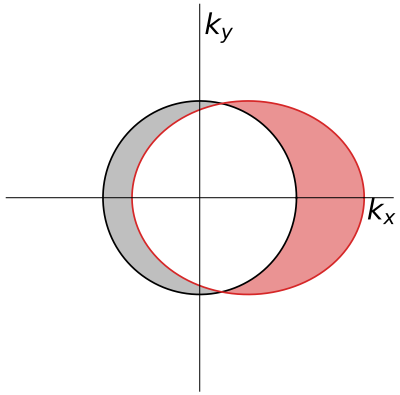




- Equilibre
- Régime linéaire
- Saturation et NDC





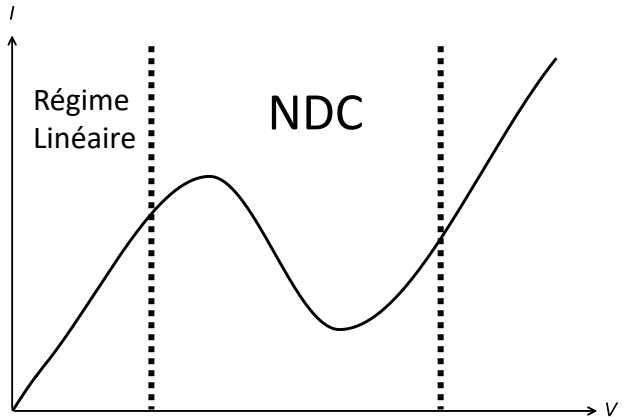


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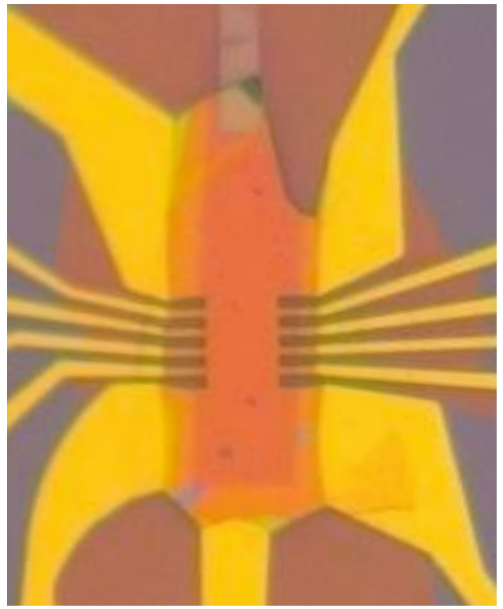
- Un champ E trop important endommage l'échantillon



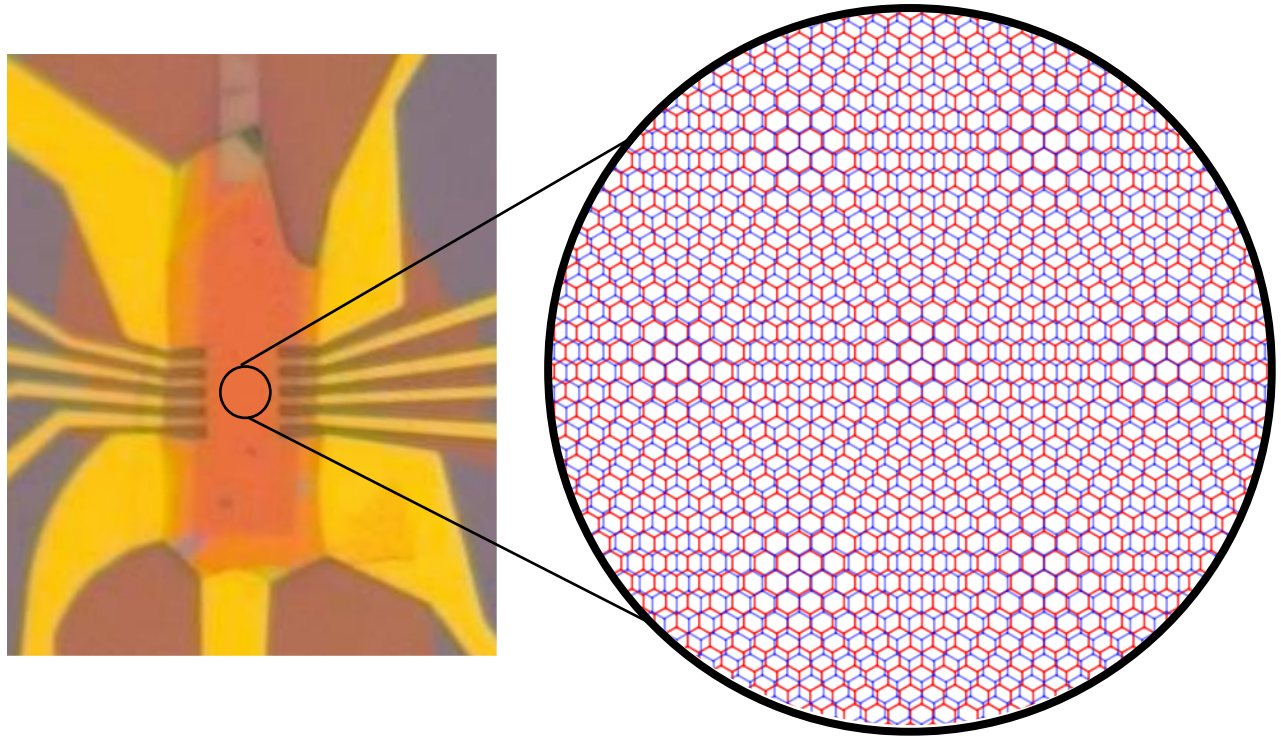
Out-of-equilibrium criticalities in graphene superlattices

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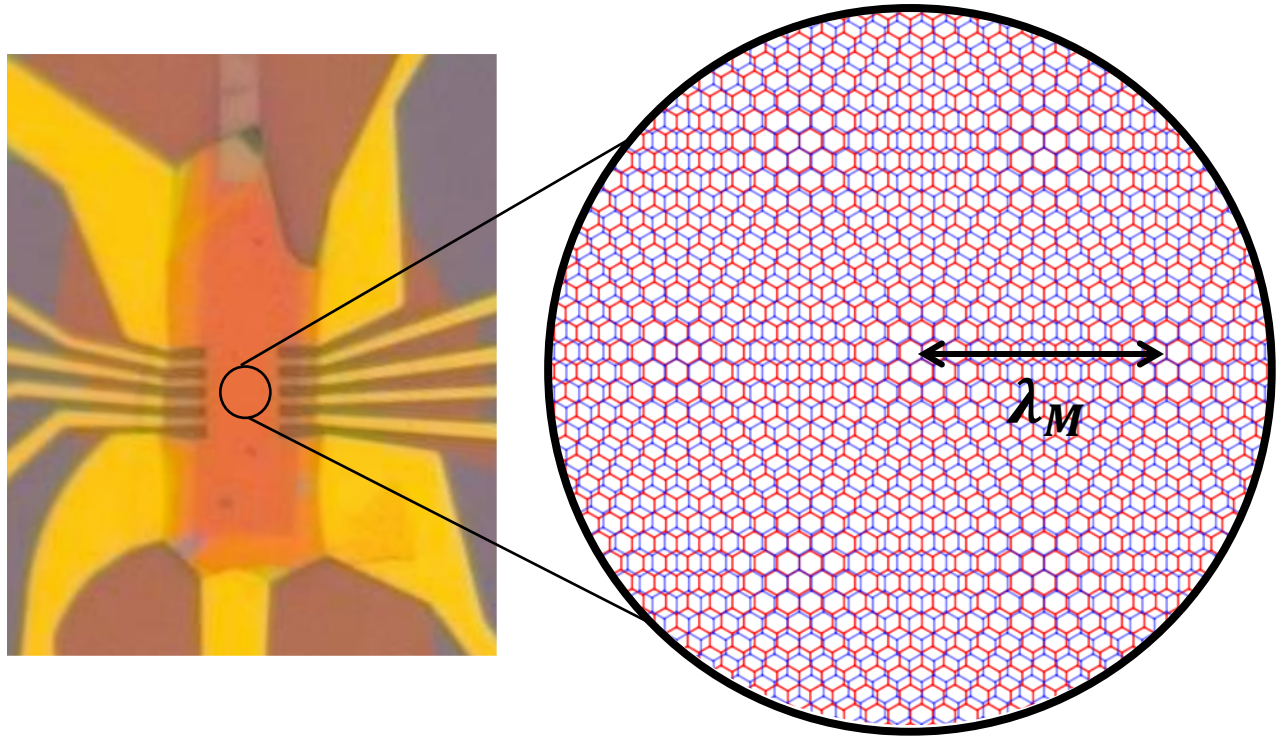
BLG/hBN: un super réseau de moiré aux bandes plates



BLG/hBN: un super réseau de moiré aux bandes plates

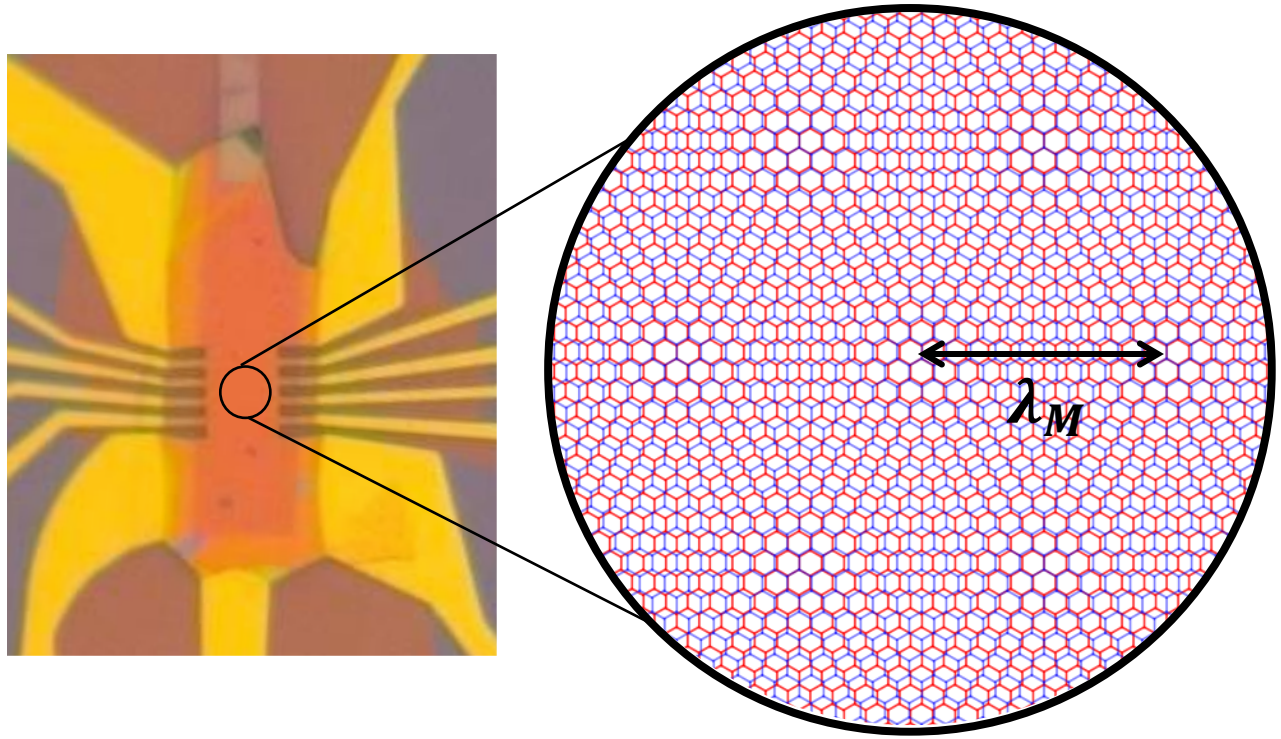


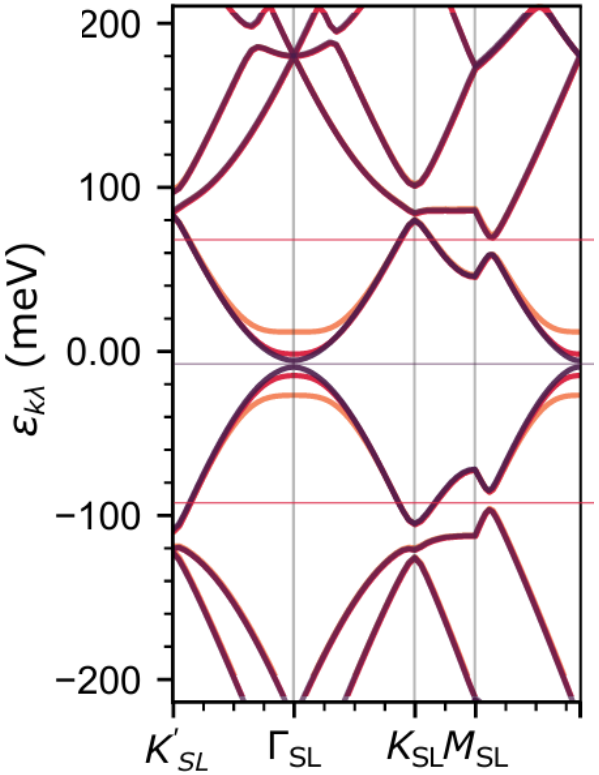
BLG/hBN: un super réseau de moiré aux bandes plates



- Moiré : nouvelle périodicité spatiale

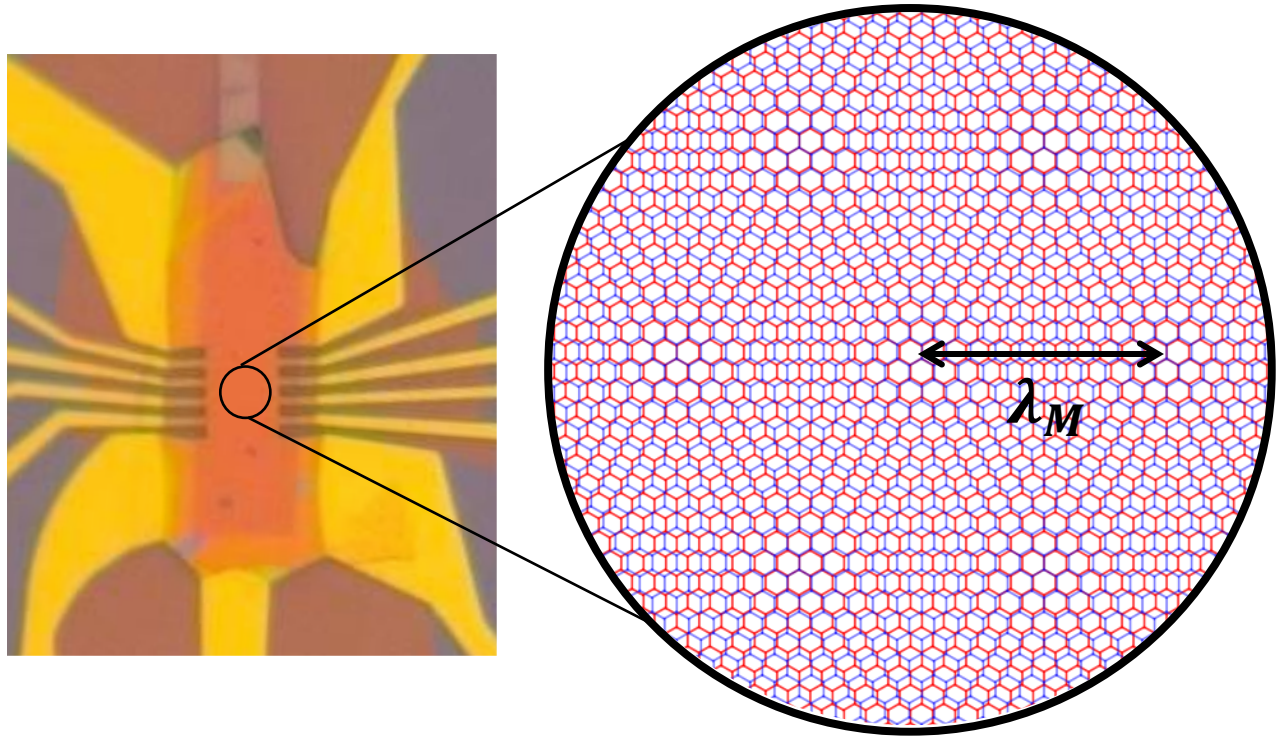
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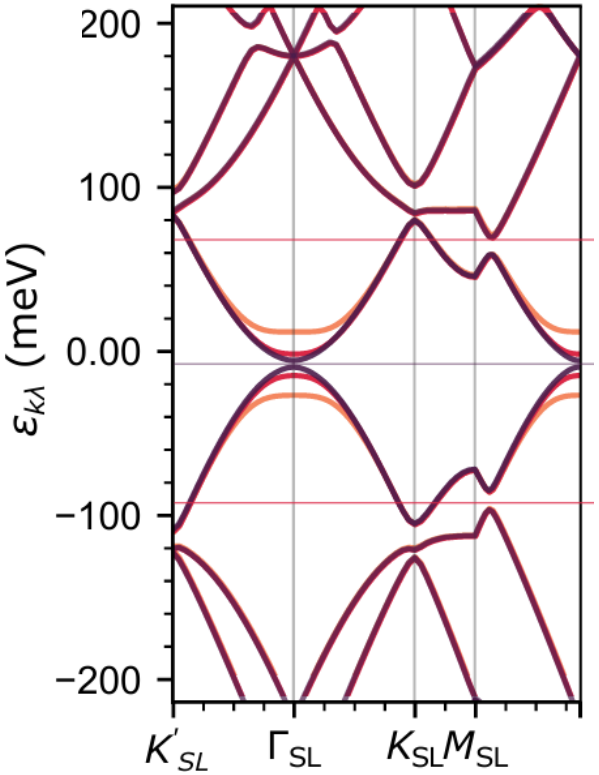




- Moiré : nouvelle périodicité spatiale
- Repliement de bandes

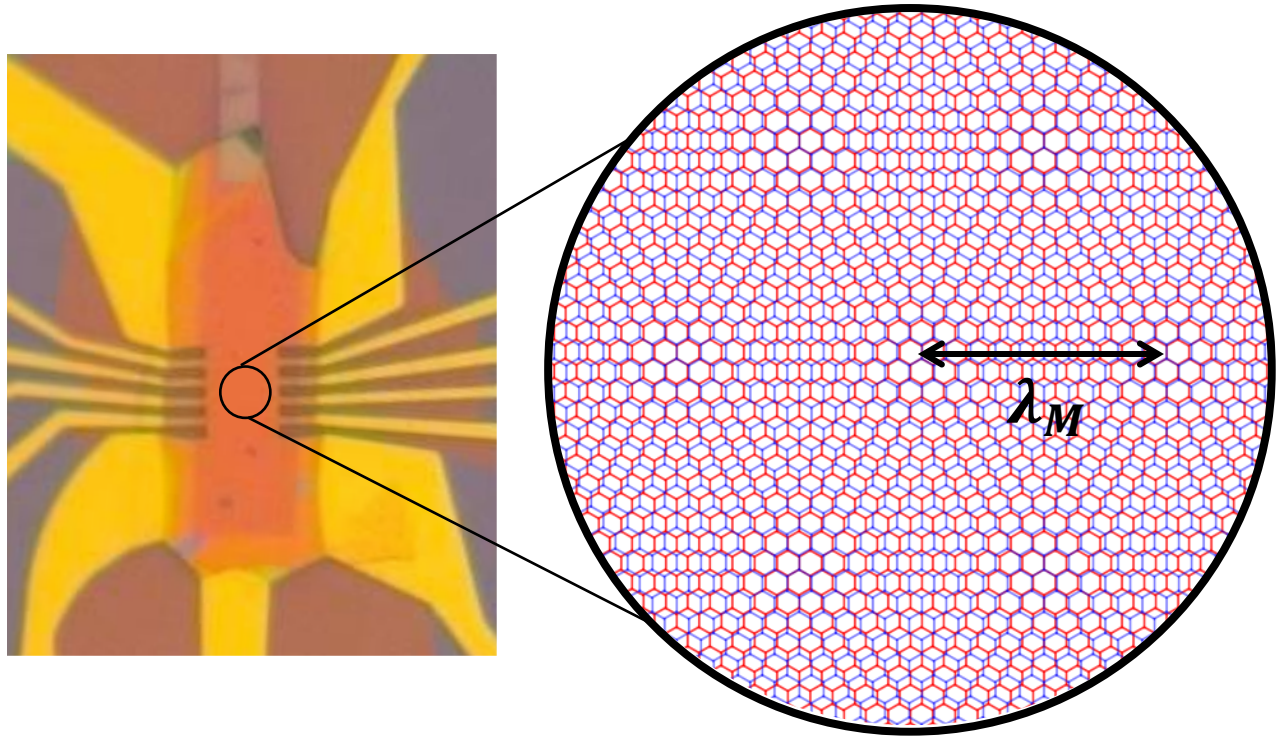
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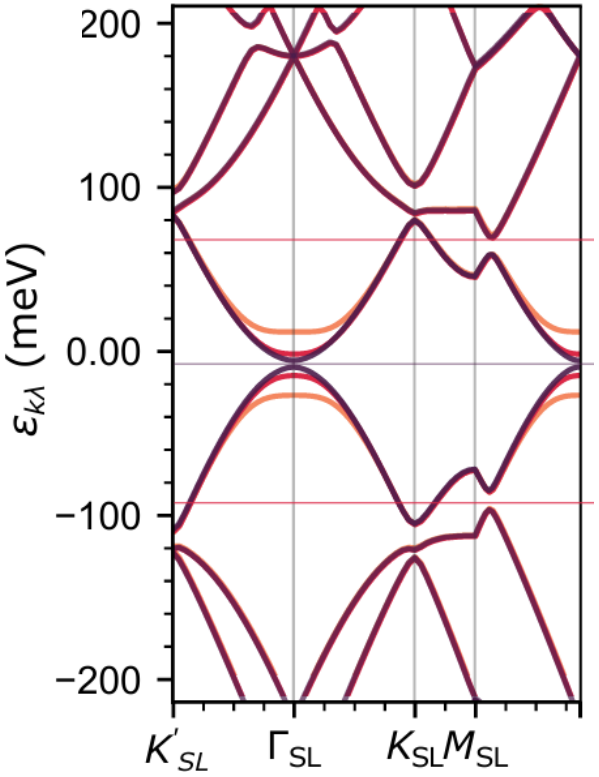




- Moiré : nouvelle périodicité spatiale
- Repliement de bandes
- Faible largeur de bande

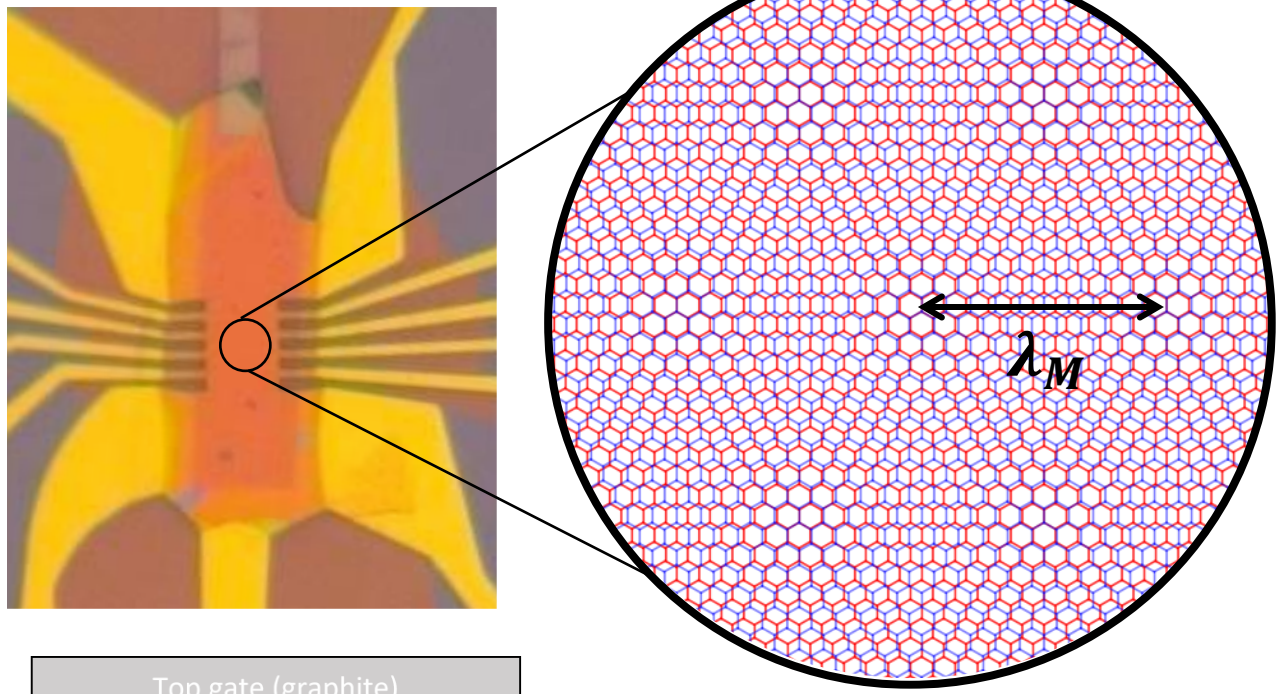
BLG/hBN: un super réseau de moiré aux bandes plates

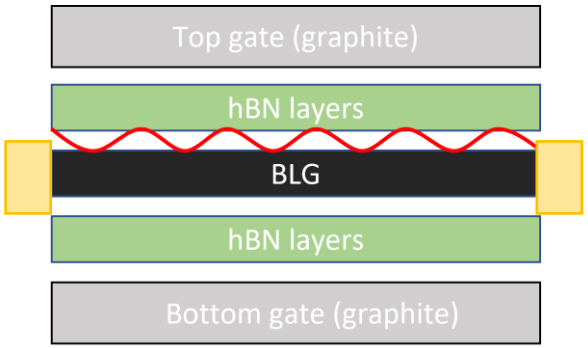




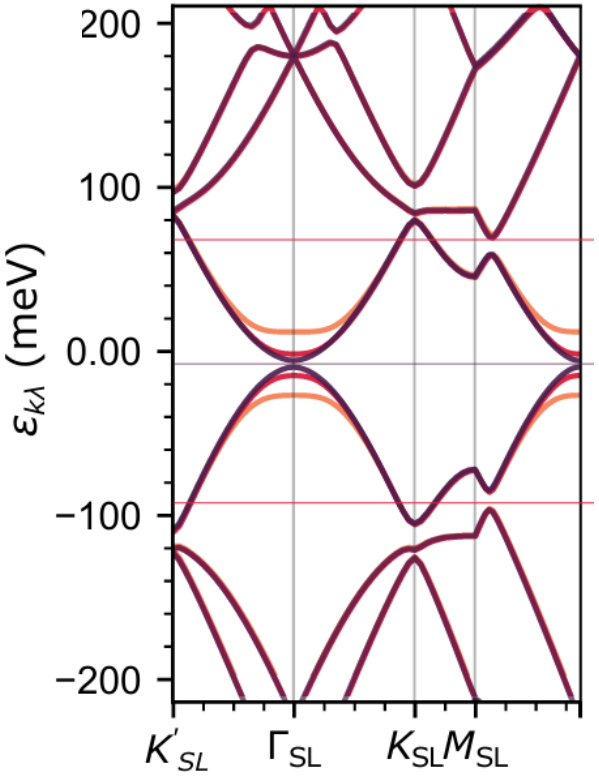
- Moiré : nouvelle périodicité spatiale
- Repliement de bandes
- Faible largeur de bande
  - Faible champ critique

BLG/hBN: un super réseau de moiré aux bandes plates



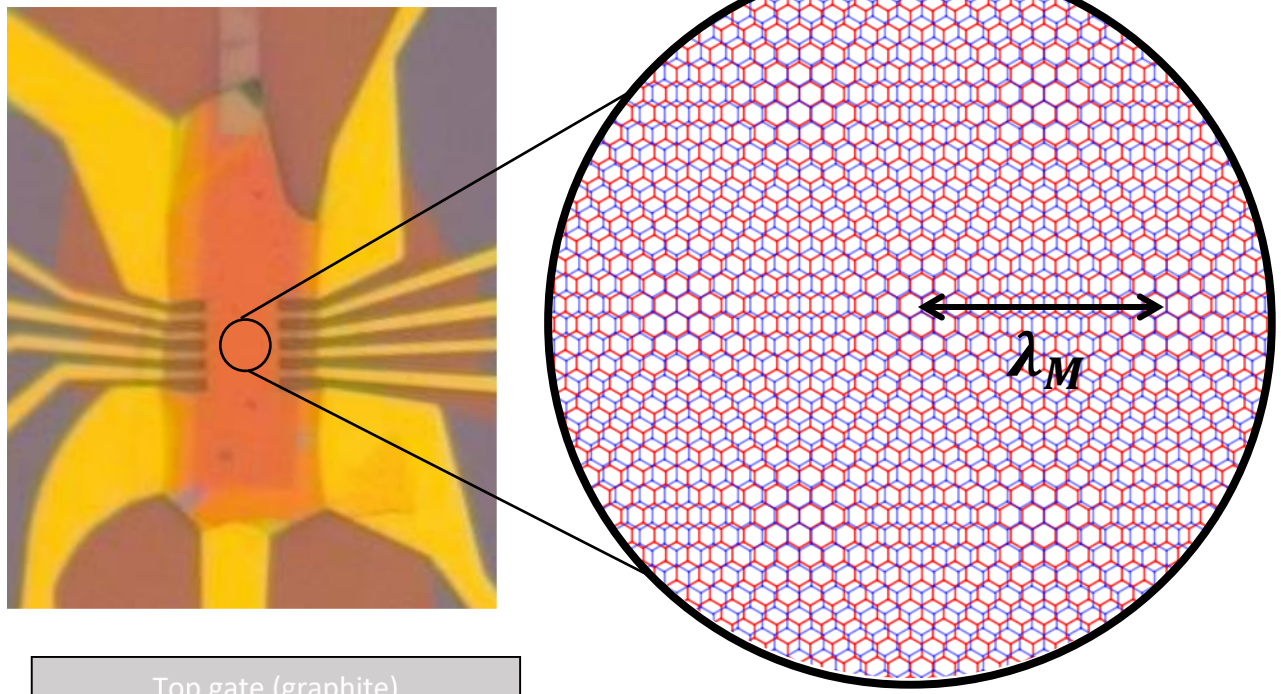


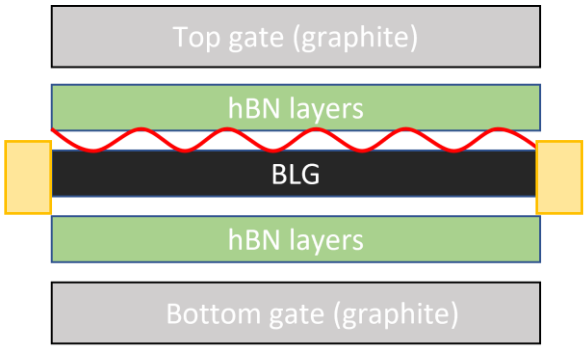
- Grilles :
  - Densité
  - Champ orthogonal



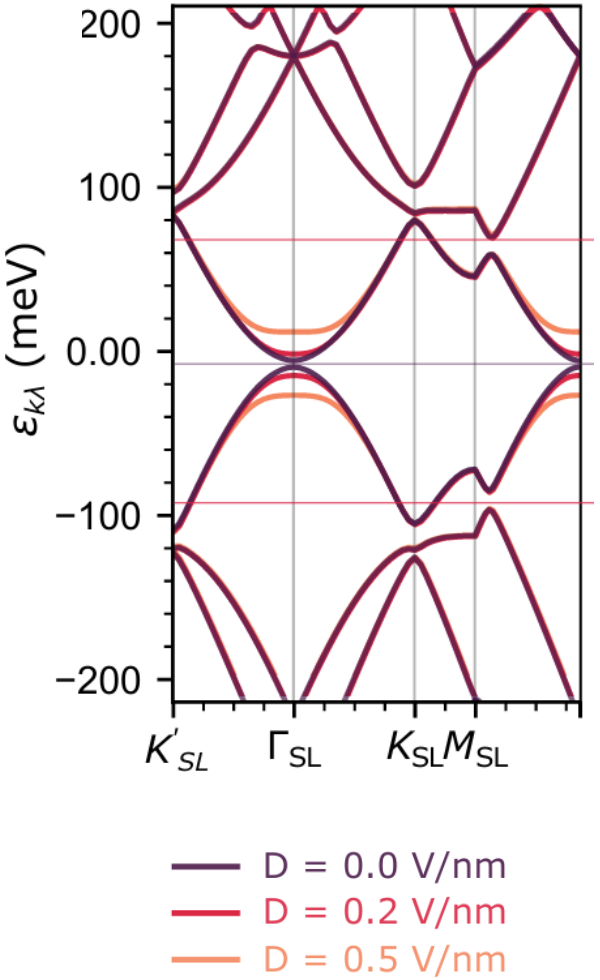
- Moiré : nouvelle périodicité spatiale
- Repliement de bandes
- Faible largeur de bande
  - Faible champ critique

BLG/hBN: un super réseau de moiré aux bandes plates





- Grilles :
  - Densité
  - Champ orthogonal

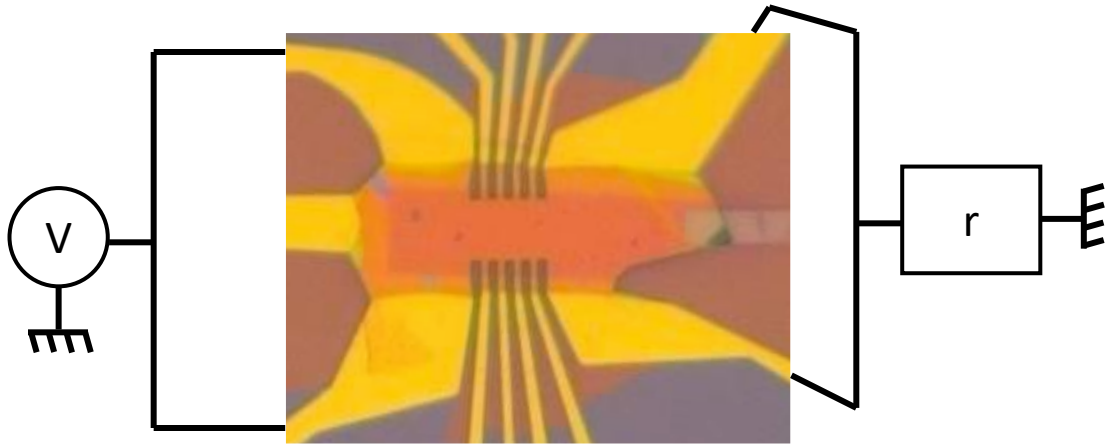


- Moiré : nouvelle périodicité spatiale
- Repliement de bandes
- Faible largeur de bande
  - Faible champ critique
- Champ orthogonal
  - Contrôle le gap
  - Largeur bande

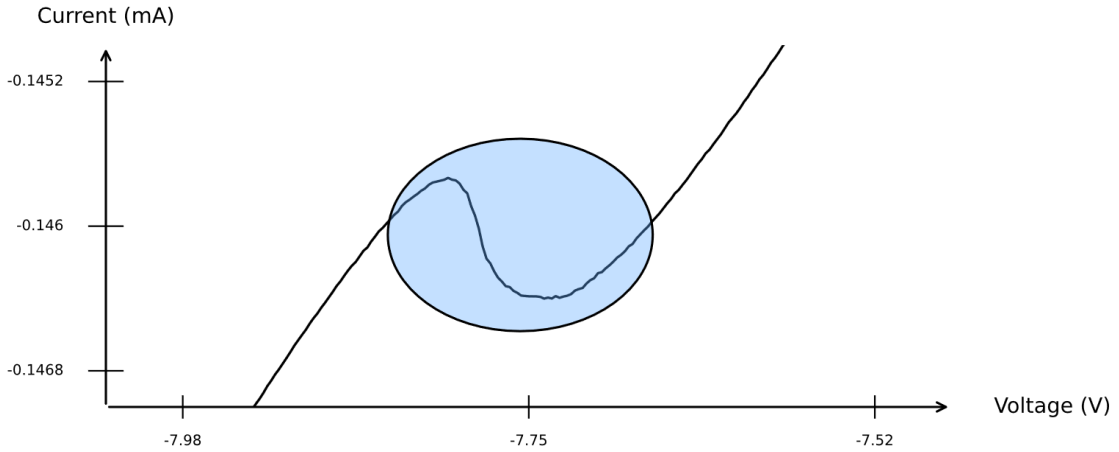
Mesures en transport

Mesures en transport de l'échantillon de BLG/hBN

Biais : tension DC

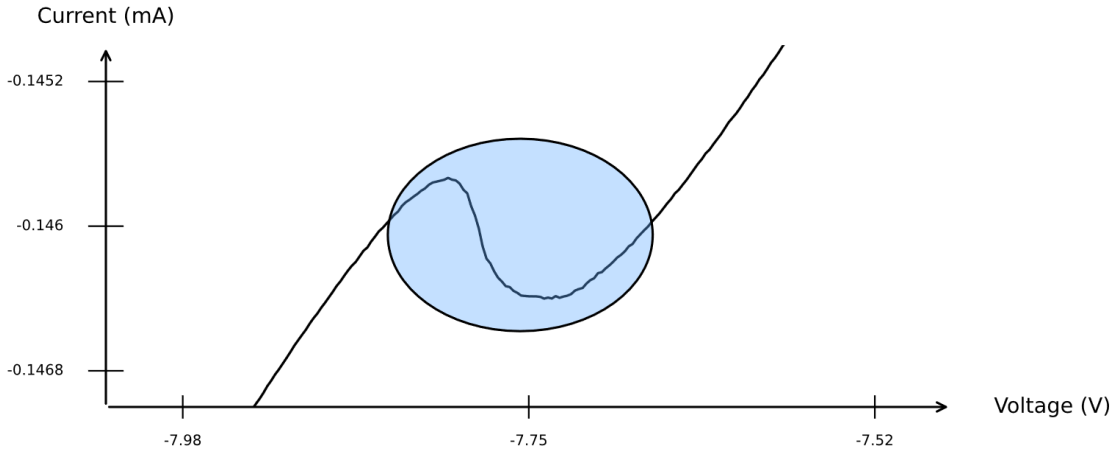


Mesures en transport de l’échantillon de BLG/hBN



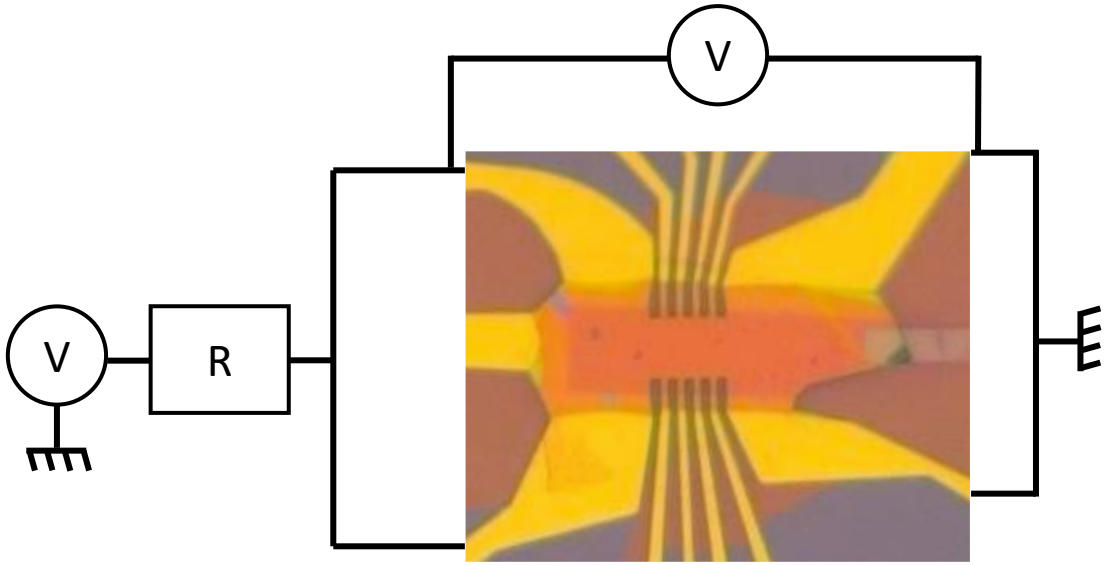
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Mesures en transport de l’échantillon de BLG/hBN

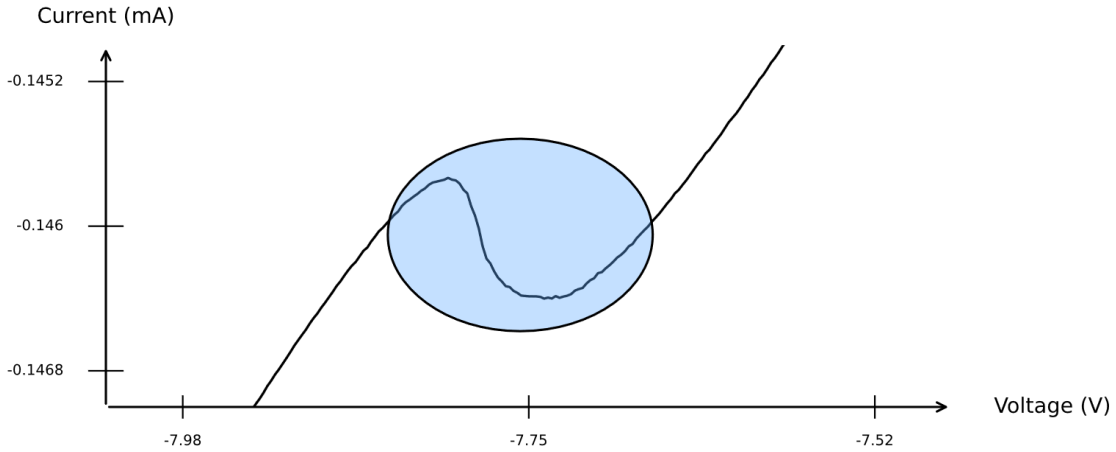


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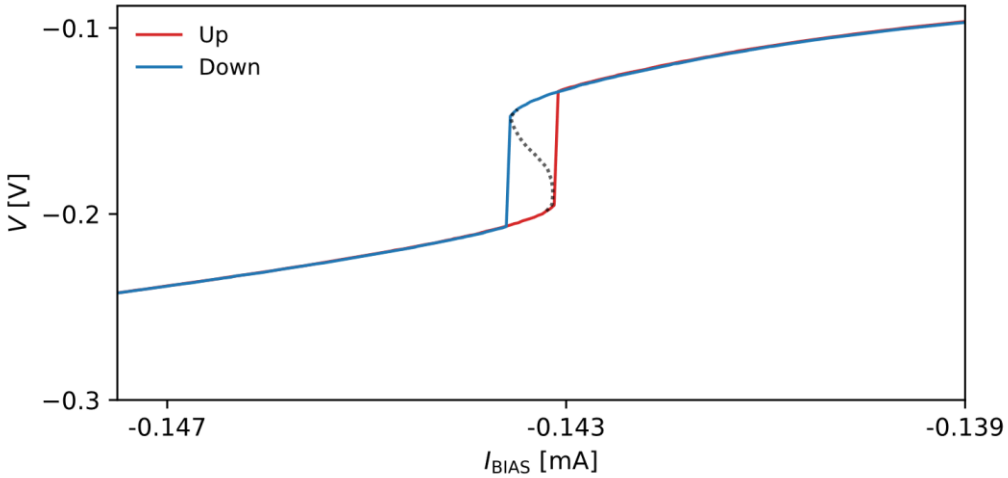
Biais : Courant DC



Mesures en transport de l'échantillon de BLG/hBN

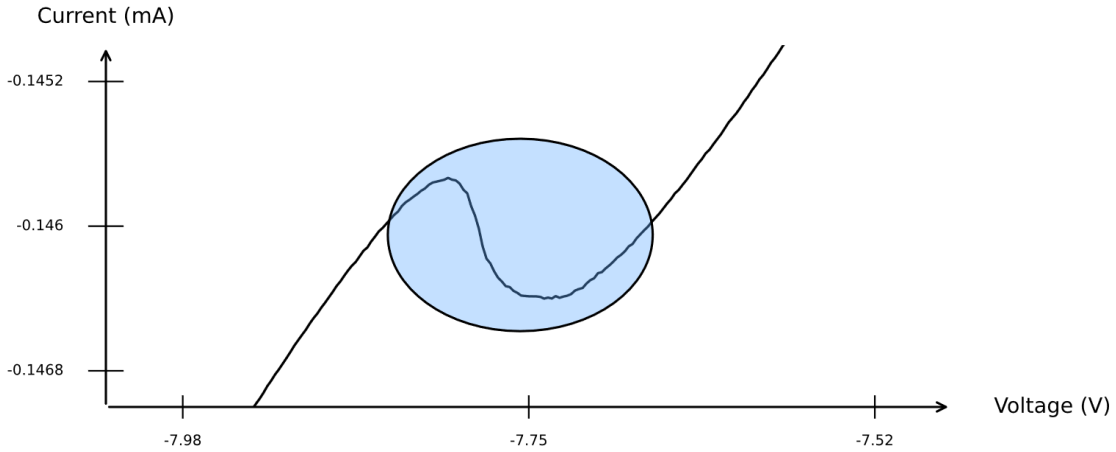


Biais : tension DC

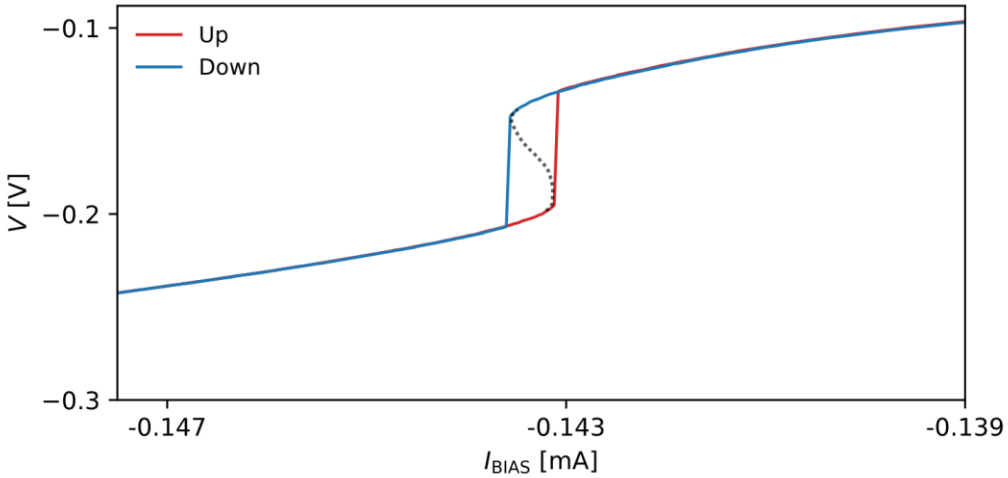


Biais : Courant DC

Mesures en transport de l'échantillon de BLG/hBN



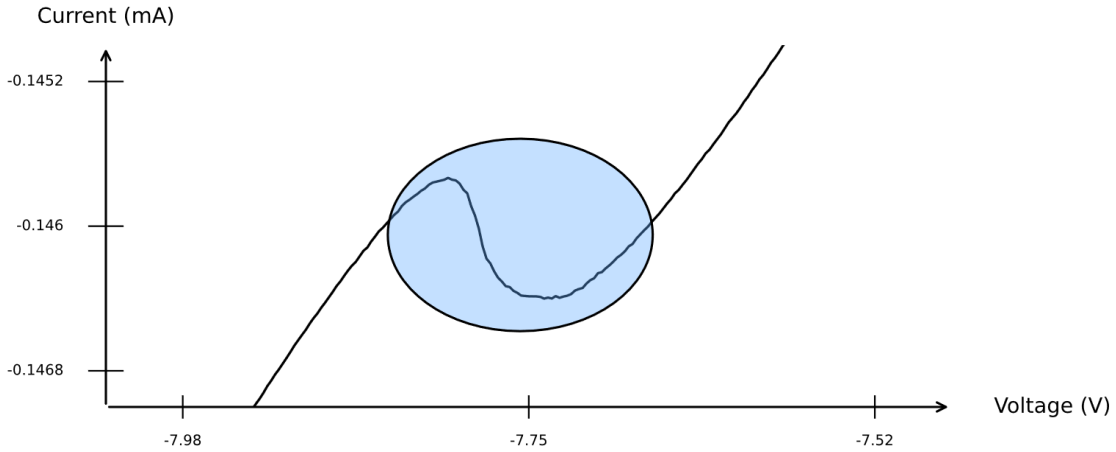
Biais : tension DC



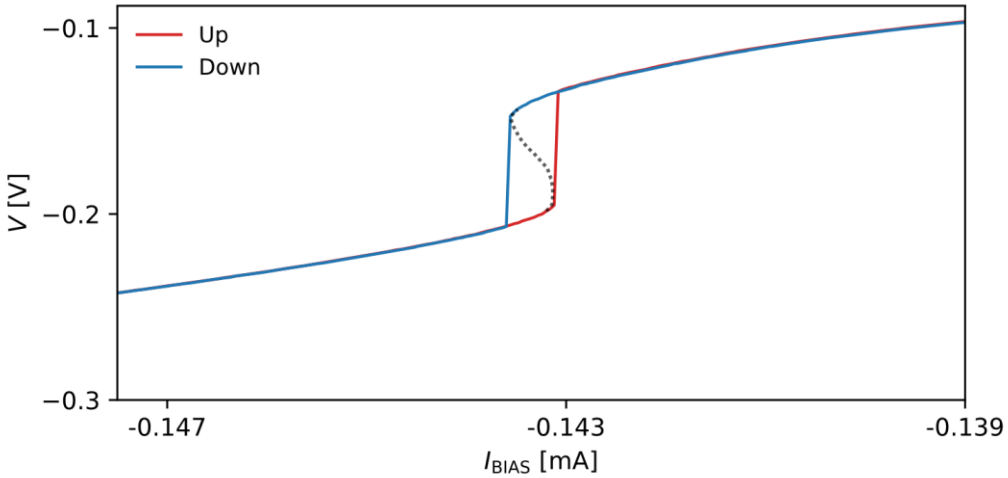
Biais : Courant DC

Comment lire ces deux sauts facilement ?

Mesures en transport de l'échantillon de BLG/hBN



Biais : tension DC



Biais : Courant DC

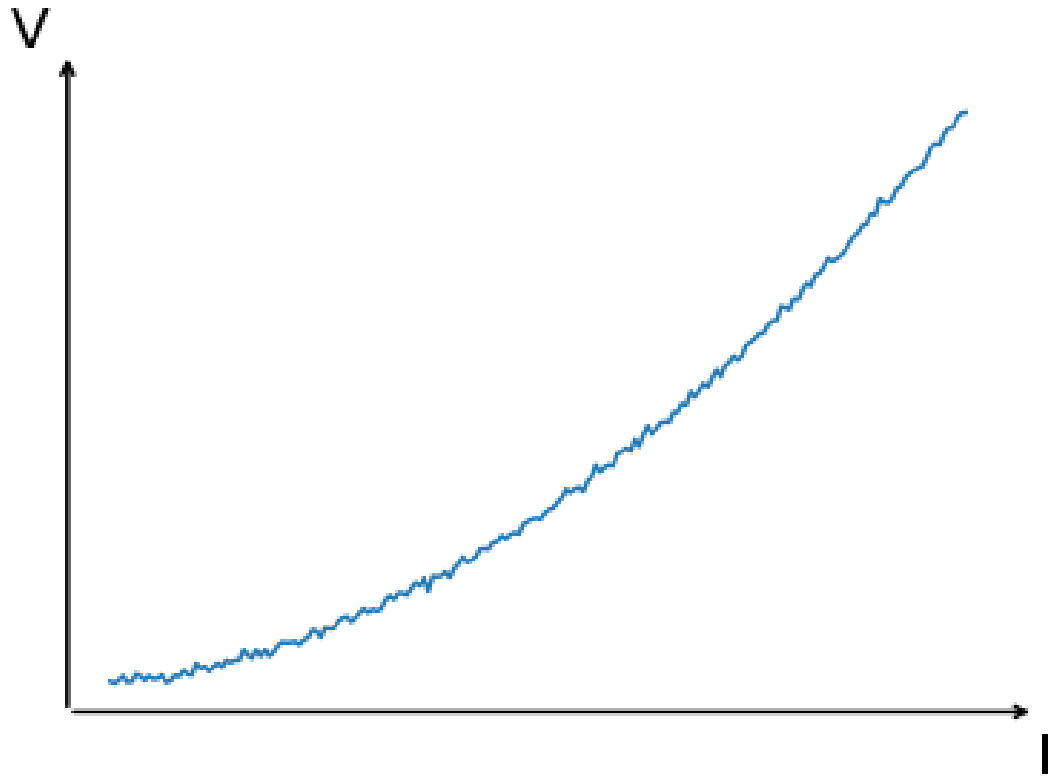
Comment lire ces deux sauts facilement ?
La différentielle

Mesures en transport de l’échantillon de BLG/hBN

Principe de fonctionnement du Lock-in :

Mesures en transport de l'échantillon de BLG/hBN

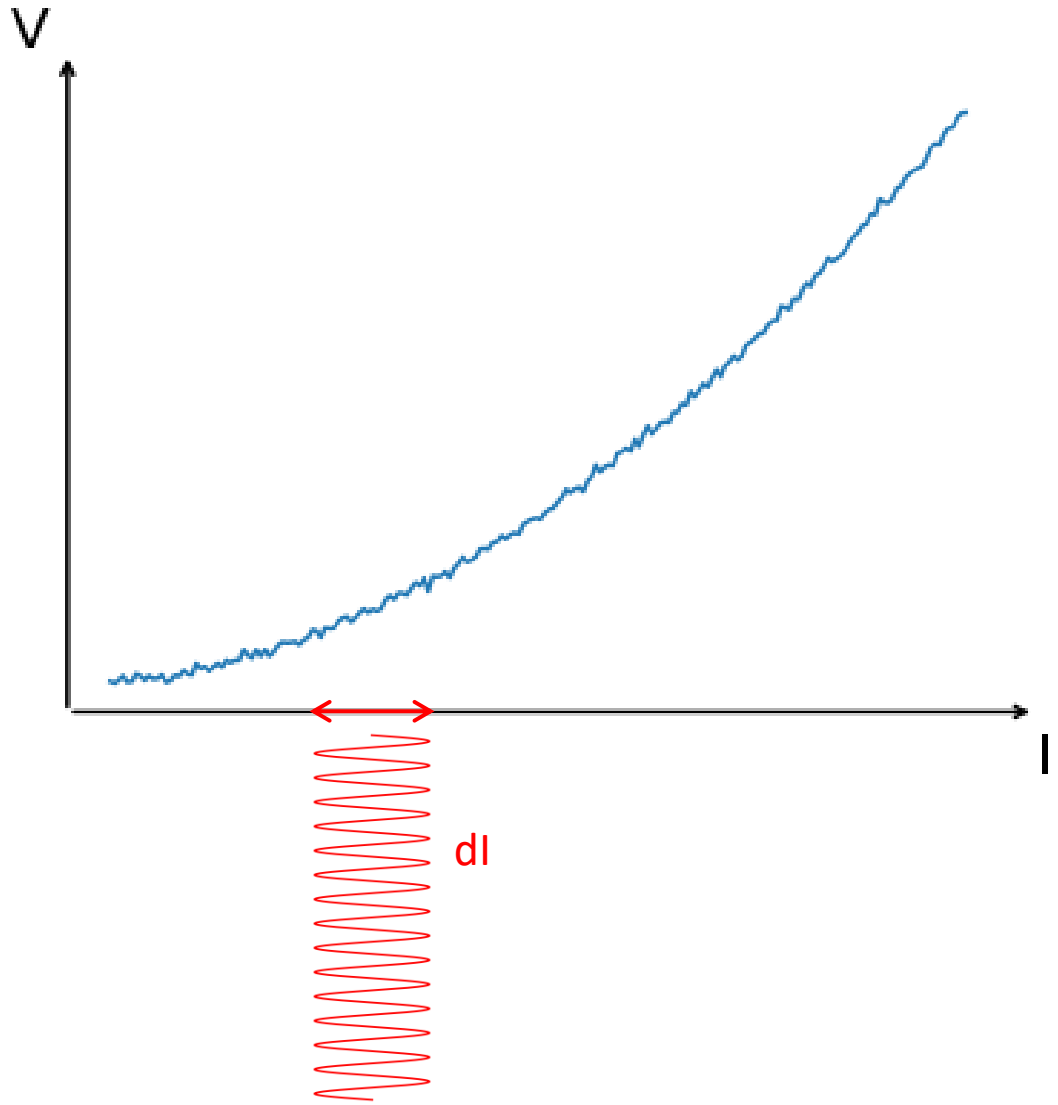
Principe de fonctionnement du Lock-in :



Mesures en transport de l'échantillon de BLG/hBN

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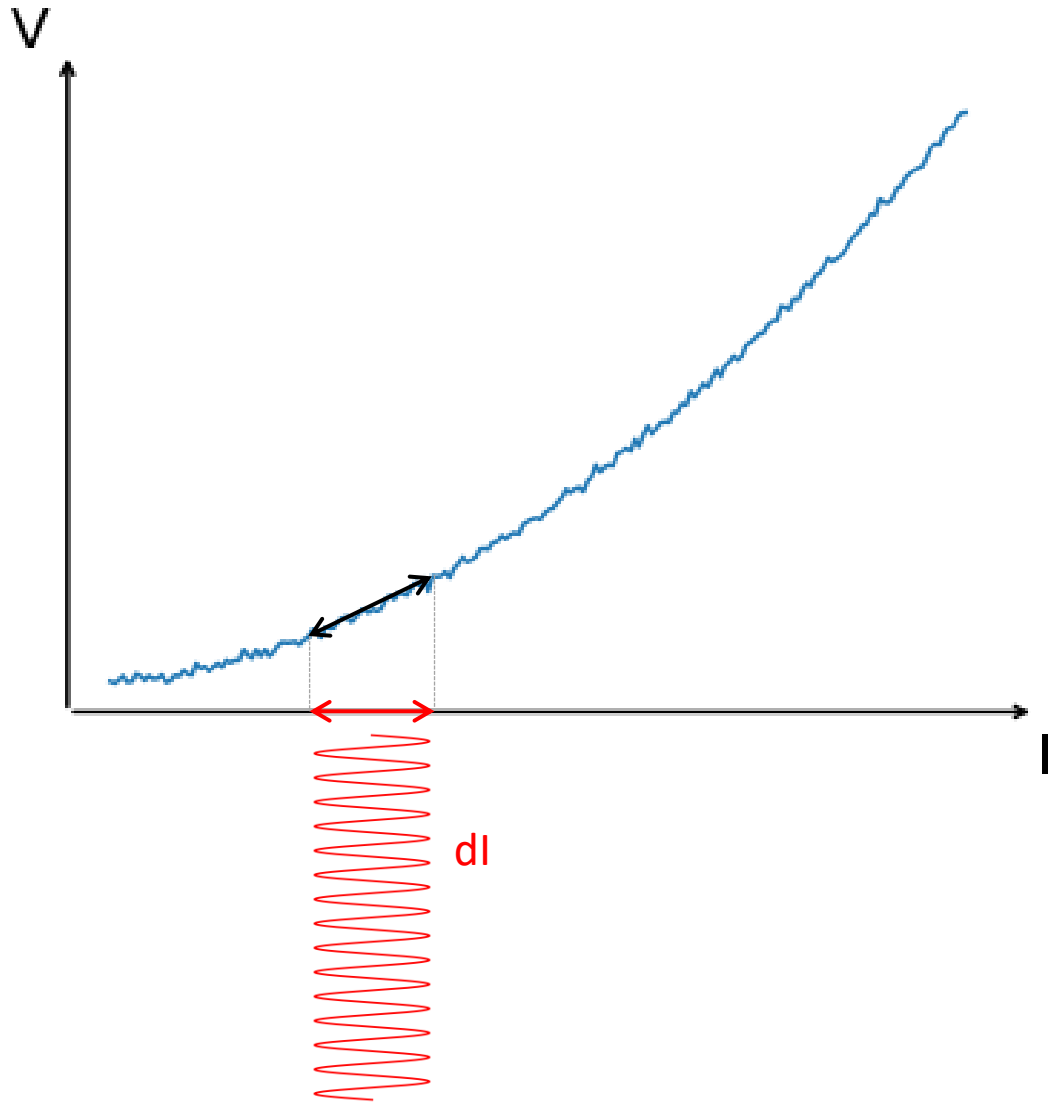
Petite perturbation AC : dI



Mesures en transport de l'échantillon de BLG/hBN

Principe de fonctionnement du Lock-in :

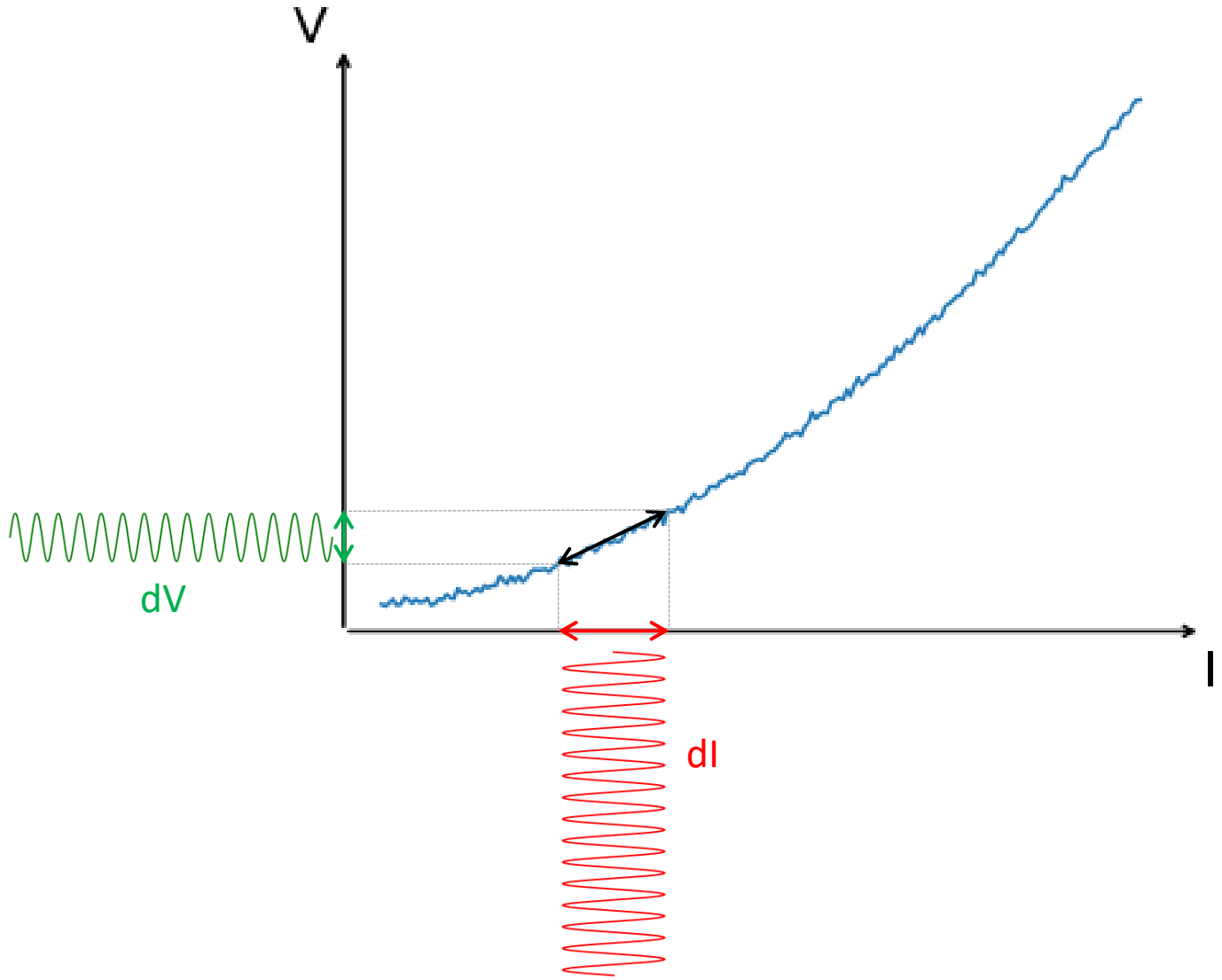
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Mesures en transport de l'échantillon de BLG/hBN

Principe de fonctionnement du Lock-in :

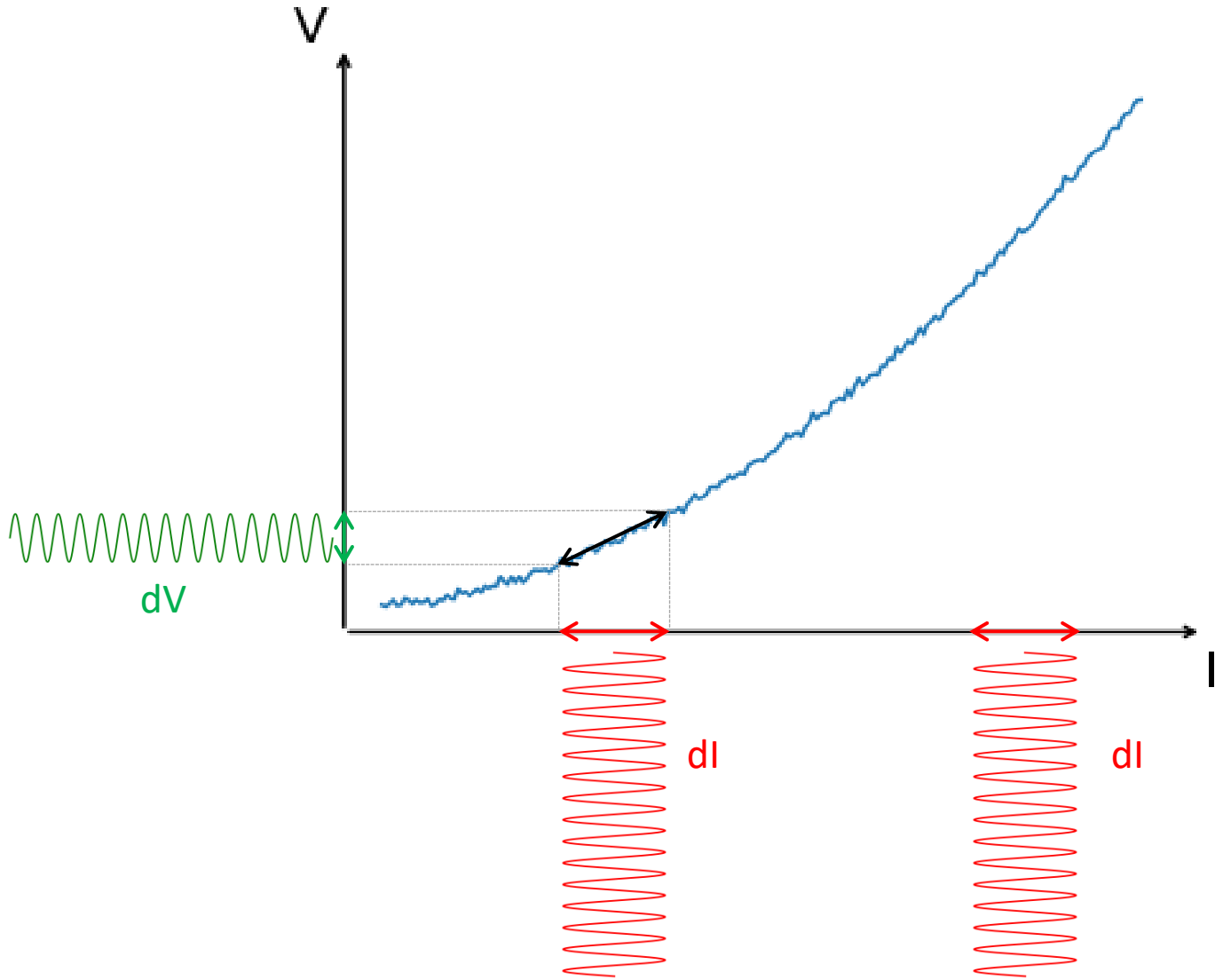
Petite perturbation AC : dI
Réponse AC du système : dV



Mesures en transport de l'échantillon de BLG/hBN

Principe de fonctionnement du Lock-in :

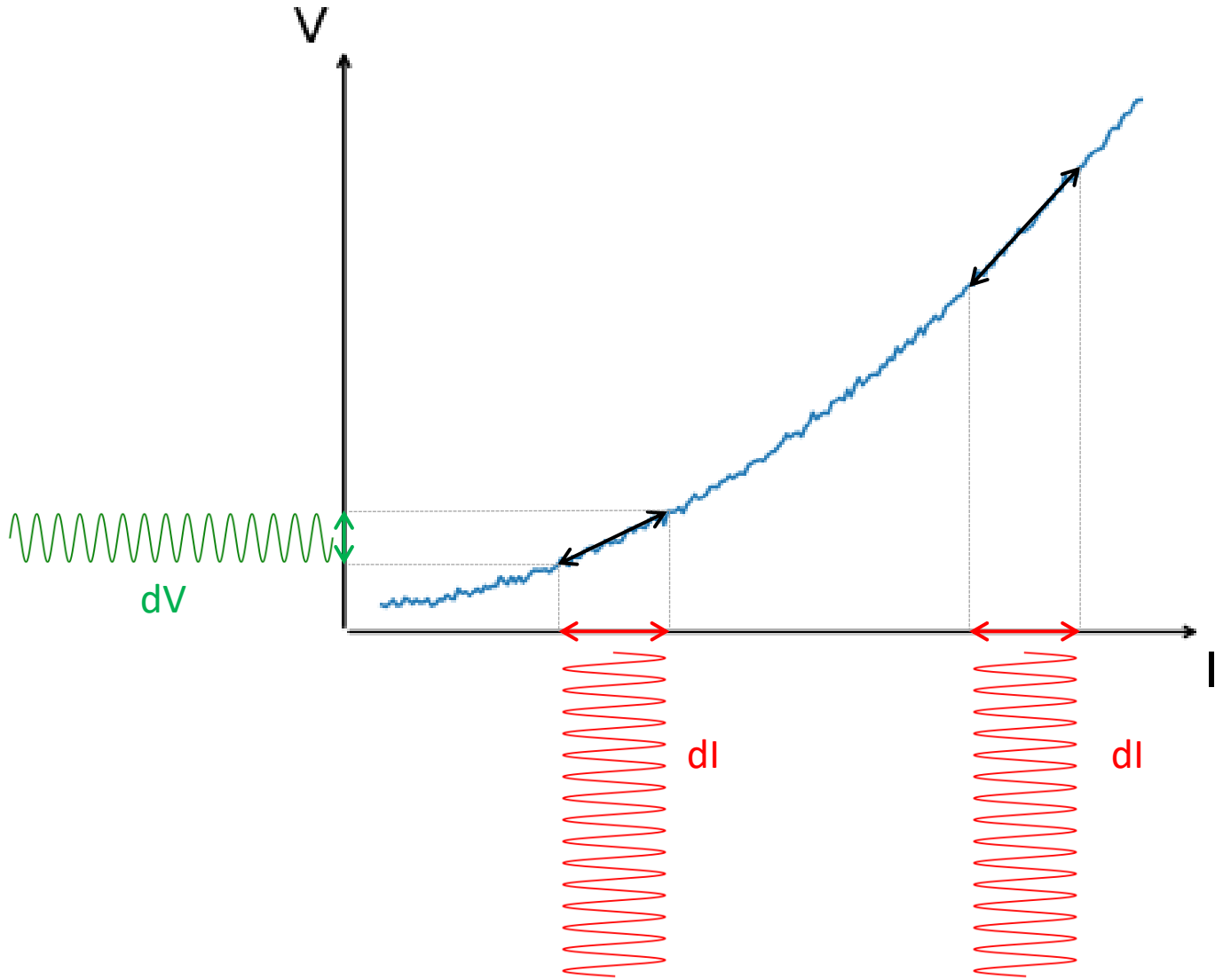
Petite perturbation AC : dI
Réponse AC du système : dV



Mesures en transport de l'échantillon de BLG/hBN

Principe de fonctionnement du Lock-in :

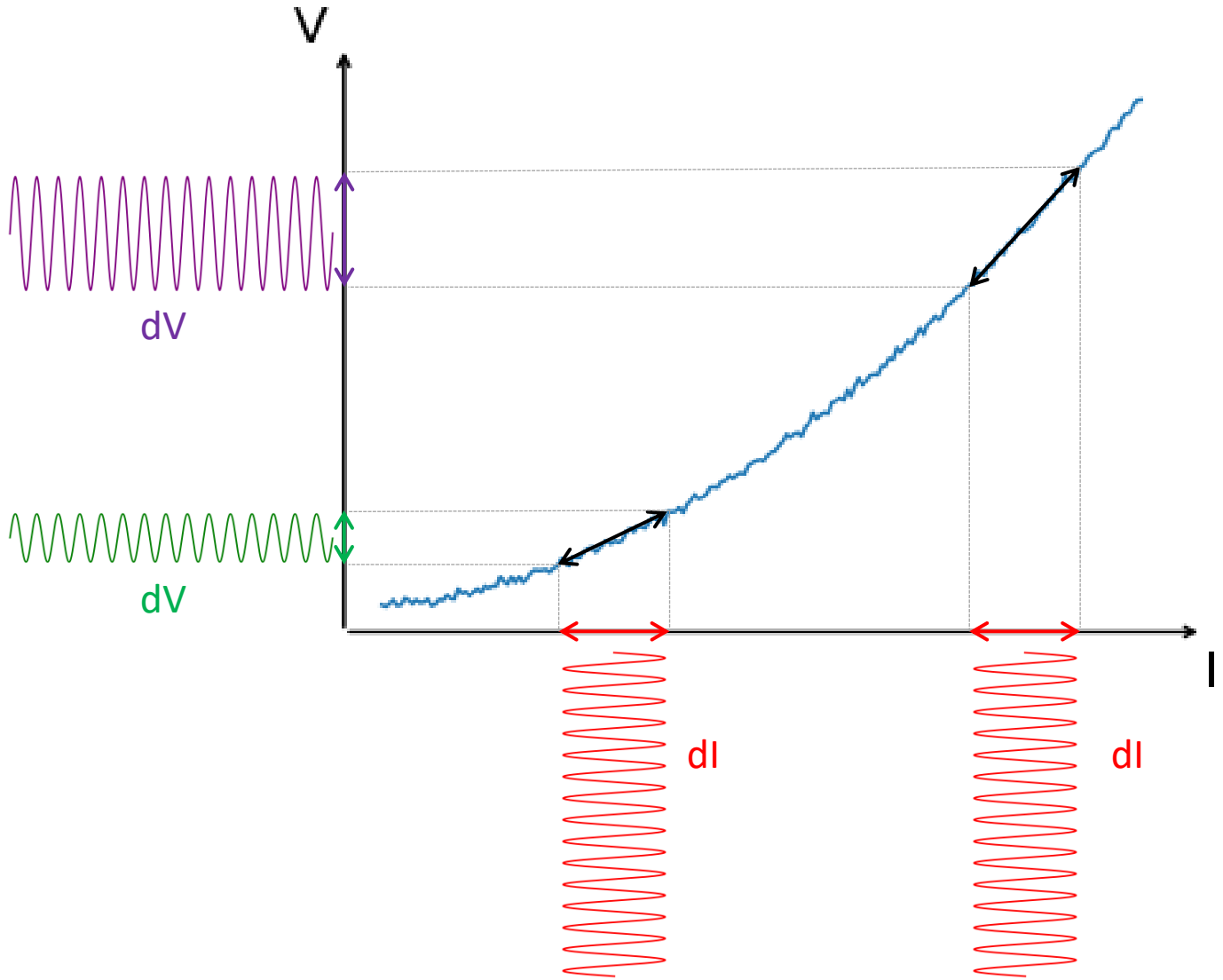
Petite perturbation AC : dI
Réponse AC du système : dV



Mesures en transport de l'échantillon de BLG/hBN

Principe de fonctionnement du Lock-in :

Petite perturbation AC : dI
Réponse AC du système : dV

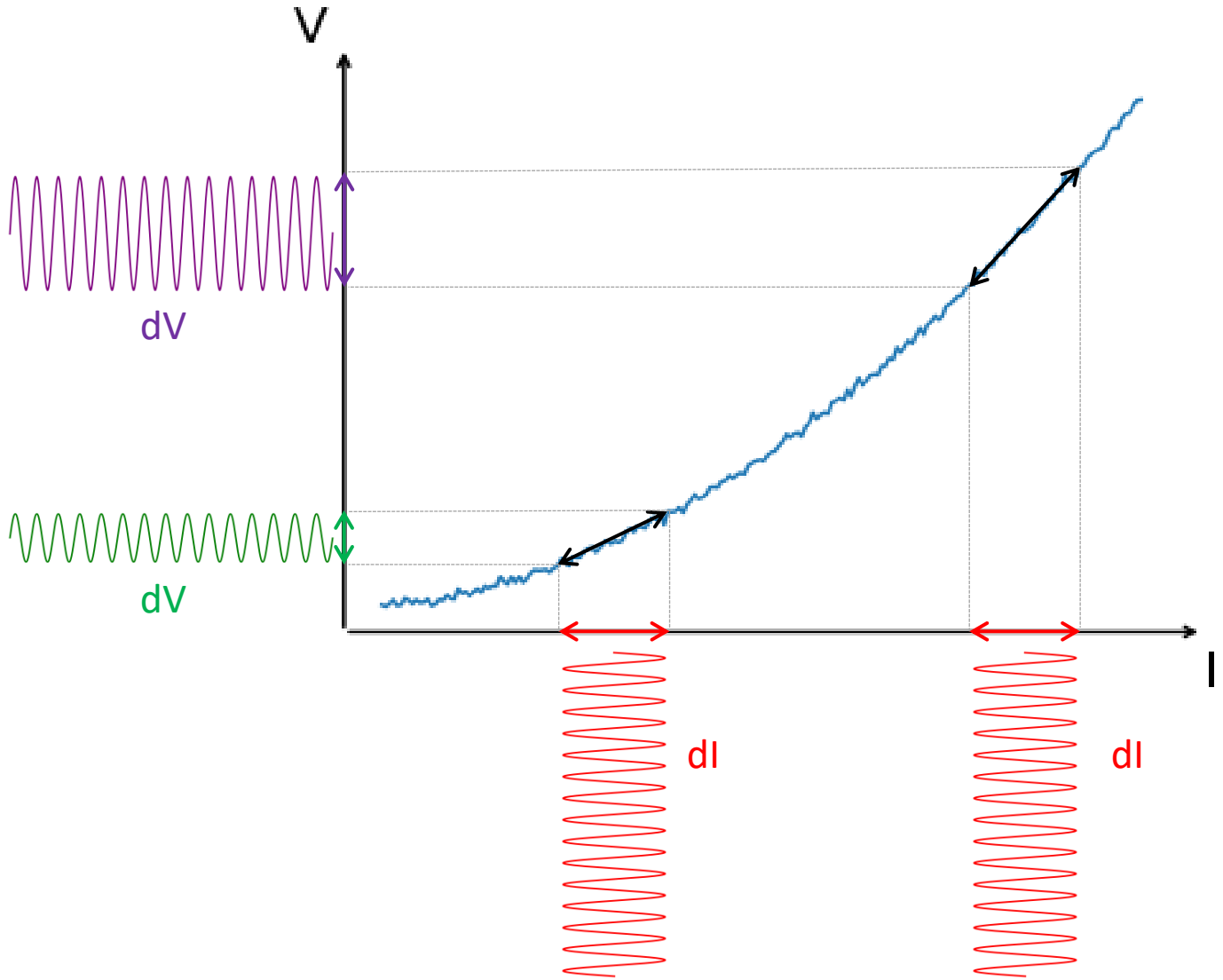


Mesures en transport de l'échantillon de BLG/hBN

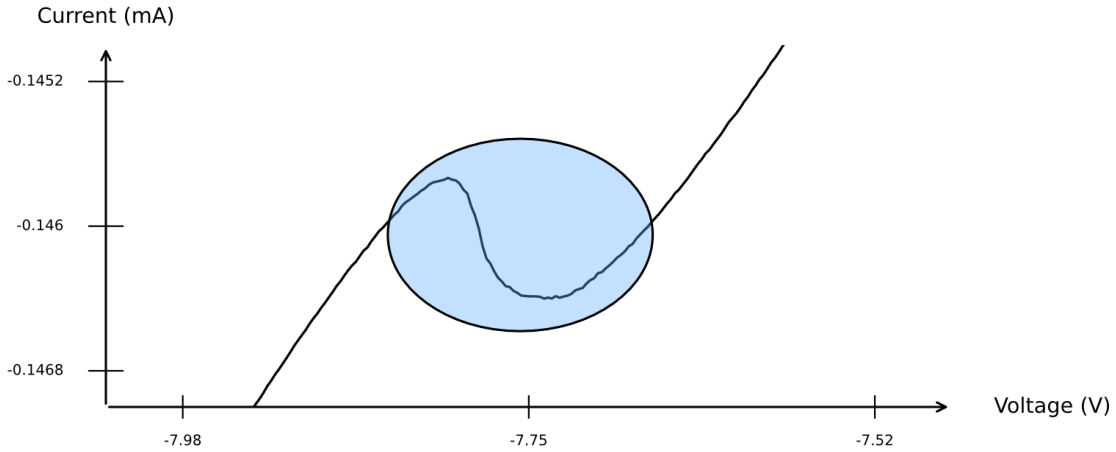
Principe de fonctionnement du Lock-in :

Petite perturbation AC : dI
Réponse AC du système : dV

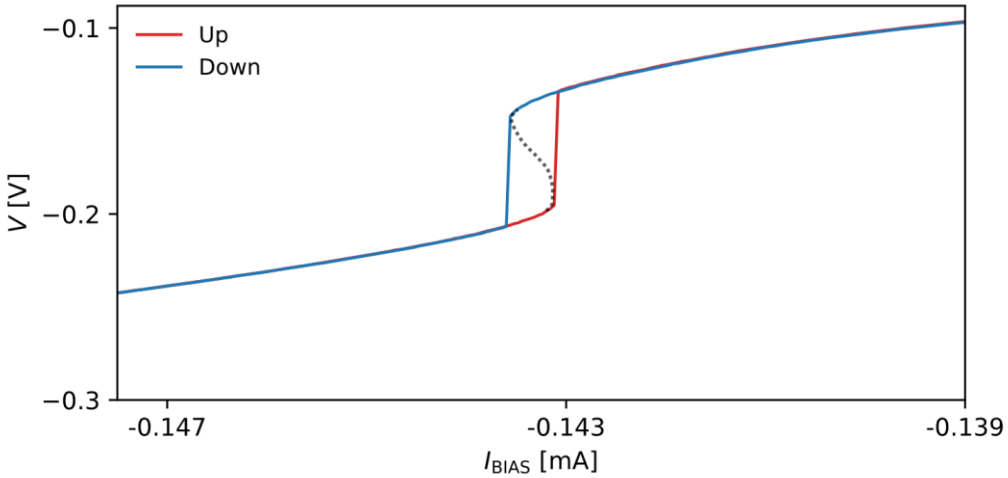
Mesure analogique de la différentielle



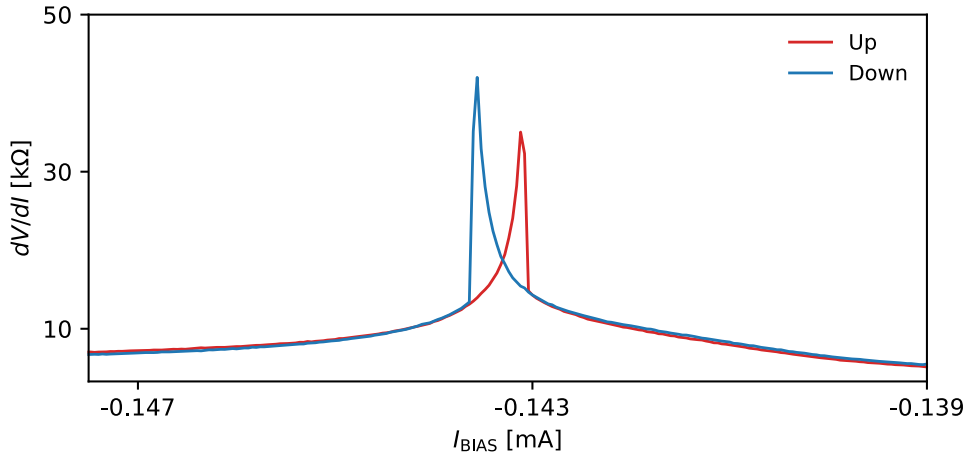
Mesures en transport de l'échantillon de BLG/hBN



Biais : tension DC

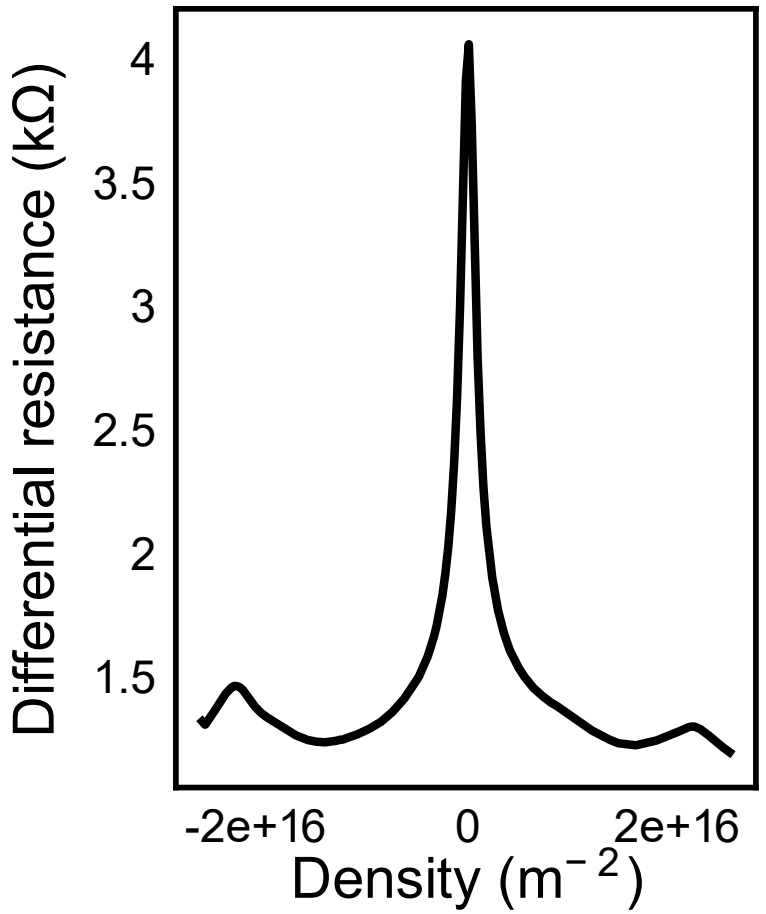


Biais : Courant DC

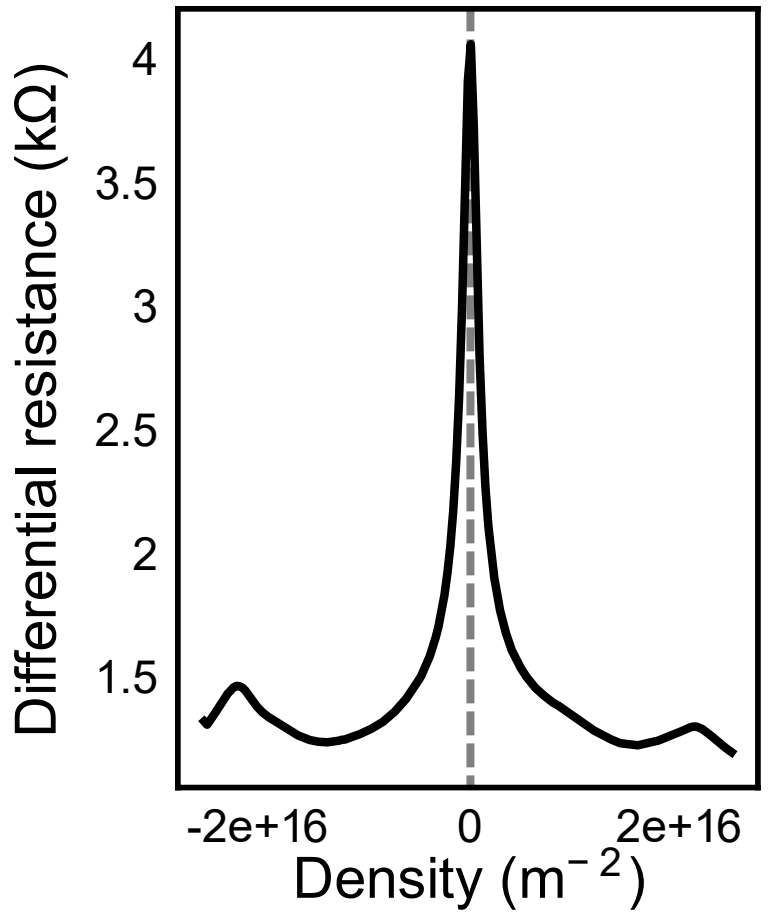


Biais : Courant DC et courant AC

Mesures en transport de l'échantillon de BLG/hBN

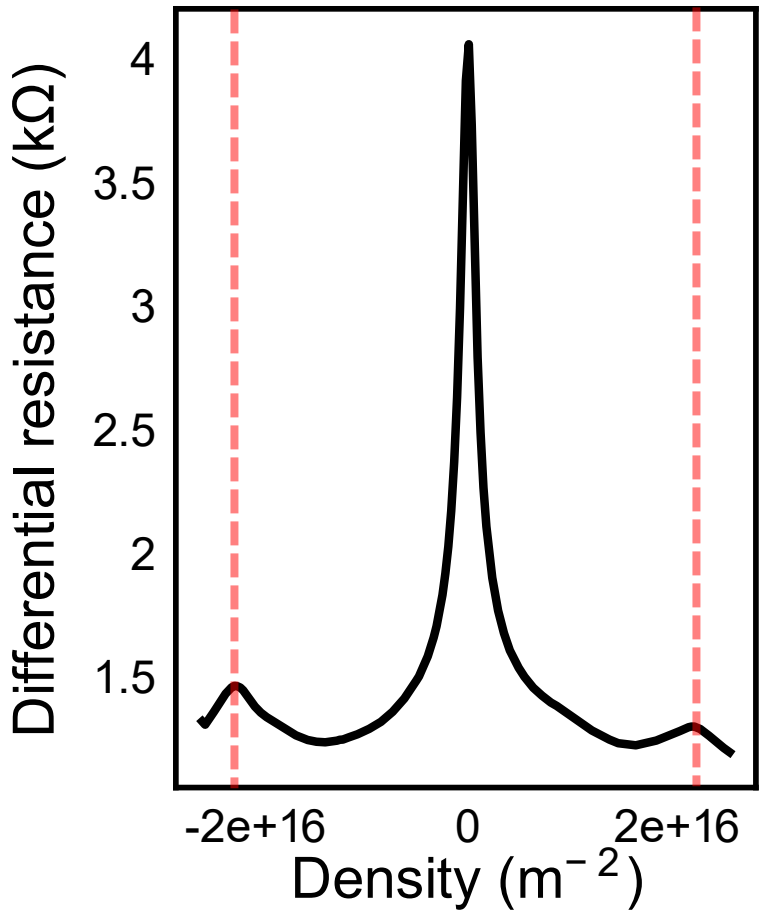


Mesures en transport de l'échantillon de BLG/hBN



- Point de neutralité de charge (CNP) :
  - n = 0
  - Gap du BLG

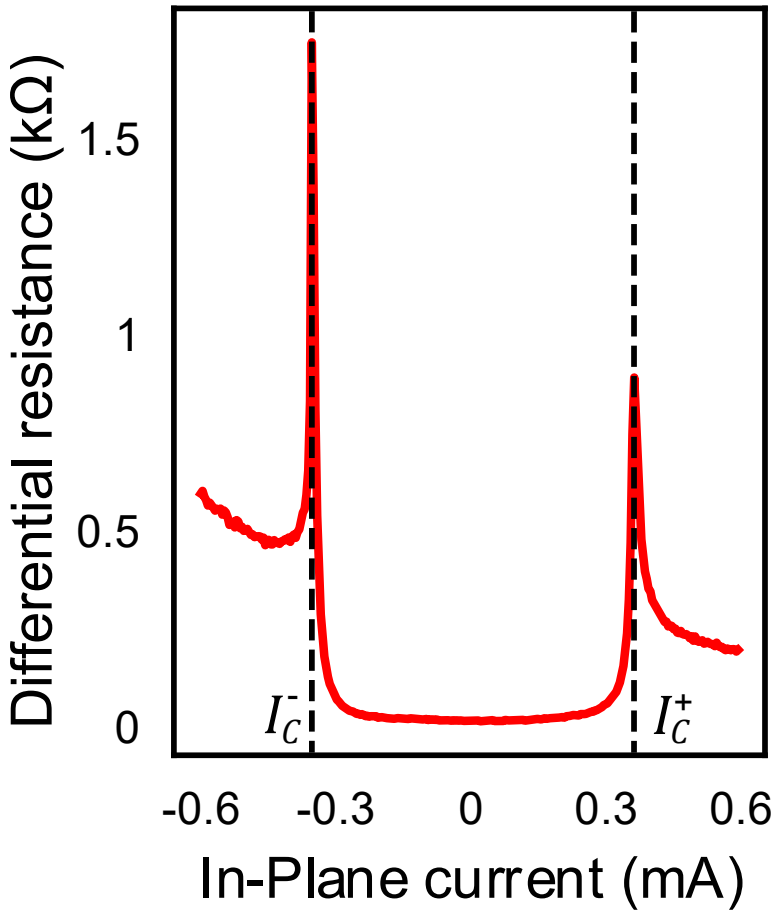
Mesures en transport de l'échantillon de BLG/hBN



- Point de neutralité de charge (CNP) :
  - n = 0
  - Gap du BLG

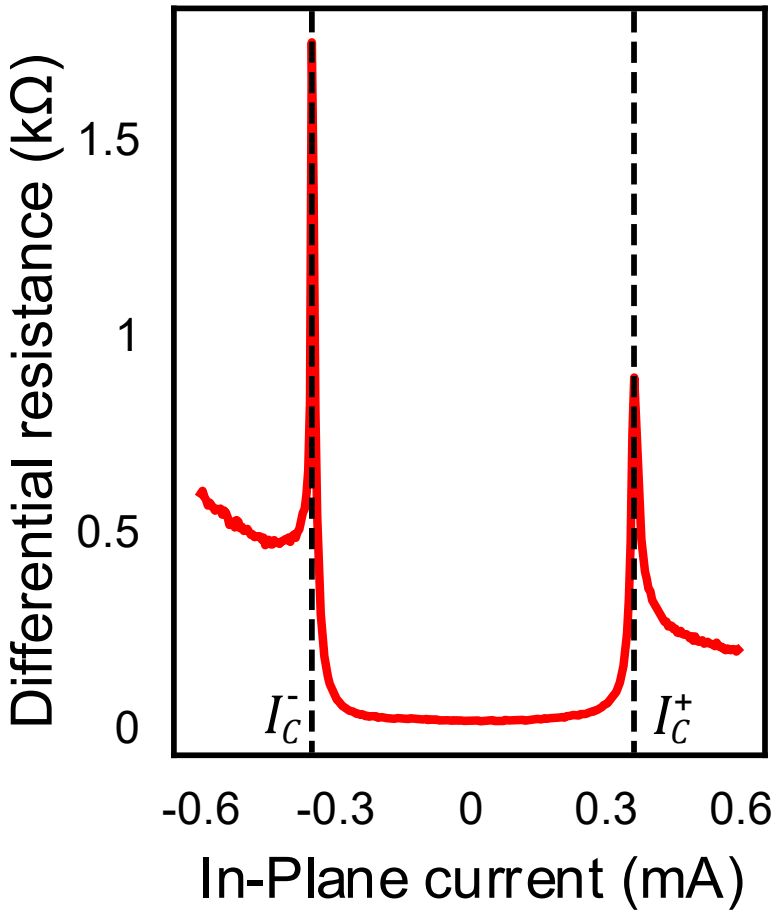
- Points de neutralité de charge secondaires (sNP) :
  - n = ± $2.2 * 10^{16}\ m^{-2}$
  - Repliement de bandes du moiré

Mesures en transport de l'échantillon de BLG/hBN



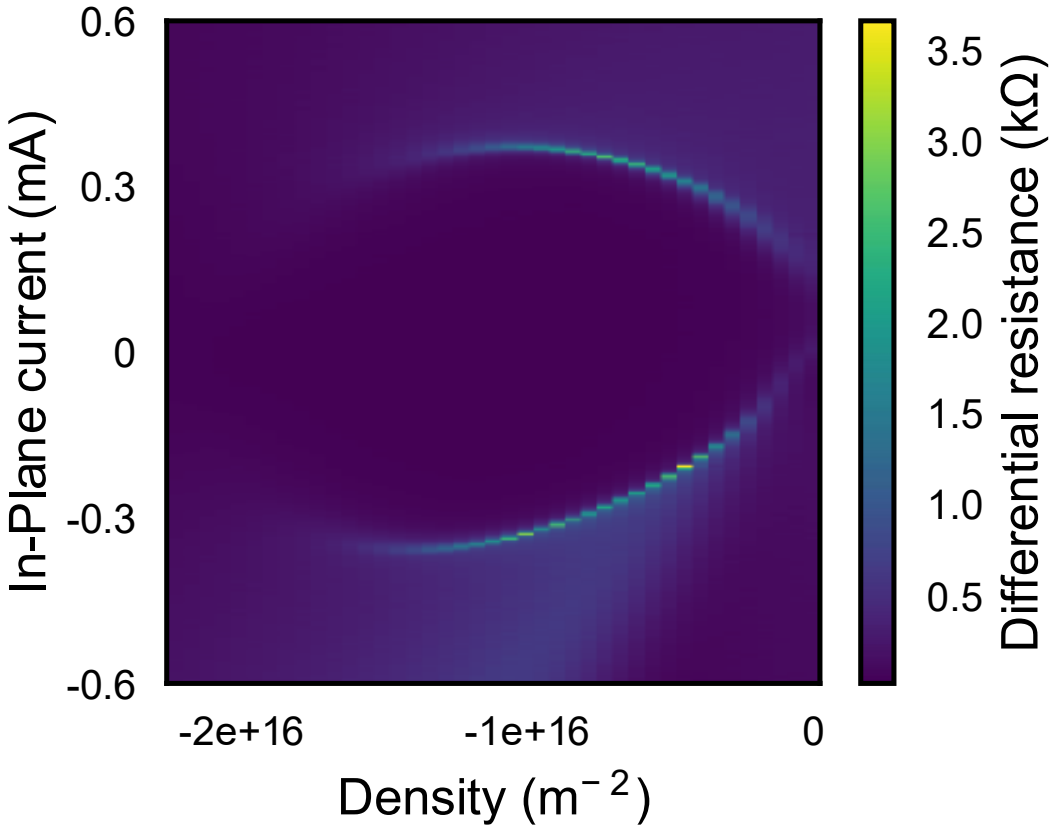
- Régime linéaire à faible biais

Mesures en transport de l'échantillon de BLG/hBN



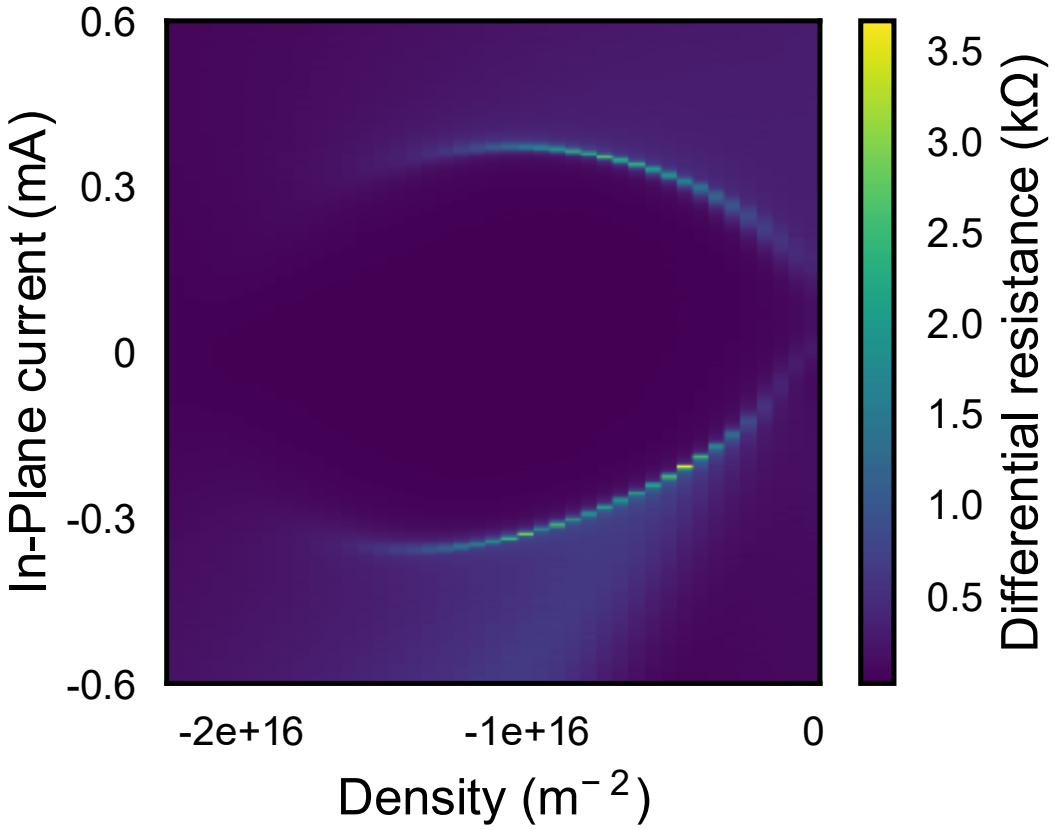
- Régime linéaire à faible biais
- Régime non-linéaire à la criticité
  - Symétrique

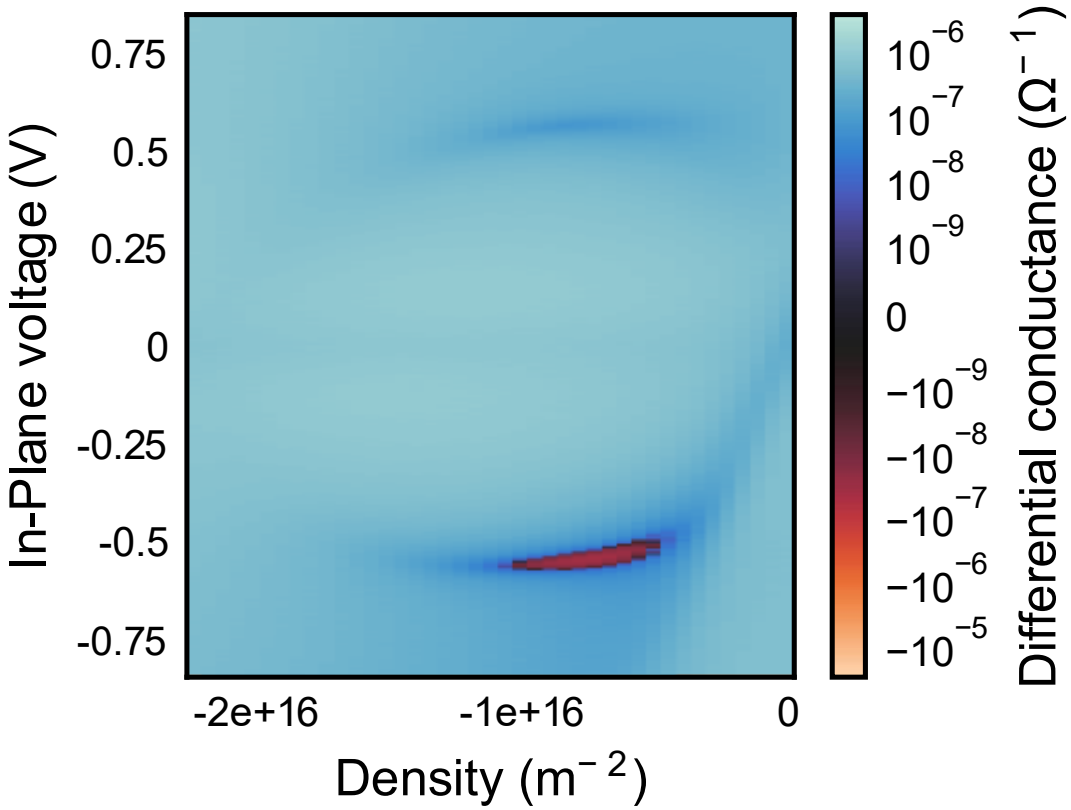
Mesures en transport de l'échantillon de BLG/hBN



- Anneau de criticité
- Le courant critique évolue avec le nombre de porteurs de charge (prévu par le modèle de Drude)

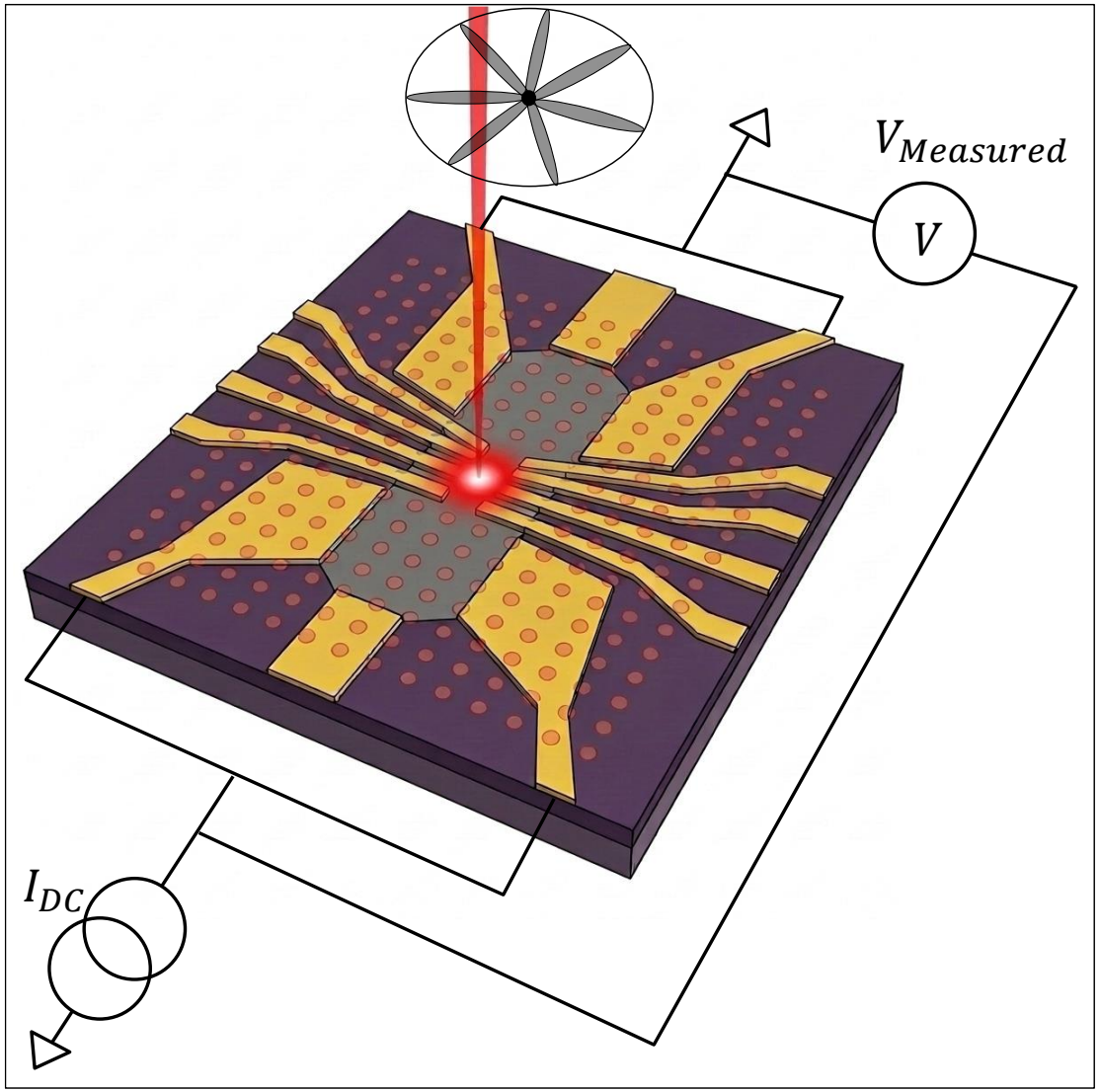
Mesures en transport de l'échantillon de BLG/hBN





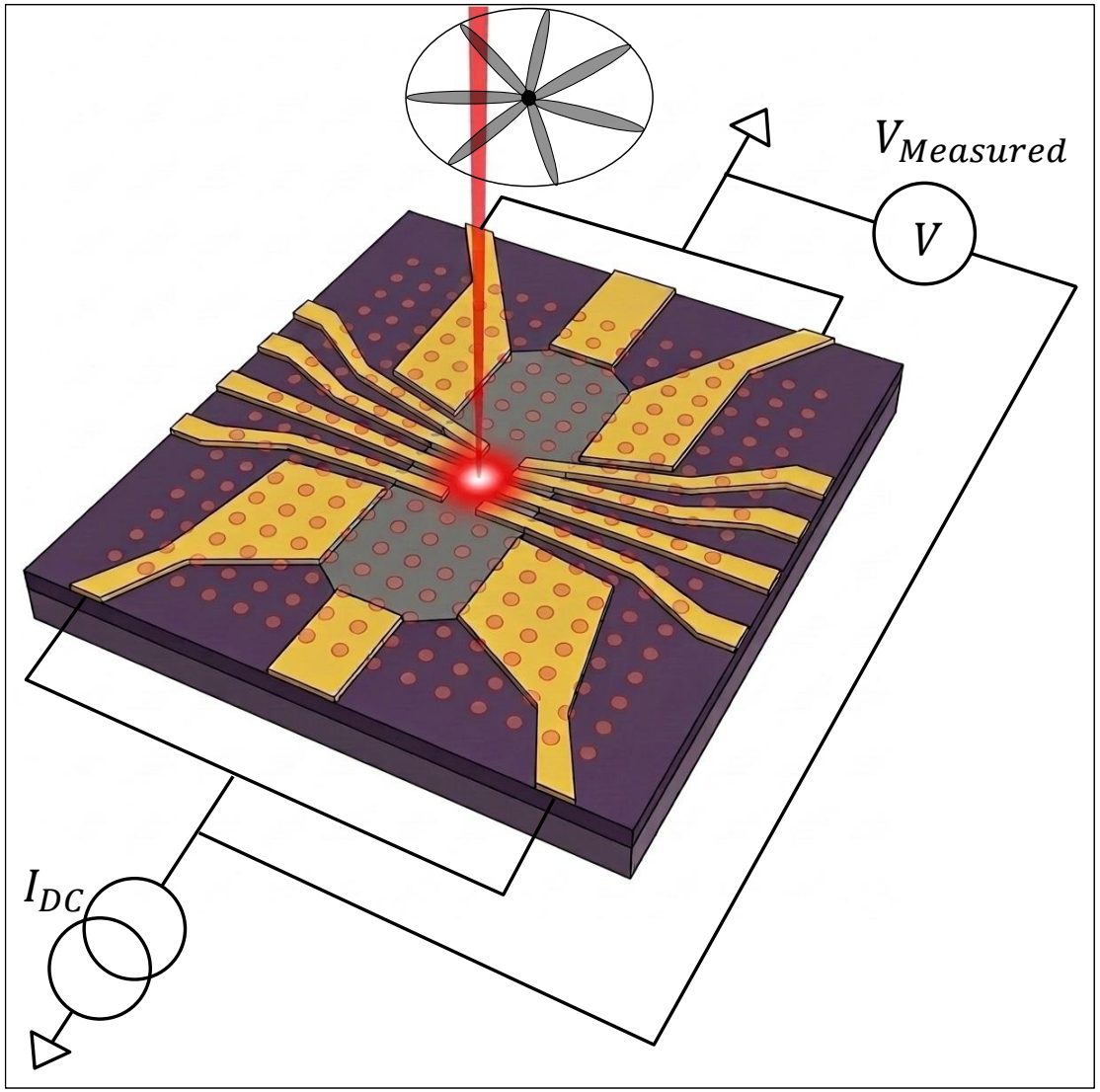
Mesure locale de photo-tension

Comment caractériser spatialement le transport ?



- Courant DC + lumière AC (hachée)

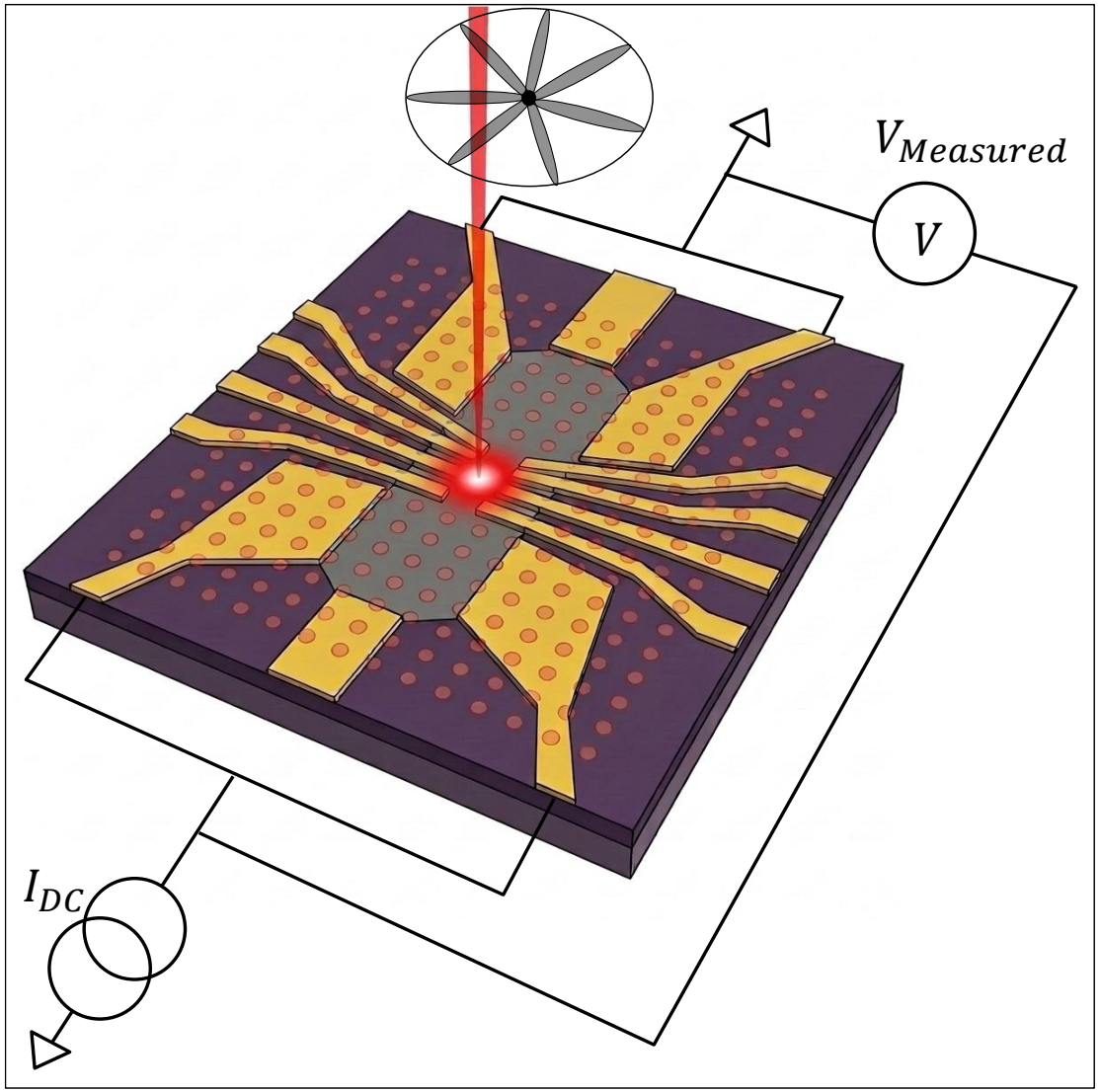
Comment caractériser spatialement le transport ?



- Courant DC + lumière AC (hachée)

- $$V_{\mathrm{PV}} \approx \frac{\mathrm{d}V_{\mathrm{dark}}}{\mathrm{d}I} \cdot \delta I.$$

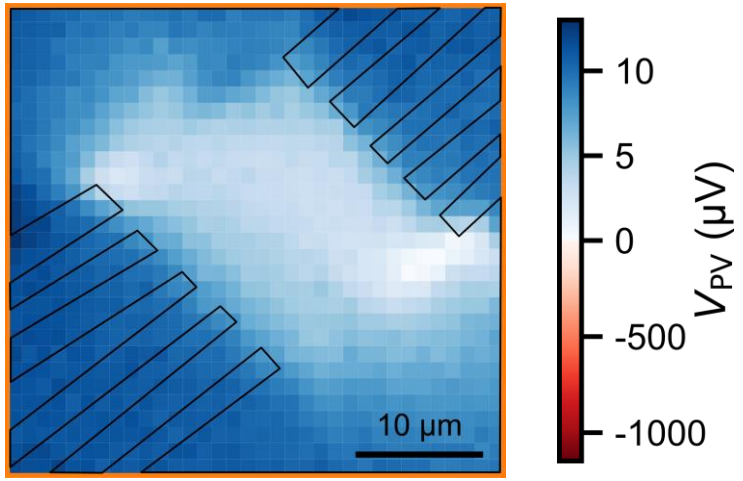
Comment caractériser spatialement le transport ?



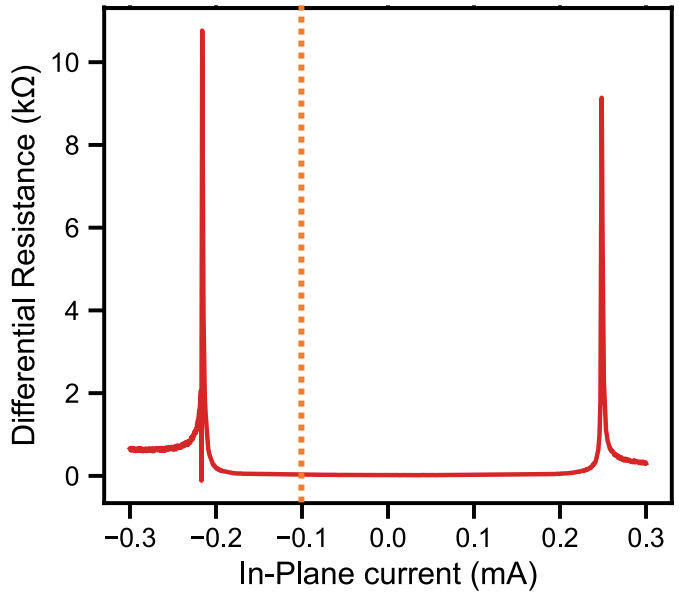
- Courant DC + lumière AC (hachée)
- $$V_{\mathrm{PV}} \approx \frac{\mathrm{d}V_{\mathrm{dark}}}{\mathrm{d}I} \cdot \delta I.$$
- Résolution spatiale : taille du faisceau laser

La photo-tension reflète dV/dI localement !

$I_{DC} = -0.1\ mA$

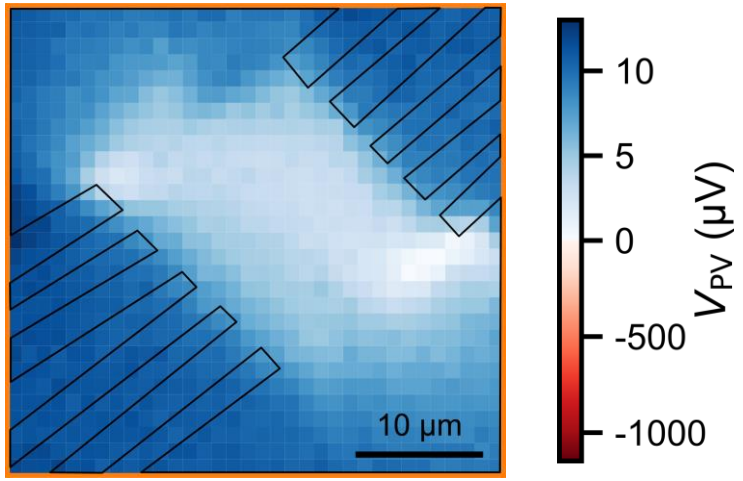


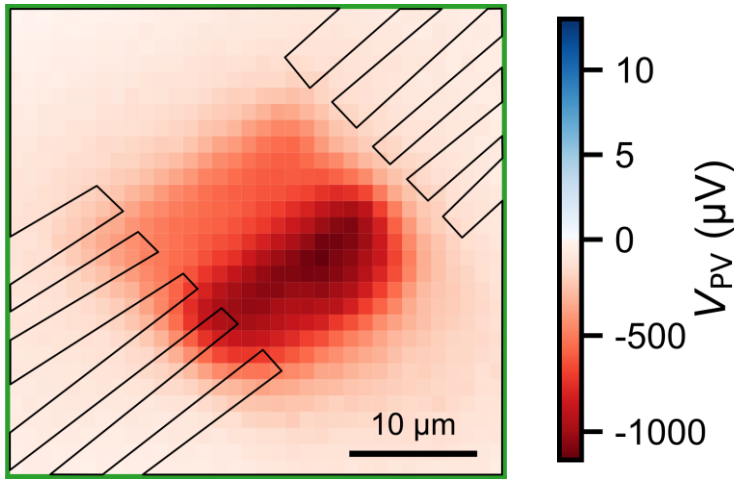
- Régime linéaire : accumulation de courant



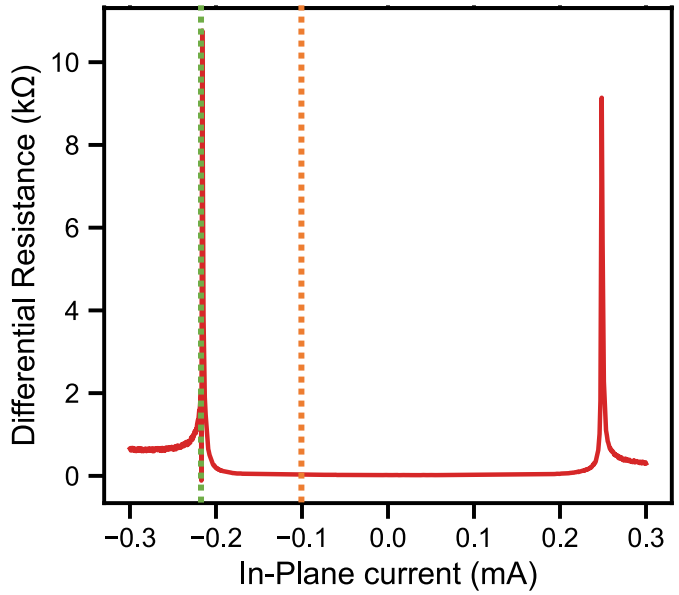
La photo-tension reflète dV/dI localement !

$I_{DC} = -0.1\ mA$





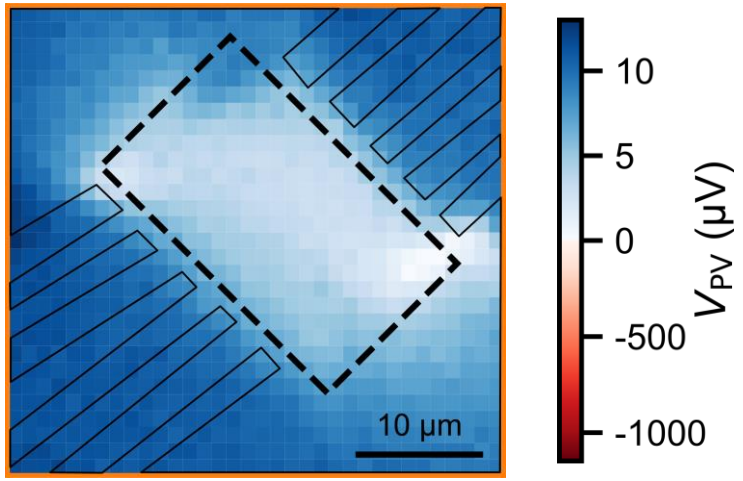
$I_{DC} = -0.21\ mA$

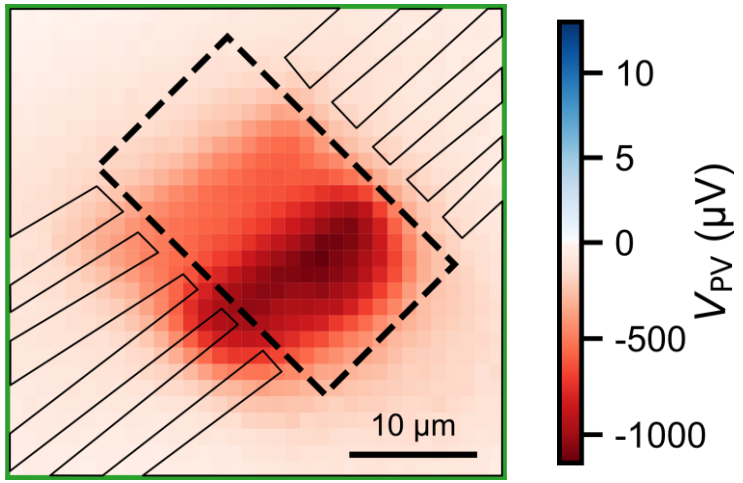


- Régime linéaire : accumulation de courant
- La criticité est localisée spatialement

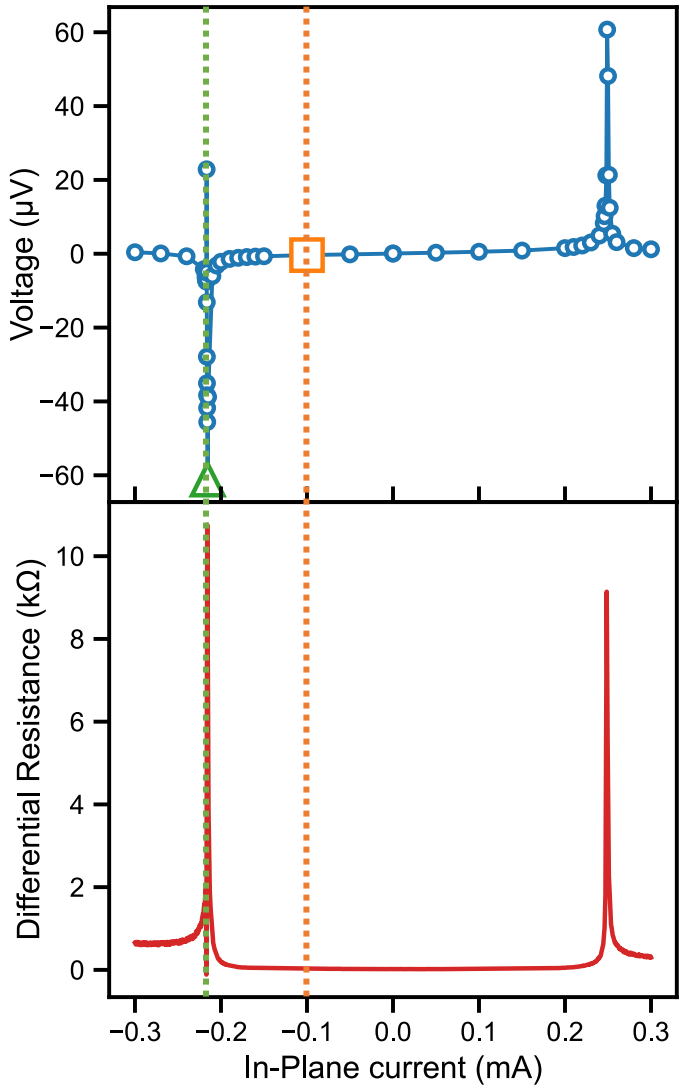
La photo-tension reflète dV/dI localement !

$I_{DC} = -0.1\ mA$



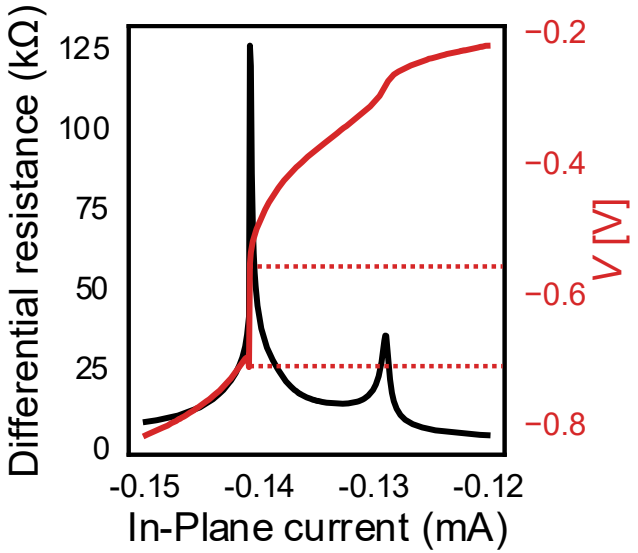


$I_{DC} = -0.21\ mA$



- Régime linéaire : accumulation de courant
- La criticité est localisée spatialement
- La photo-tension est bien proportionnelle a la résistance différentielle !

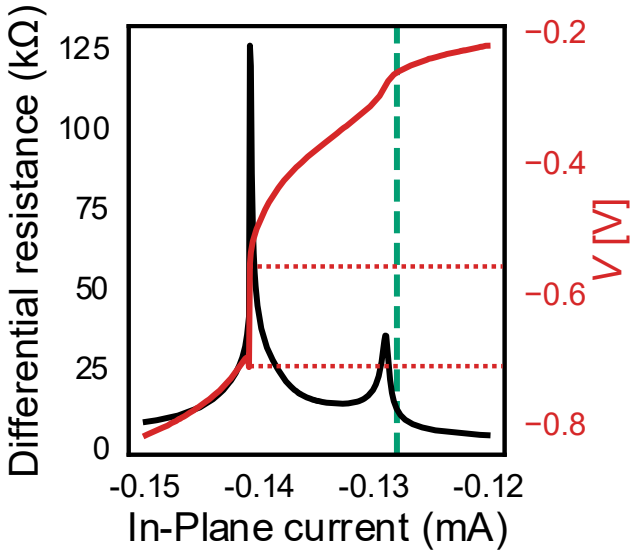
L’impact de l’effet d’auto-grille sur la NDC

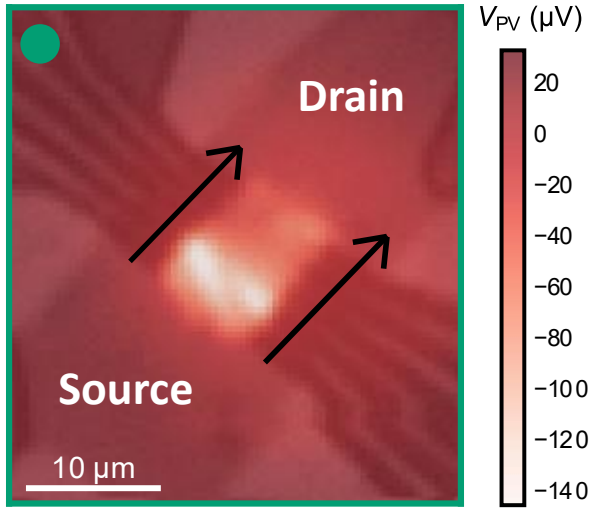


Source-Drain : gauche-droite

- NDC en transport

L’impact de l’effet d’auto-grille sur la NDC



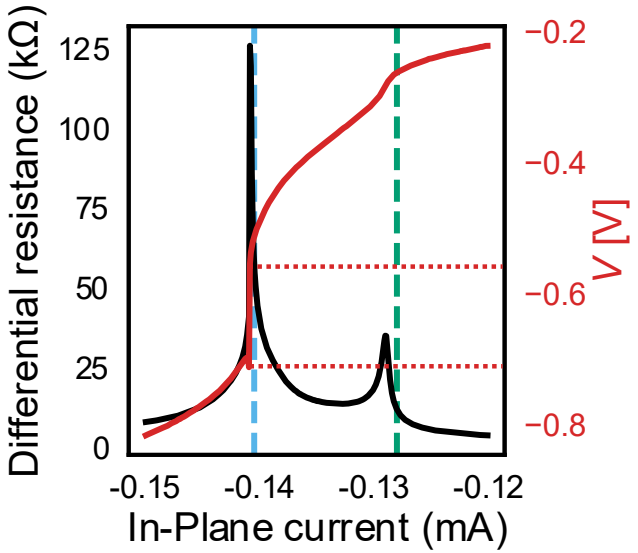


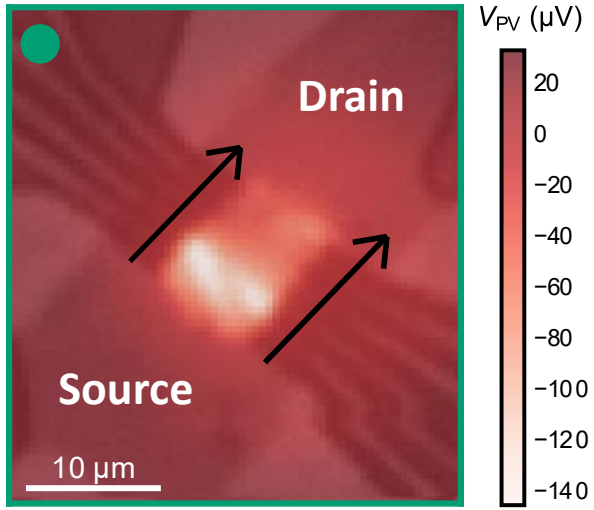
$I_{DC} = -0.128\ mA$

Source-Drain : gauche-droite

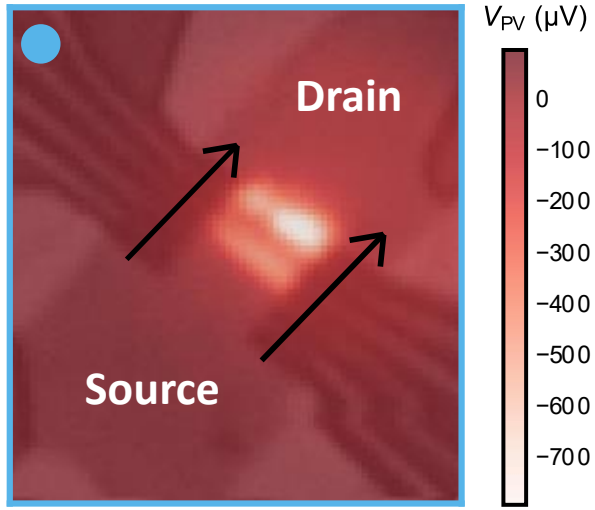
- NDC en transport
- -0.128 mA pic : côté **gauche**

L’impact de l’effet d’auto-grille sur la NDC





$I_{DC} = -0.128\ mA$

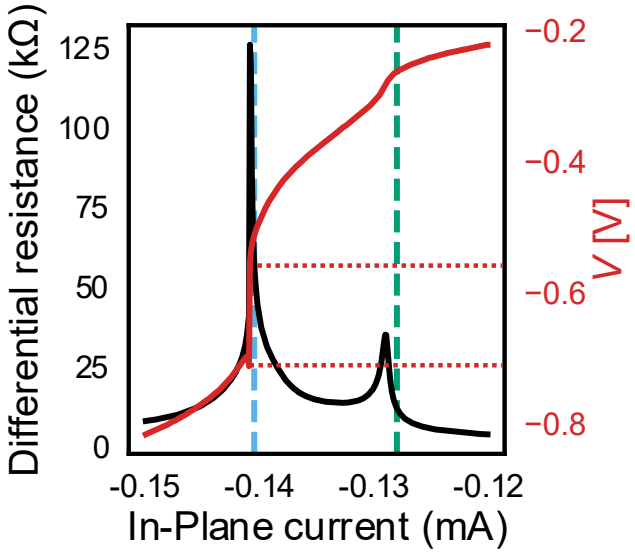


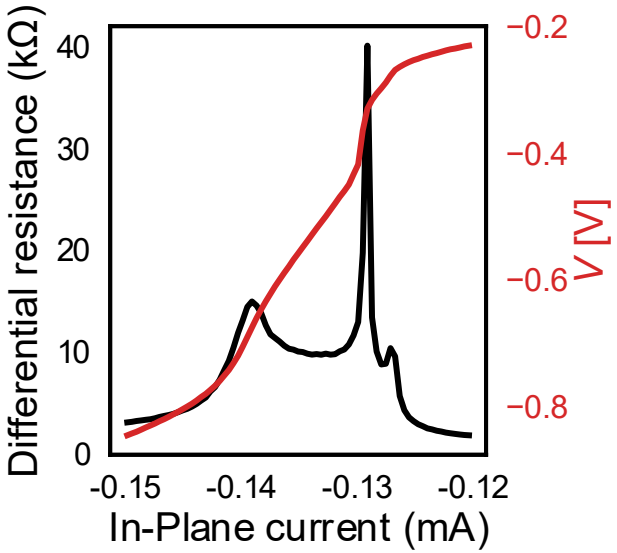
$I_{DC} = -0.14\ mA$

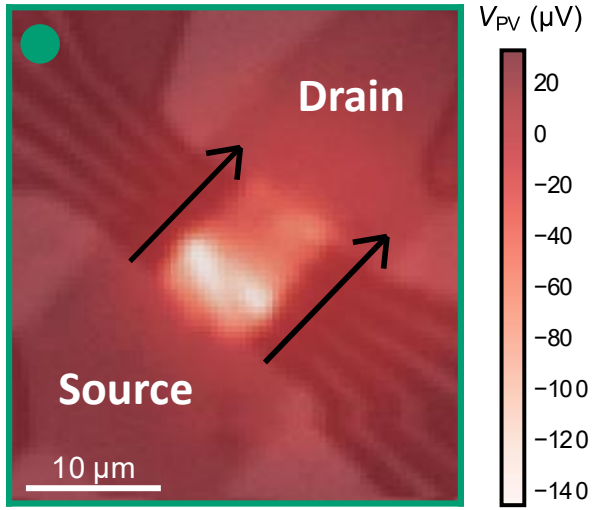
Source-Drain : gauche-droite

- NDC en transport
- -0.128 mA pic : côté **gauche**
- -0.14 mA pic (NDC): côté **droit**

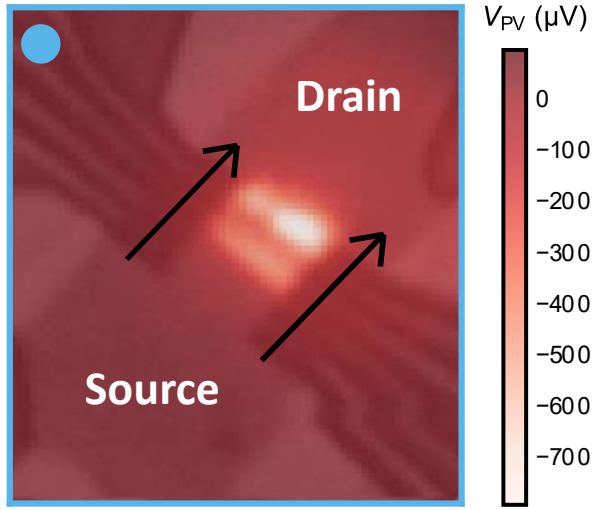
L’impact de l’effet d’auto-grille sur la NDC







$I_{DC} = -0.128\ mA$



$I_{DC} = -0.14\ mA$

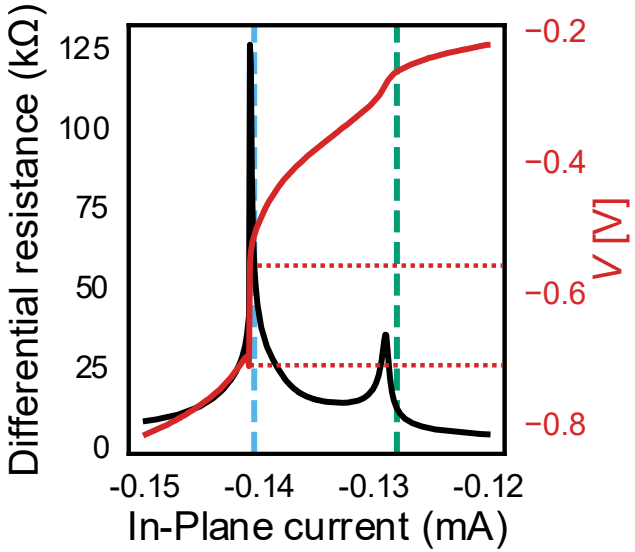
Source-Drain : gauche-droite

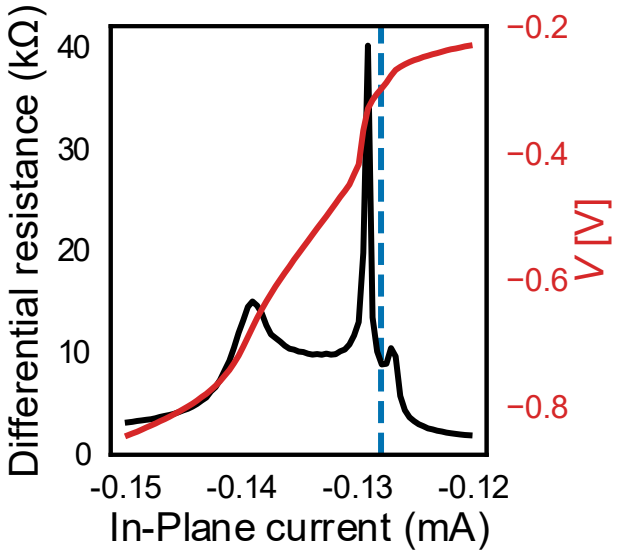
- NDC en transport
- -0.128 mA pic : côté **gauche**
- -0.14 mA pic (NDC): côté **droit**

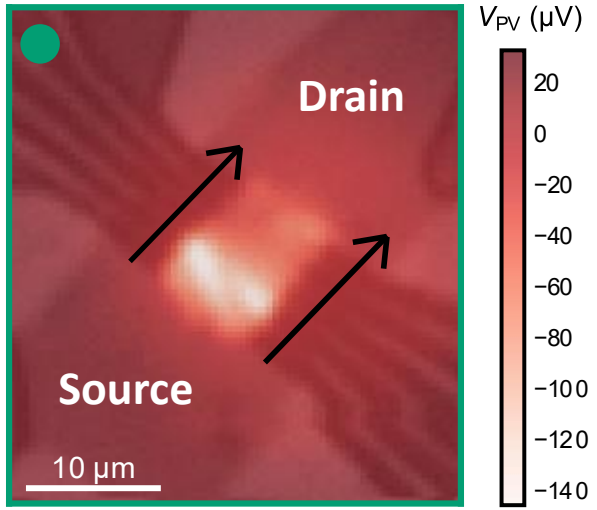
Source-Drain : droite-gauche

- Pas de NDC en transport

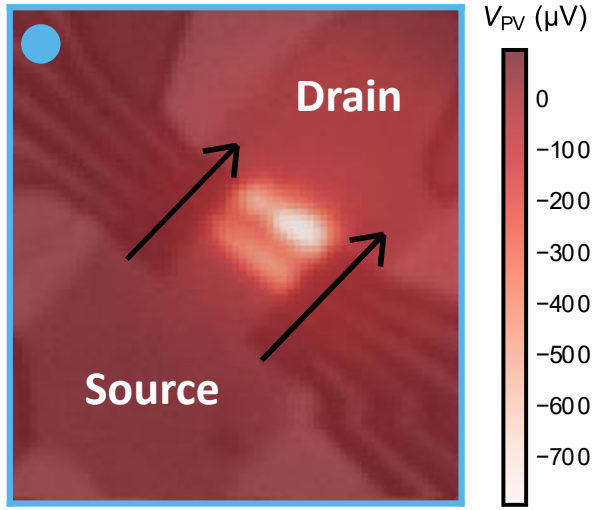
L’impact de l’effet d’auto-grille sur la NDC



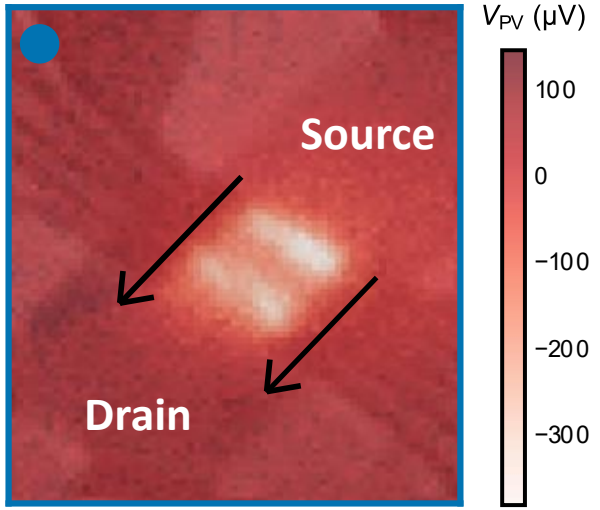




$I_{DC} = -0.128\ mA$



$I_{DC} = -0.14\ mA$



$I_{DC} = -0.128\ mA$

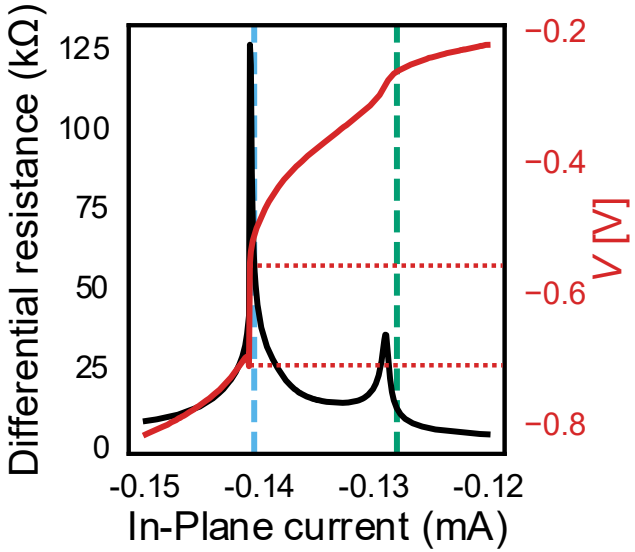
Source-Drain : gauche-droite

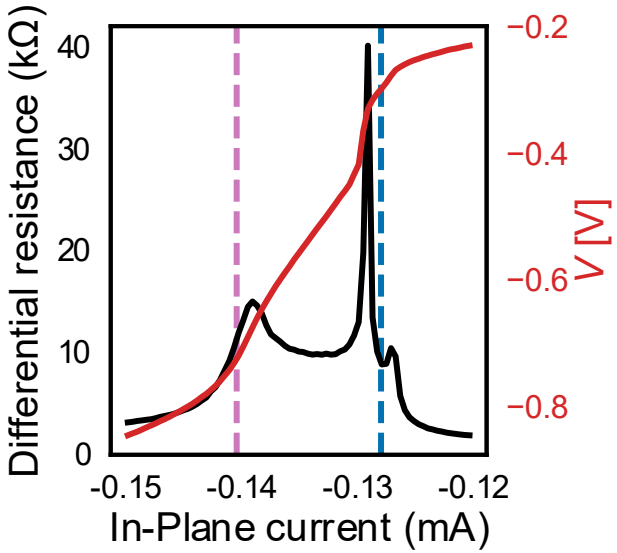
- NDC en transport
- -0.128 mA pic : côté **gauche**
- -0.14 mA pic (NDC): côté **droit**

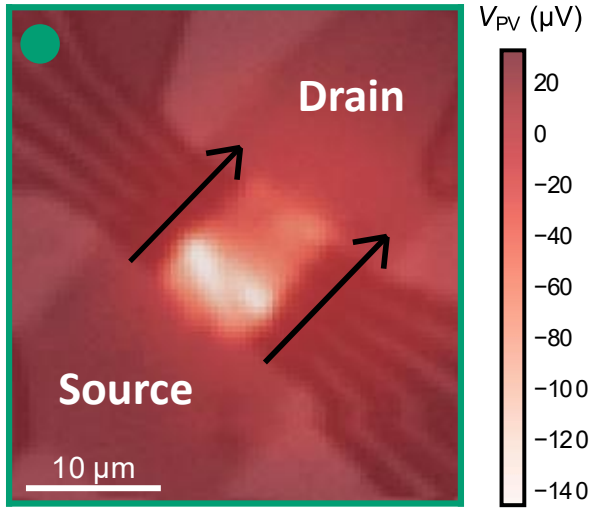
Source-Drain : droite-gauche

- Pas de NDC en transport
- -0.128 mA pic : côté **droit**

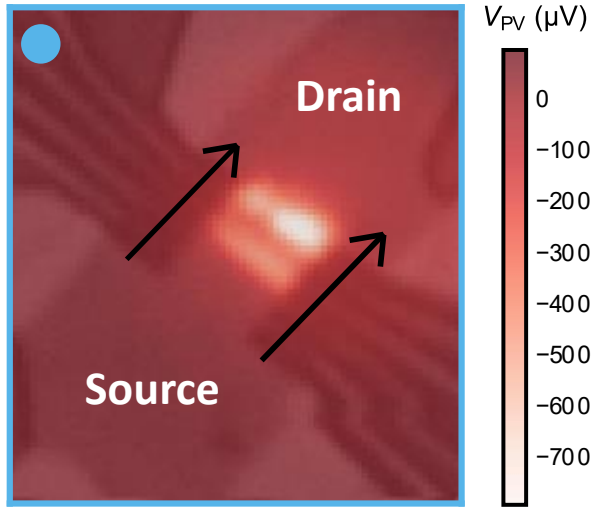
L’impact de l’effet d’auto-grille sur la NDC



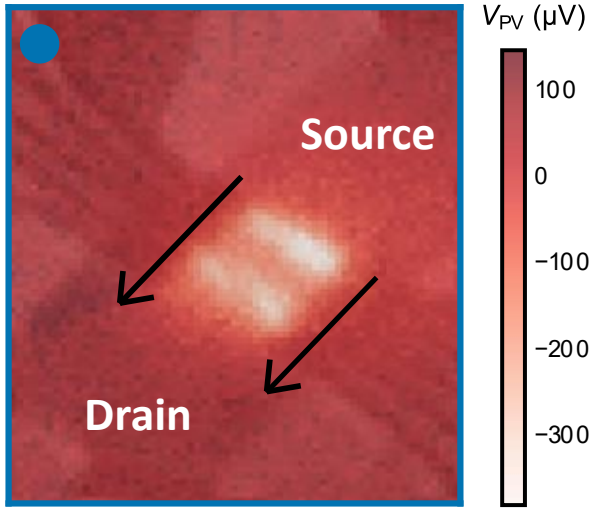




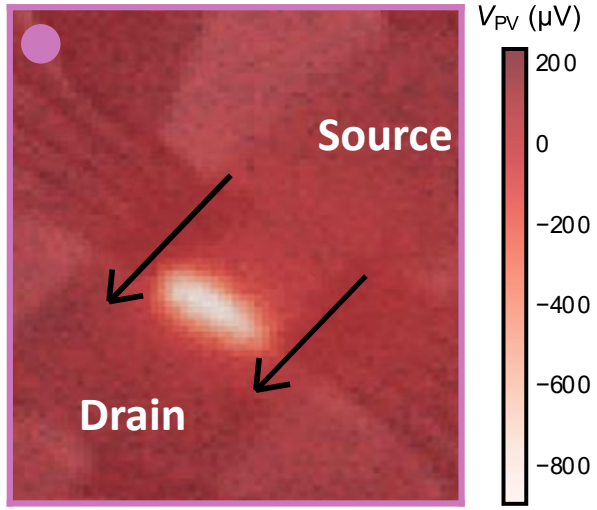
$I_{DC} = -0.128\ mA$



$I_{DC} = -0.14\ mA$



$I_{DC} = -0.128\ mA$



$I_{DC} = -0.14\ mA$

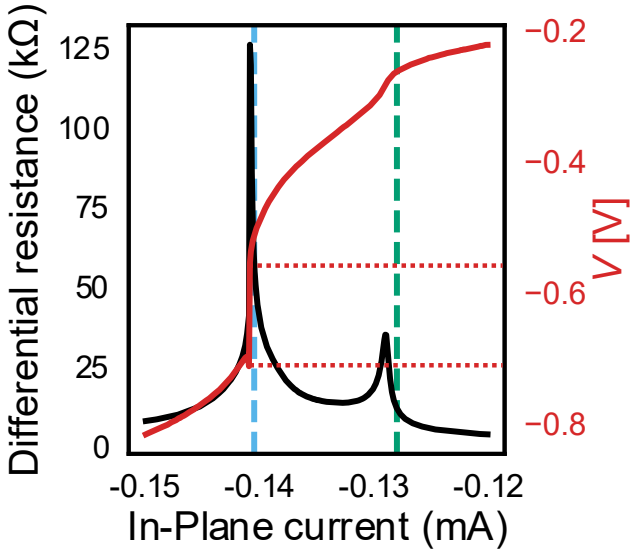
Source-Drain : gauche-droite

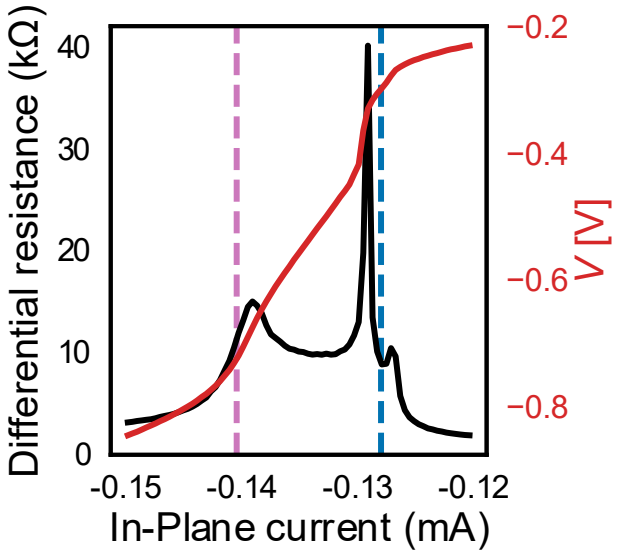
- NDC en transport
- -0.128 mA pic : côté **gauche**
- -0.14 mA pic (NDC): côté **droit**

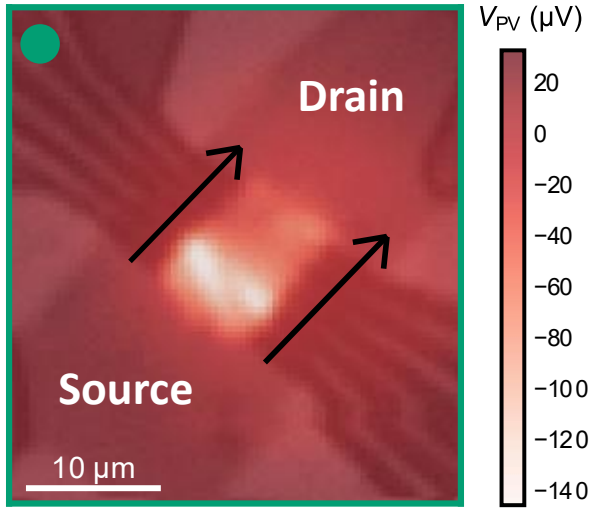
Source-Drain : droite-gauche

- Pas de NDC en transport
- -0.128 mA pic : côté **droit**
- -0.14 mA pic : côté **gauche**

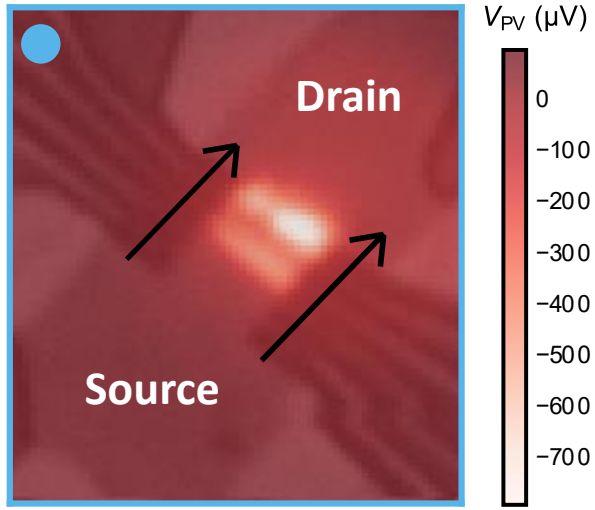
L'impact de l'effet d'auto-grille sur la NDC



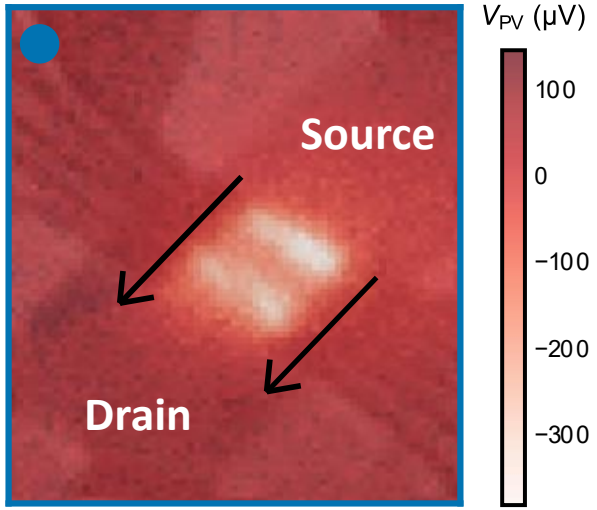




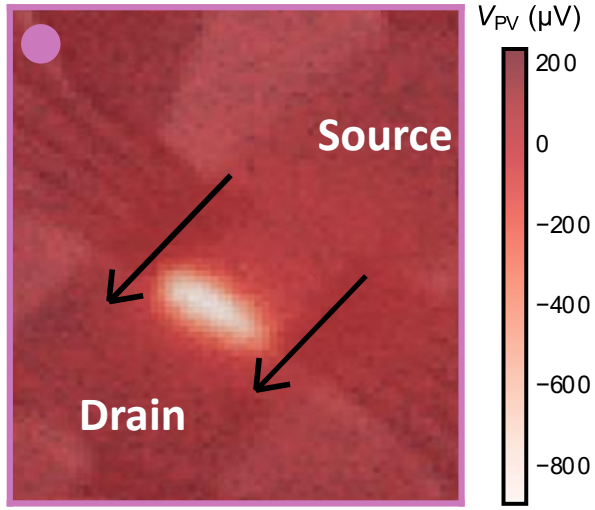
$I_{DC} = -0.128\ mA$



$I_{DC} = -0.14\ mA$



$I_{DC} = -0.128\ mA$



$I_{DC} = -0.14\ mA$

Source-Drain : gauche-droite

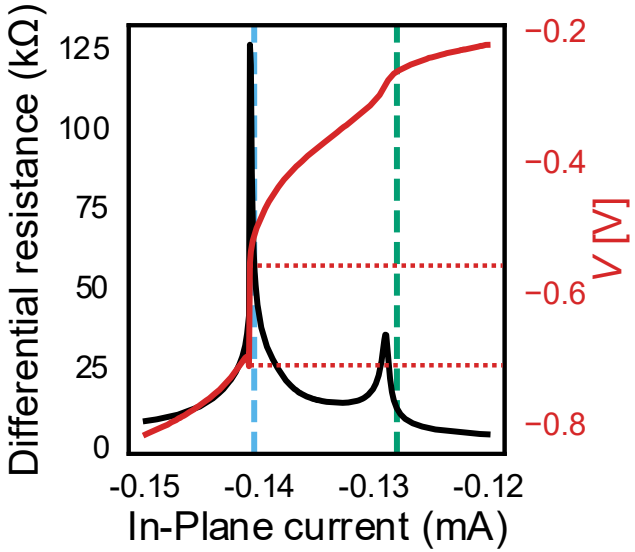
- NDC en transport
- -0.128 mA pic : côté **gauche**
- -0.14 mA pic (NDC): côté **droit**

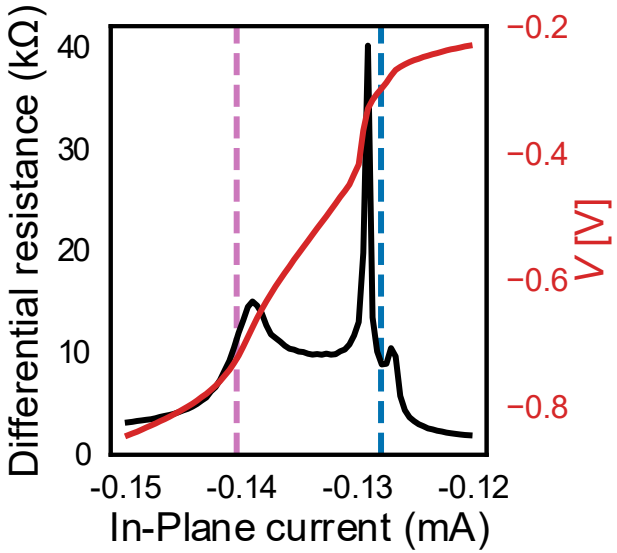
Source-Drain : droite-gauche

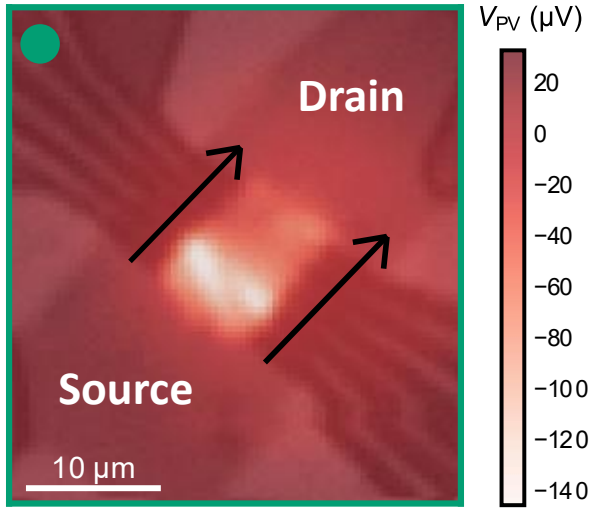
- Pas de NDC en transport
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- $n = n(x)$: Effet d'auto-grille

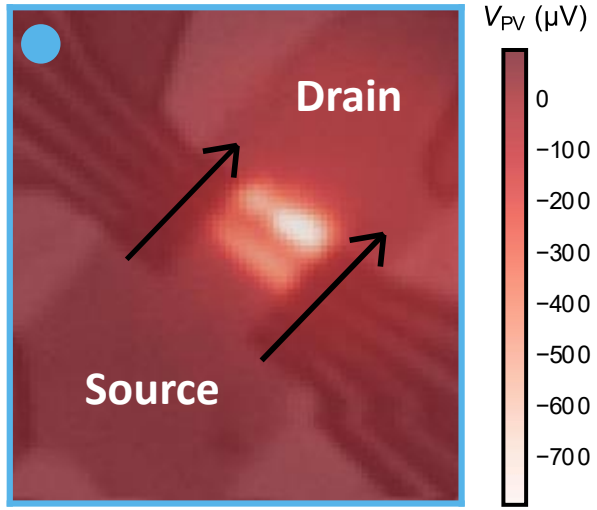
L'impact de l'effet d'auto-grille sur la NDC



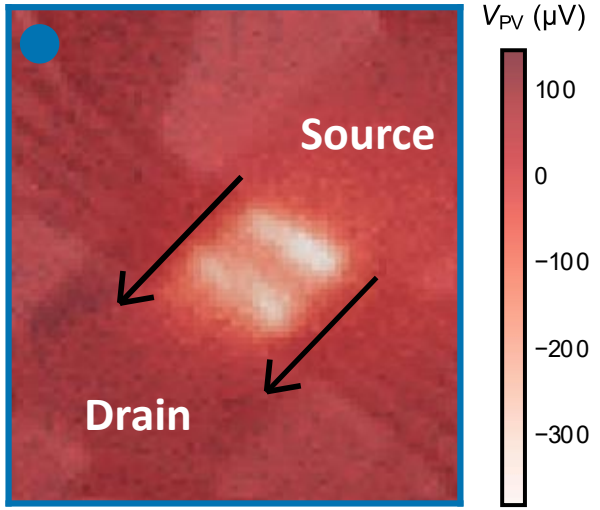




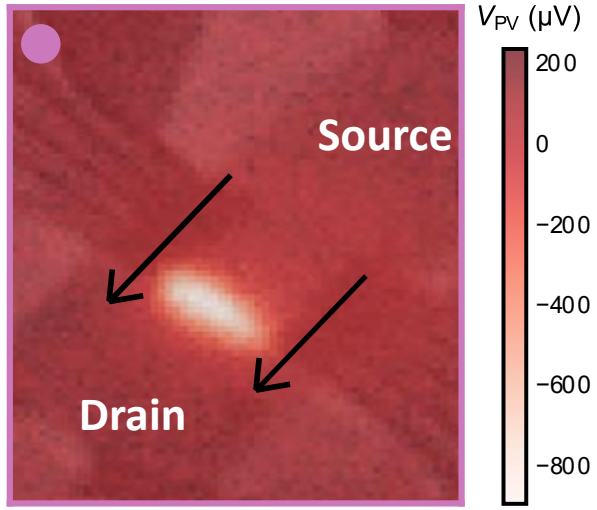
$I_{DC} = -0.128\ mA$



$I_{DC} = -0.14\ mA$



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Source-Drain : gauche-droite

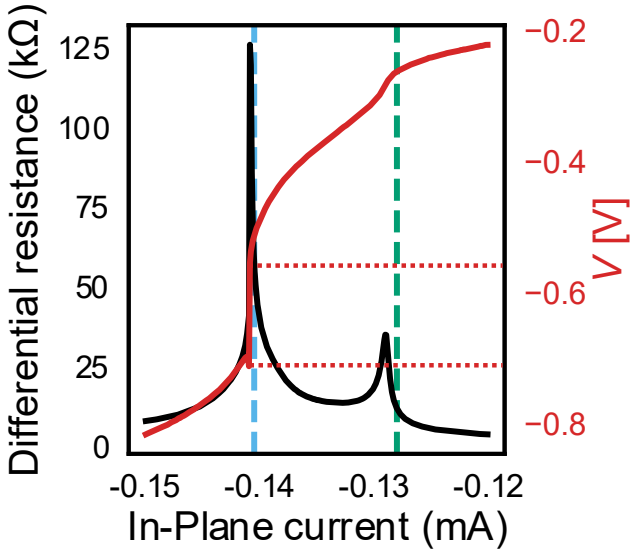
- NDC en transport
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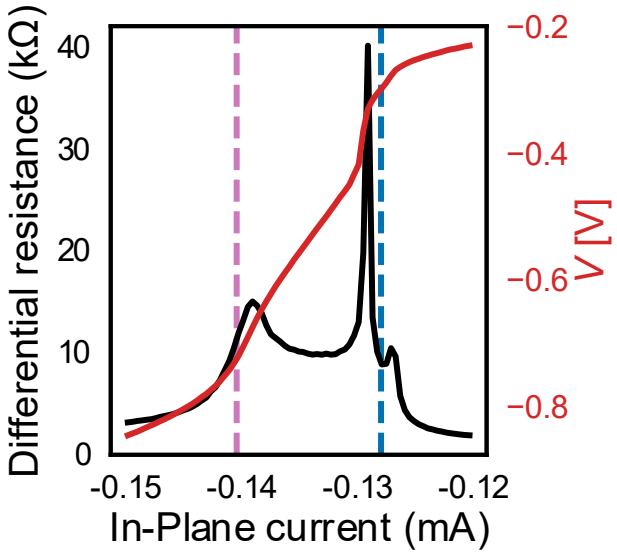
Source-Drain : droite-gauche

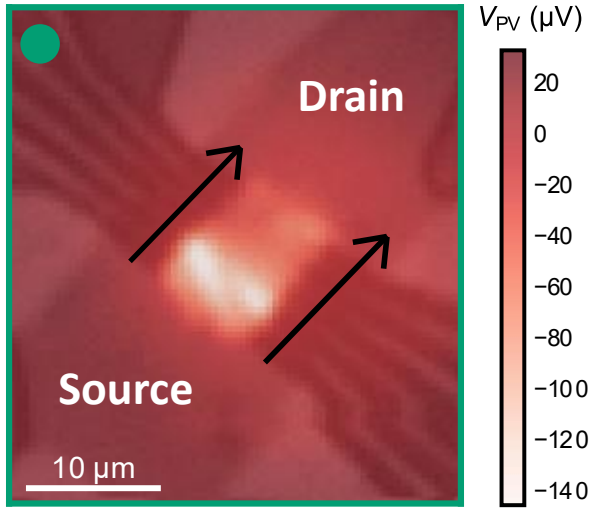
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- $n = n(x)$: Effet d'auto-grille
- $n(x) = n_0 - \frac{C_G}{e} * V(x)$

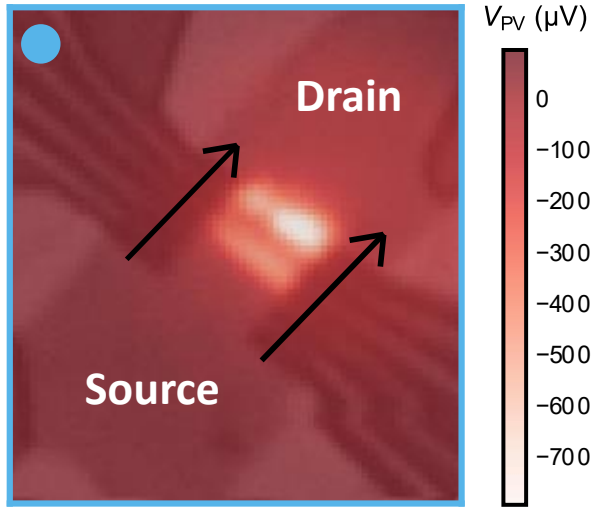
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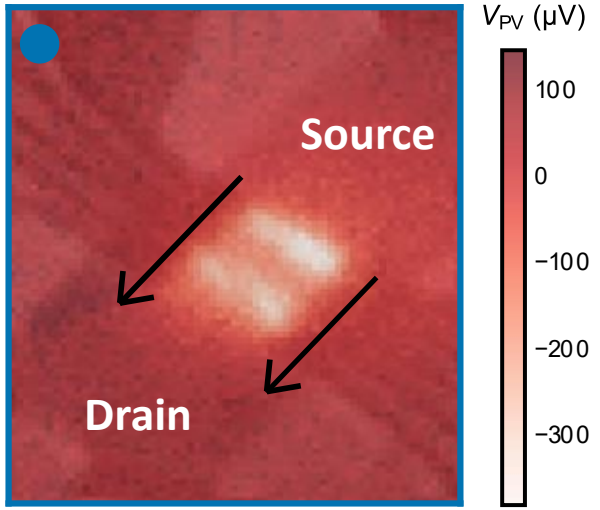




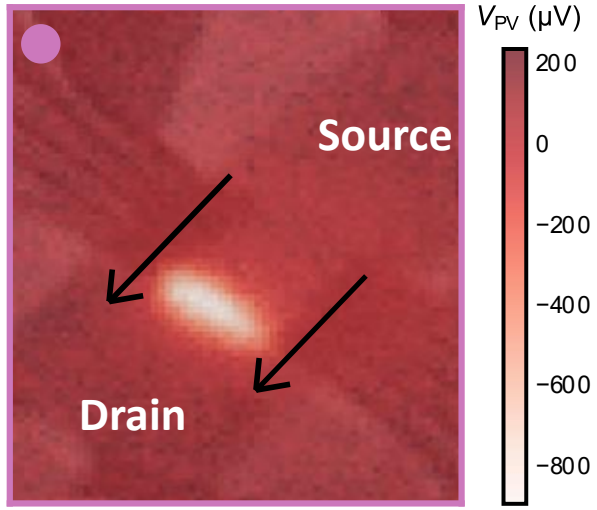
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Source-Drain : gauche-droite

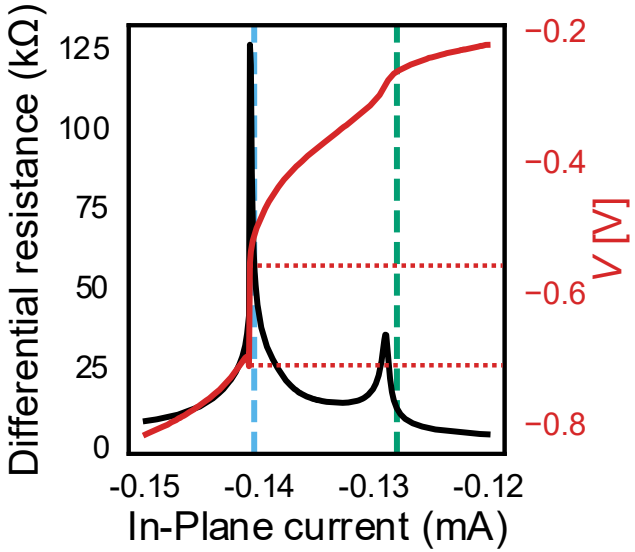
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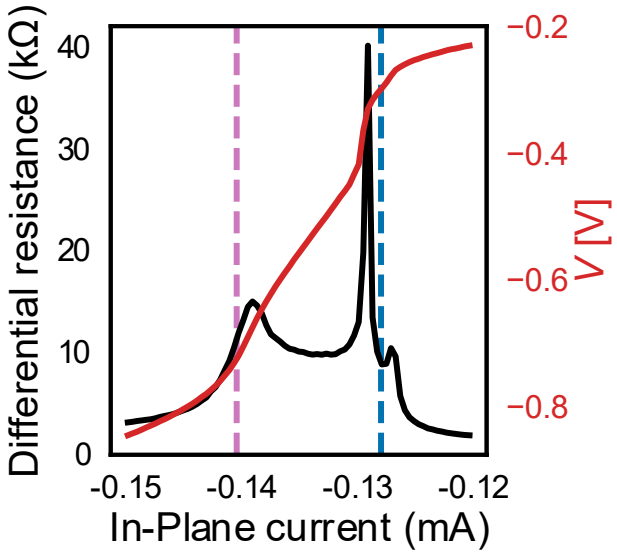
Source-Drain : droite-gauche

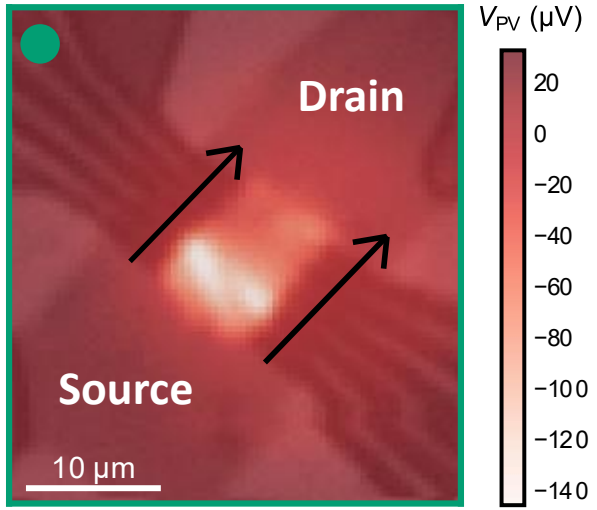
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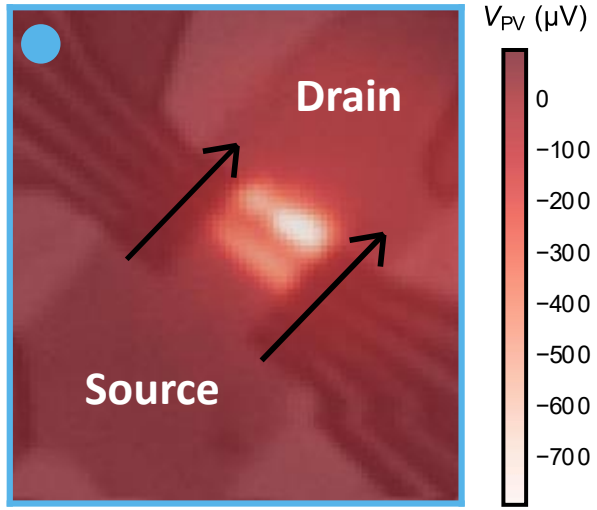
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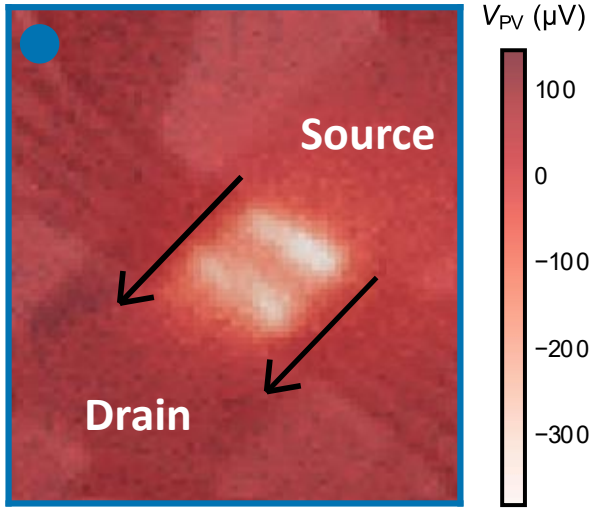




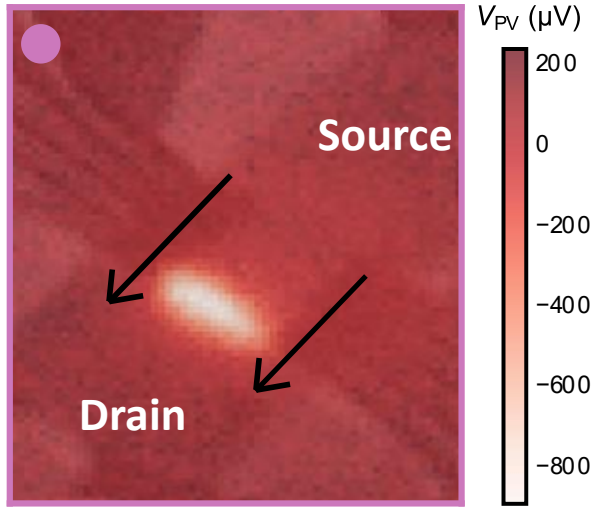
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Source-Drain : gauche-droite

- NDC en transport
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Echantillon géométriquement complexe !

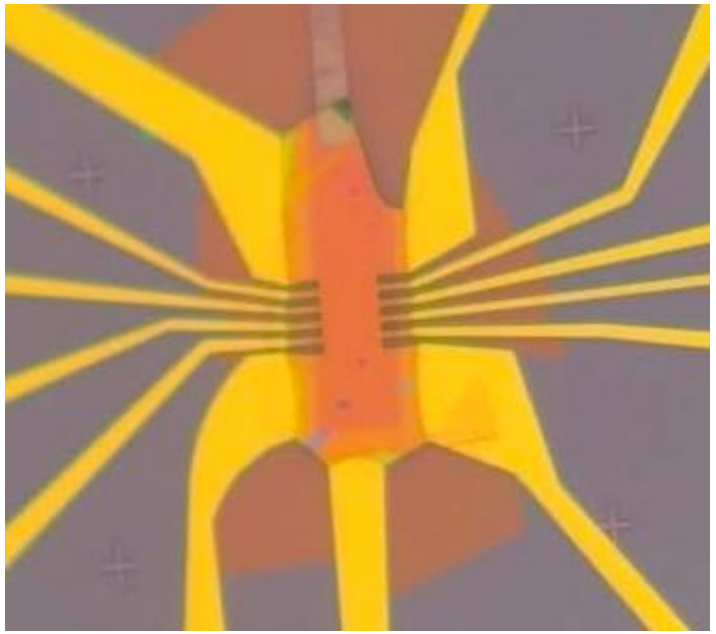
Simulation

Simulation

- Nouvel échantillon :
  - Forme rectangulaire
  - Même matériau
  - Densité de courant constante

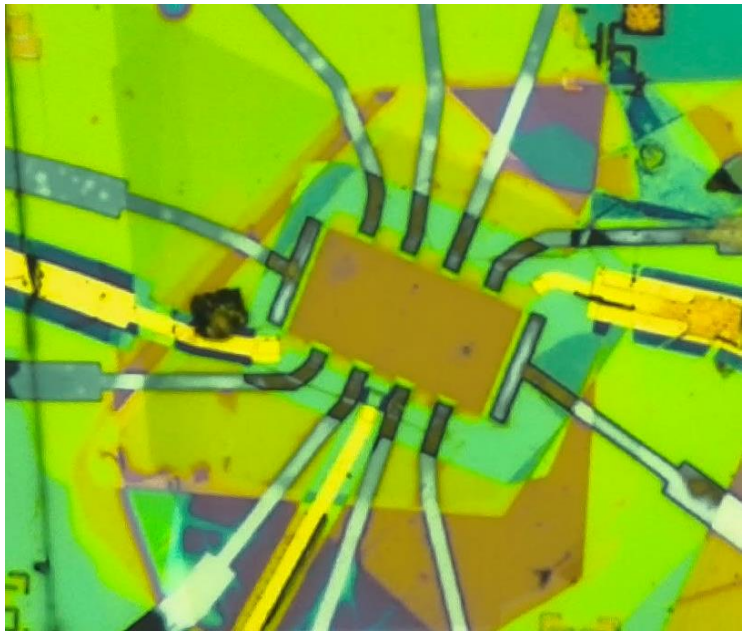
Simulation

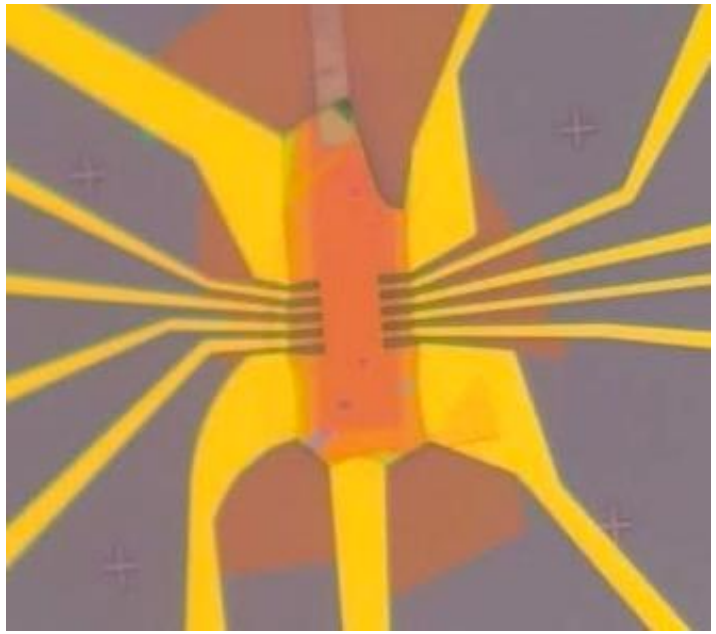
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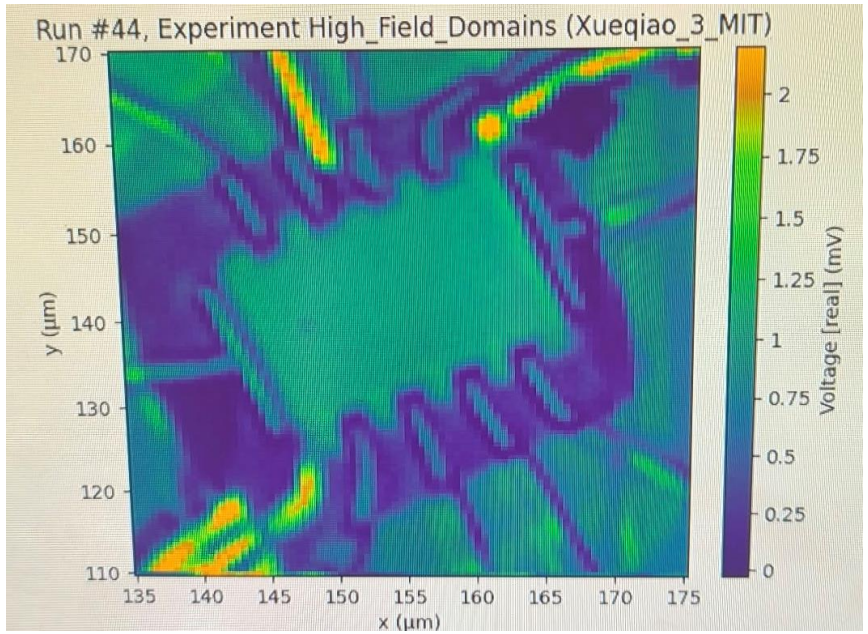
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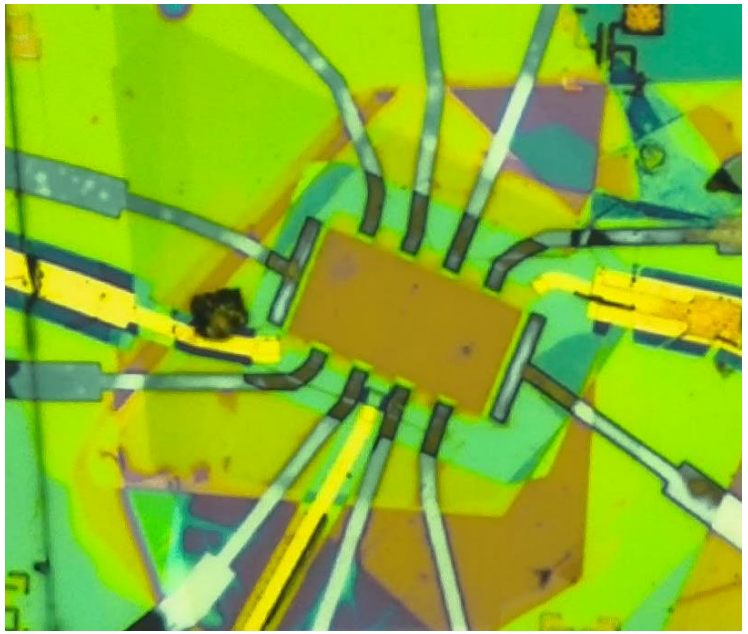


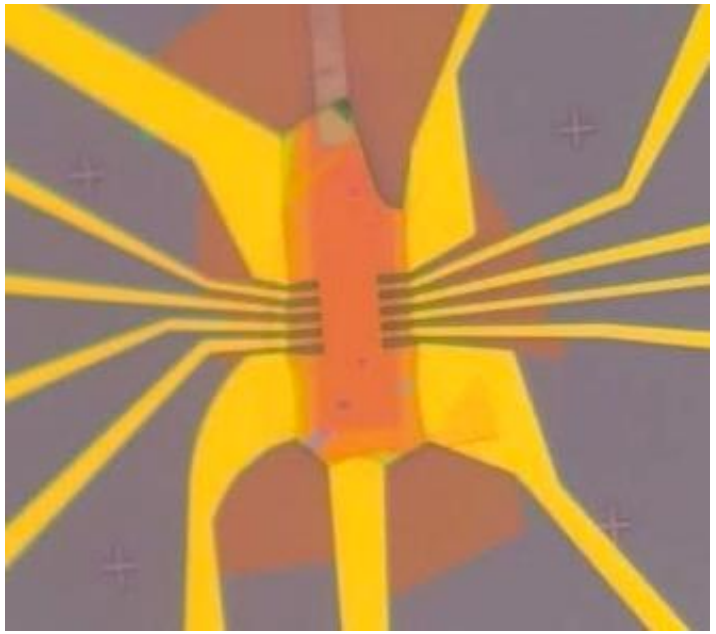


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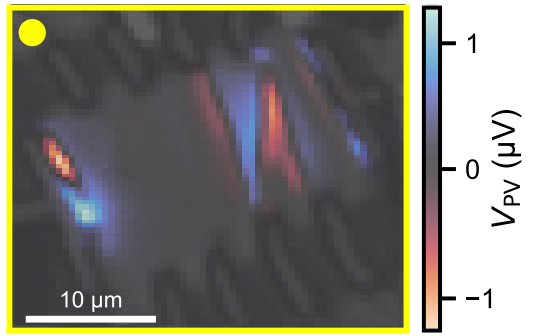
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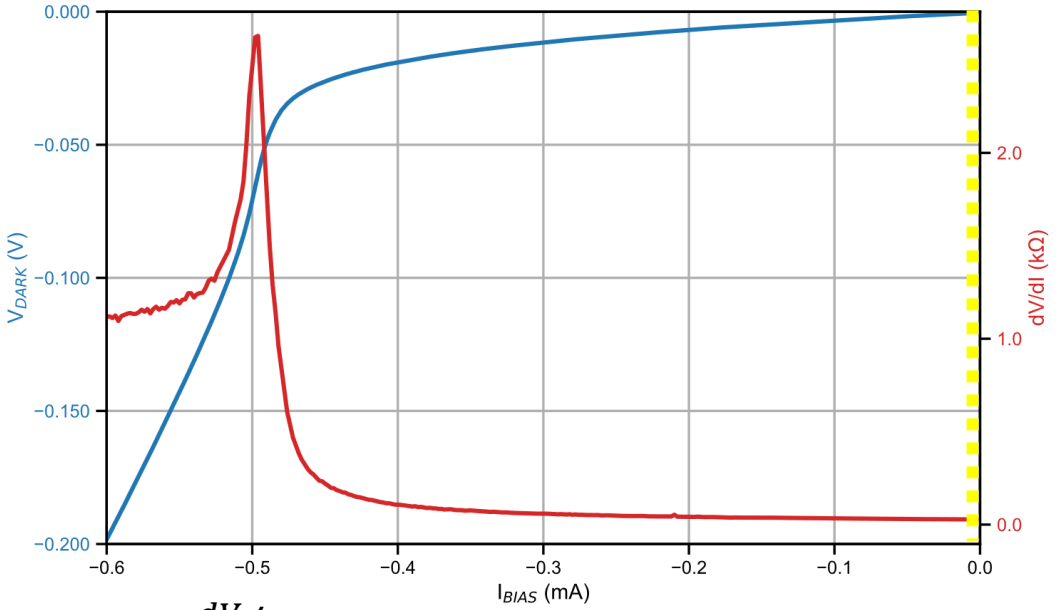




La criticité voyage à travers l'échantillon

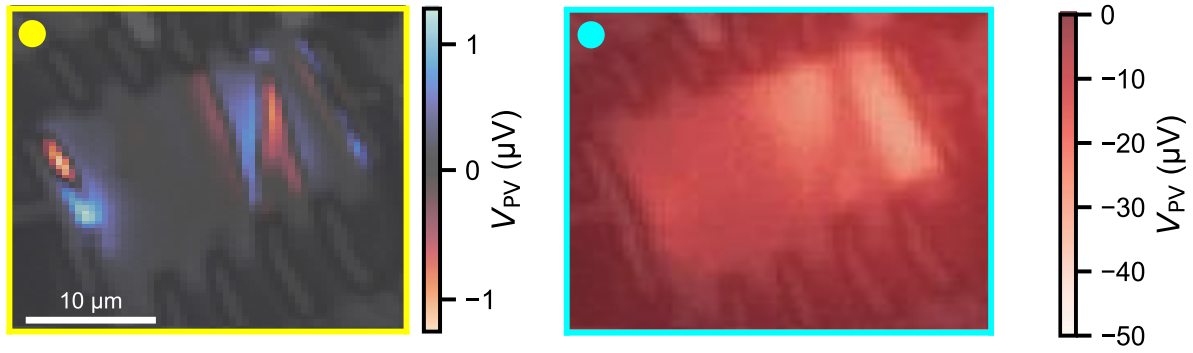


0 mA DC



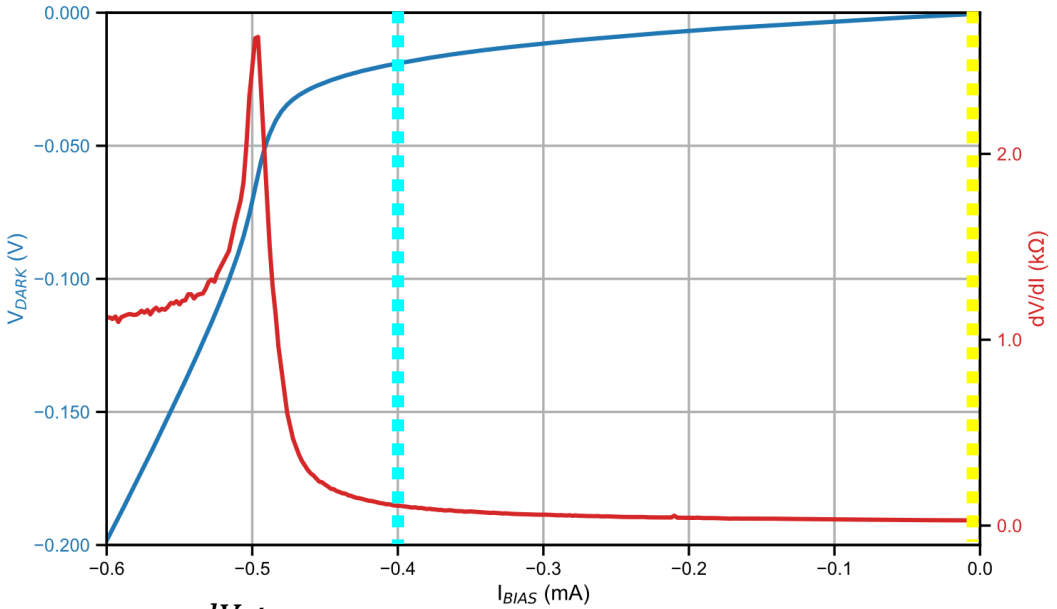
dV/dI : Transport measurement

La criticité voyage à travers l'échantillon



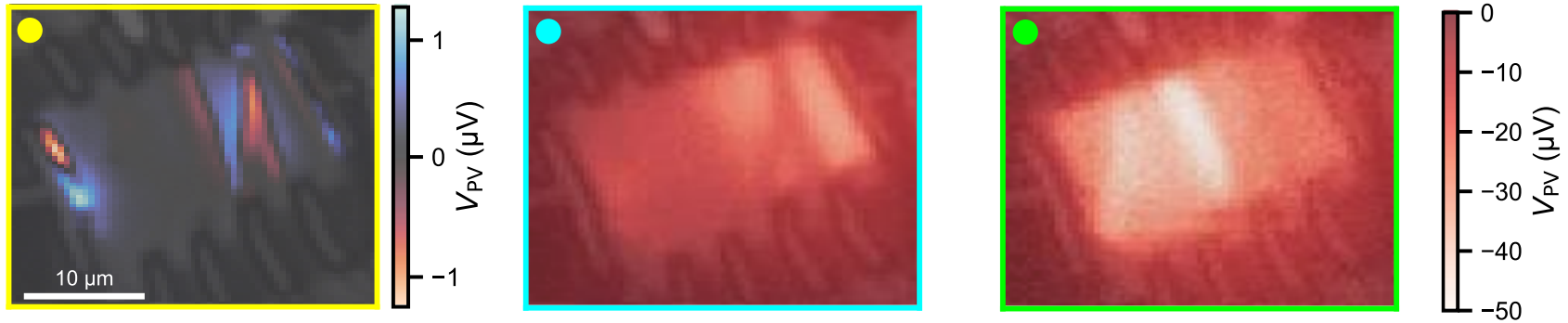
0 mA DC

-0.4 mA DC



dV/dI : Transport measurement

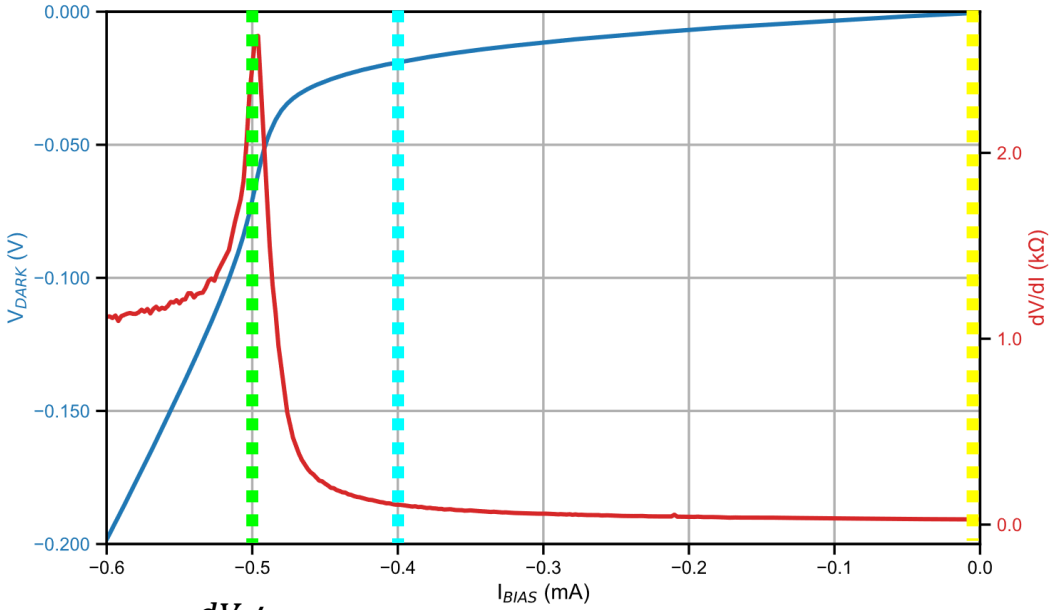
La criticité voyage à travers l'échantillon



0 mA DC

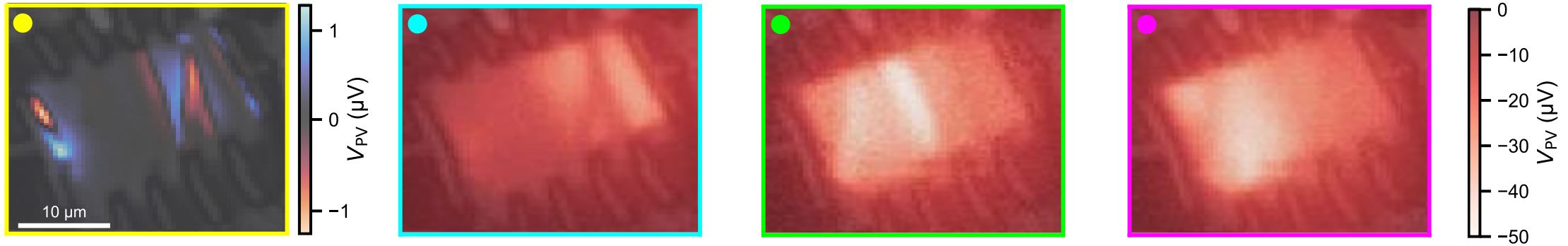
-0.4 mA DC

-0.5 mA DC



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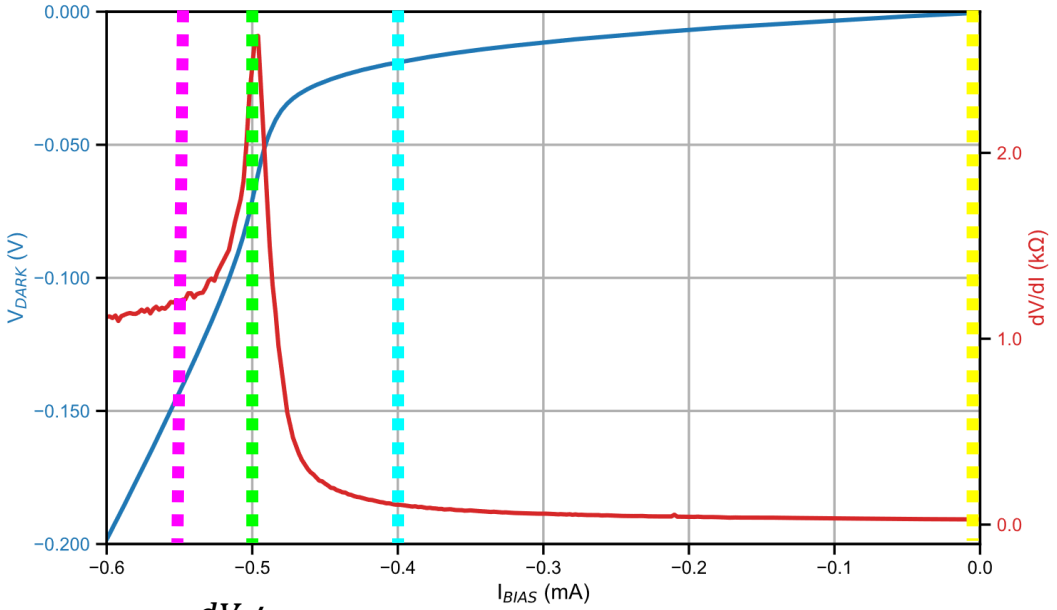


0 mA DC

-0.4 mA DC

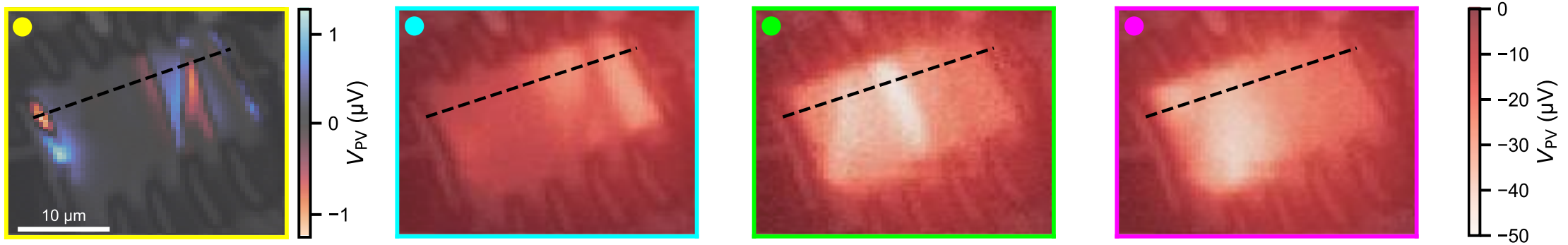
-0.5 mA DC

-0.55 mA DC

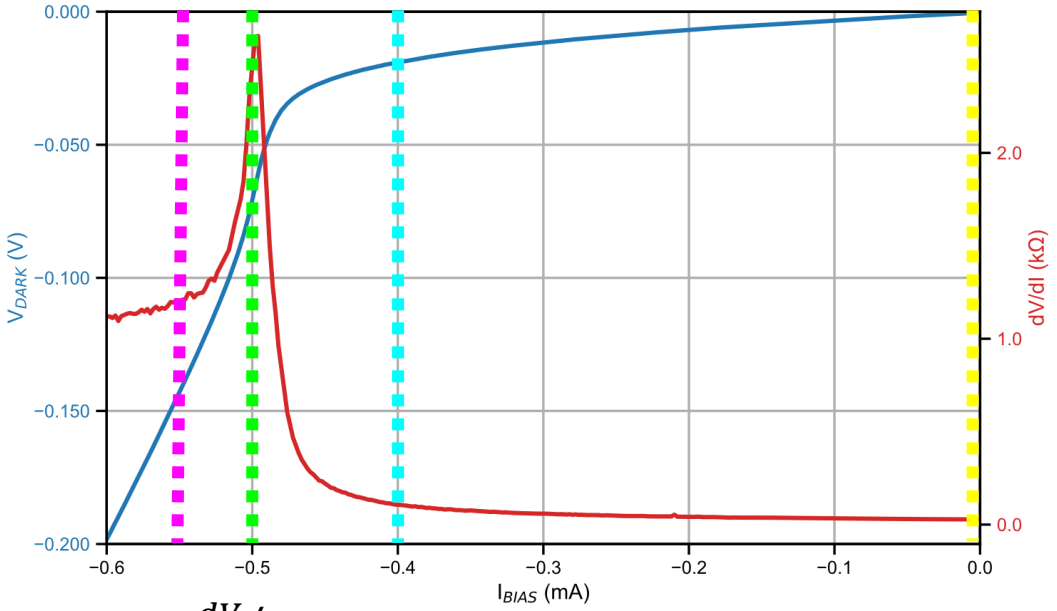


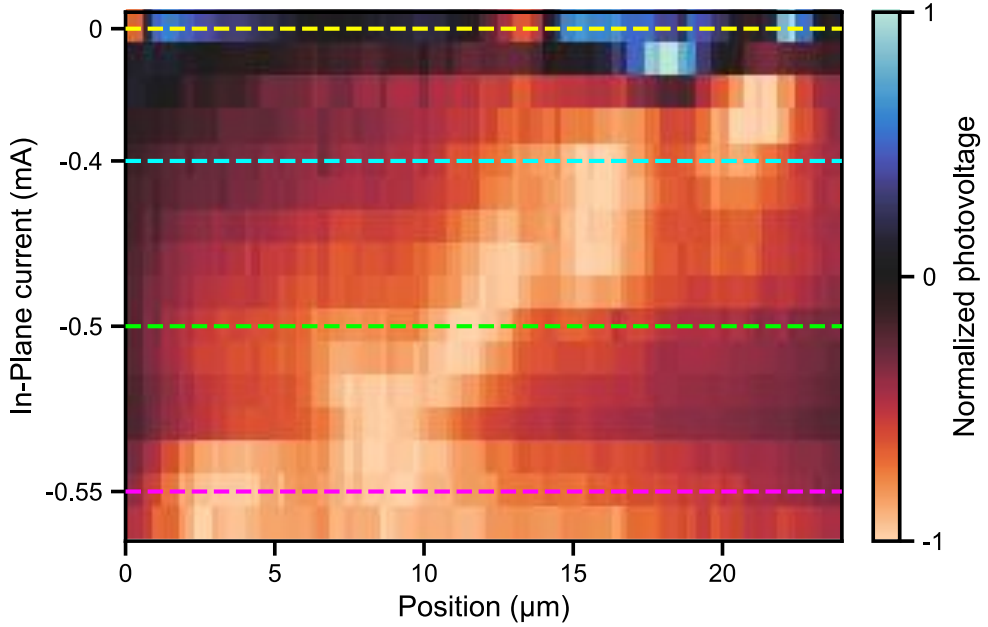
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La criticité voyage à travers l'échantillon



0 mA DC -0.4 mA DC -0.5 mA DC -0.55 mA DC

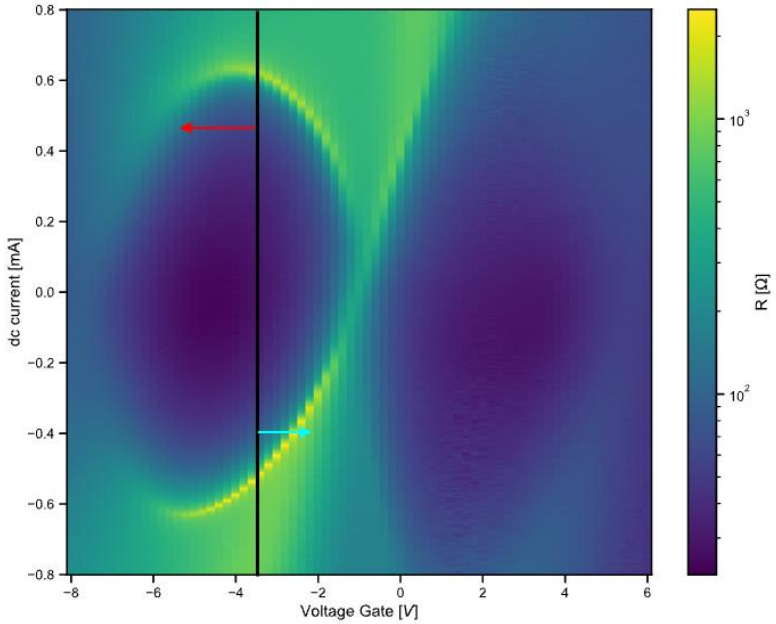




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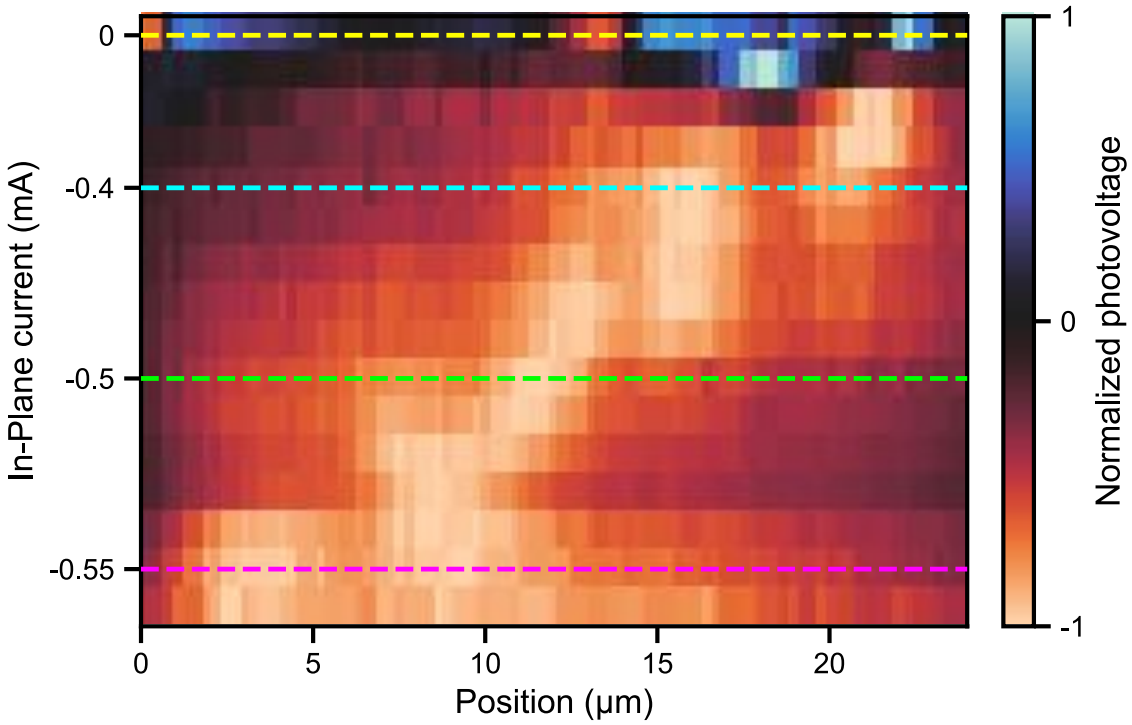
Contexte de la simulation

- Echantillon discrétisé en pixel :
  - Chaque pixel a sa propre :
    - Resistance DC
    - Tension
    - Densité de charges

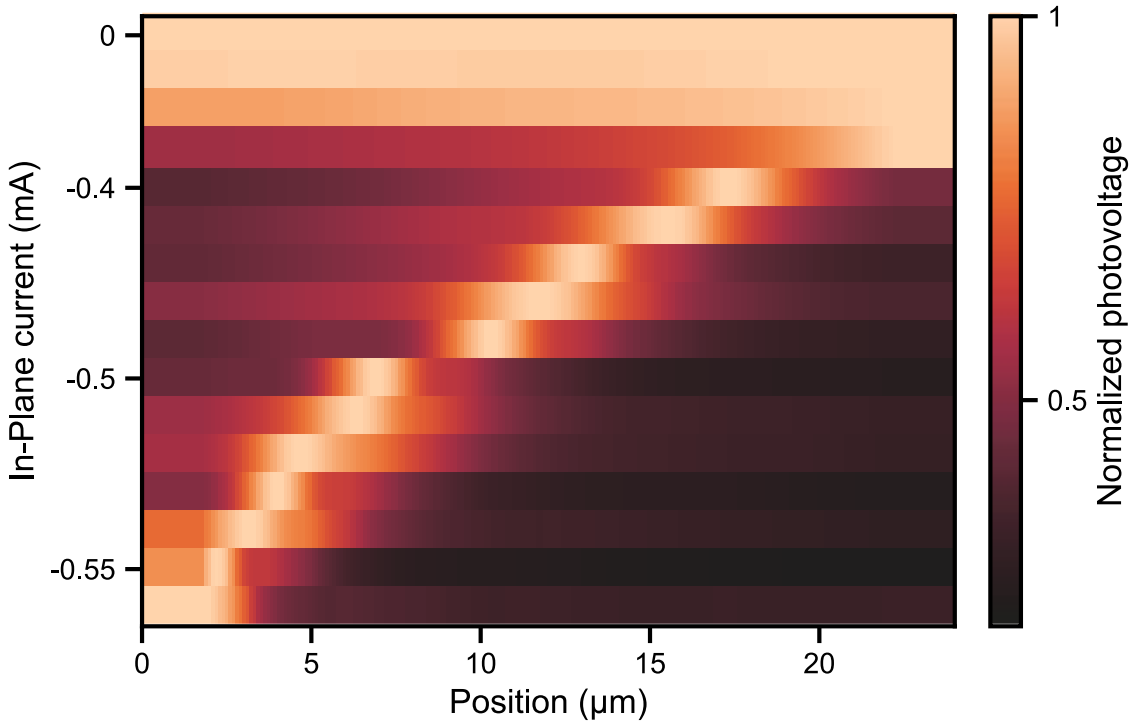


- On utilise les mesures en transport
- On considère l'échantillon homogène
- On inclut l'effet d'auto-grille et on itère sur l'équation de Laplace : $\nabla \cdot [\sigma(x,y)\,\nabla\phi_{\text{unit}}(x,y)] = 0$
- Convergence quand le potentiel est stabilisé
- Photo-tension grâce à dV/dI

Comparaison mesure / simulation



Mesure expérimentale



Simulations

- $n(x,y) = n_0 - \frac{C_G}{e} * V(x,y)$

Conclusion et perspective

- Grâce à la spectroscopie de photo-tension, les mesures de transport peuvent être caractérisées localement !

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- Perspectives :
  - Utiliser moins de hBN pour augmenter l'effet d'auto-grille
  - Utiliser des grilles en silice pour tuer l'effet d'auto-grille

Merci de votre attention !



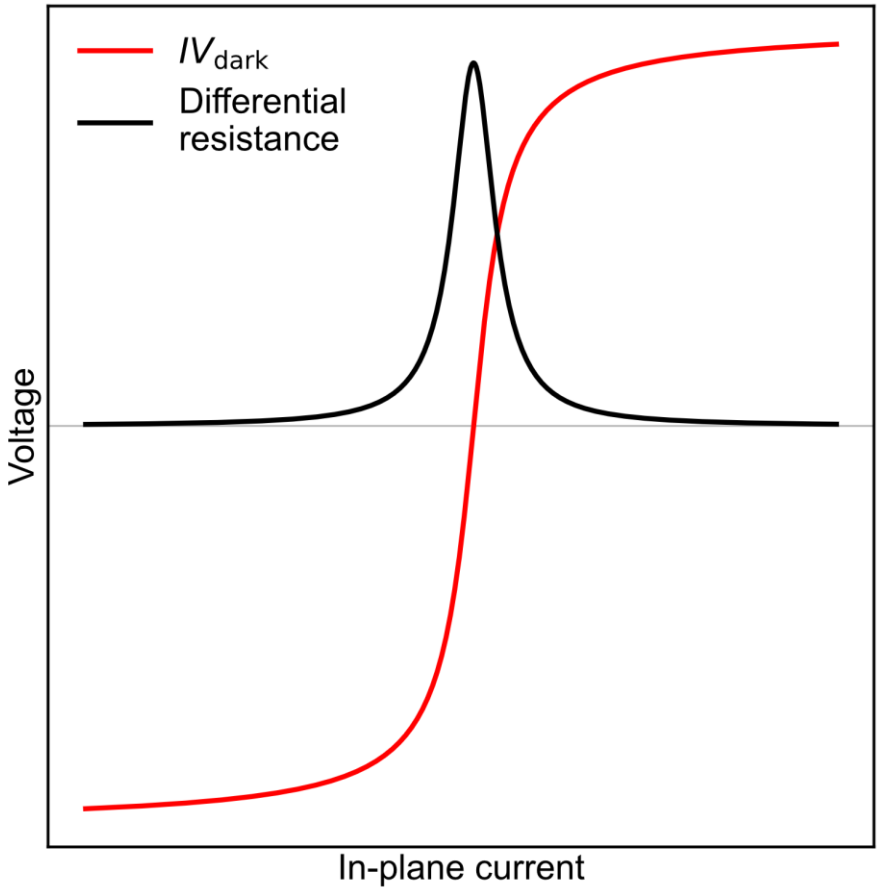
Supplementary slides

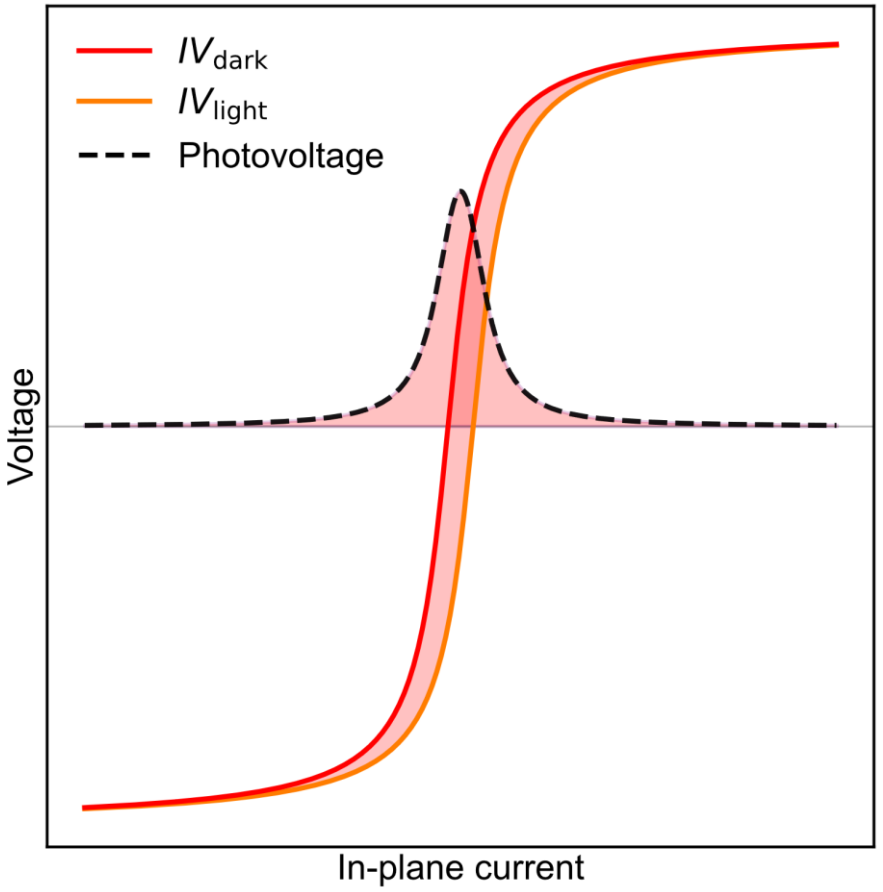
Out-of-equilibrium transport :

- Fermi velocity is the local slope and depends on n for a non Dirac material :
  - The fermi surface is hard to define properly
- Critical current is only ¼ of what it's supposed to be.

- More realistic explanation :
  - Dominant scattering mechanism : Umklapp
    - e-e interaction
  - Bad e-phonon coupling in BLG

Photovoltage vs dV/dI





- AC bias
- AC light

Photovoltage is proportional to the local differential resistance !

Photovoltage vs dV/dI

$$R_{\text{light}}(I) = R_{\text{dark}}(I + \delta I) \approx R_{\text{dark}}(I) + \delta I \cdot \frac{\mathrm{d}R_{\text{dark}}}{\mathrm{d}I}. \tag{1}$$

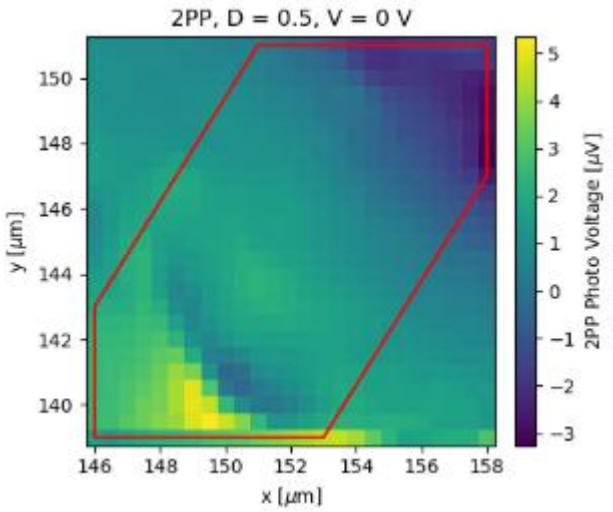
The photovoltage is then:

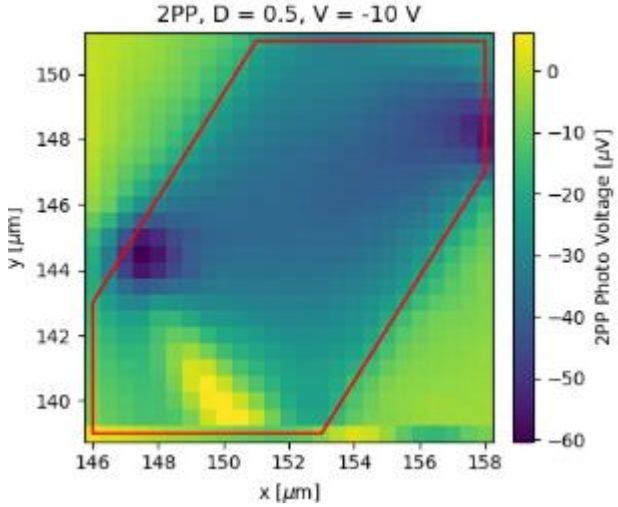
$$V_{\text{PV}} = R_{\text{light}} \cdot I - R_{\text{dark}} \cdot I = \frac{\mathrm{d}R_{\text{dark}}}{\mathrm{d}I} \cdot \delta I \cdot I. \tag{2}$$

Near the NDC criticality, the resistance varies rapidly with current ($I\,\mathrm{d}R/\mathrm{d}I \gg R$), so that $\frac{\mathrm{d}R}{\mathrm{d}I} \cdot I \approx \frac{\mathrm{d}(RI)}{\mathrm{d}I} = \frac{\mathrm{d}V_{\text{dark}}}{\mathrm{d}I}$, and the photovoltage reduces to:

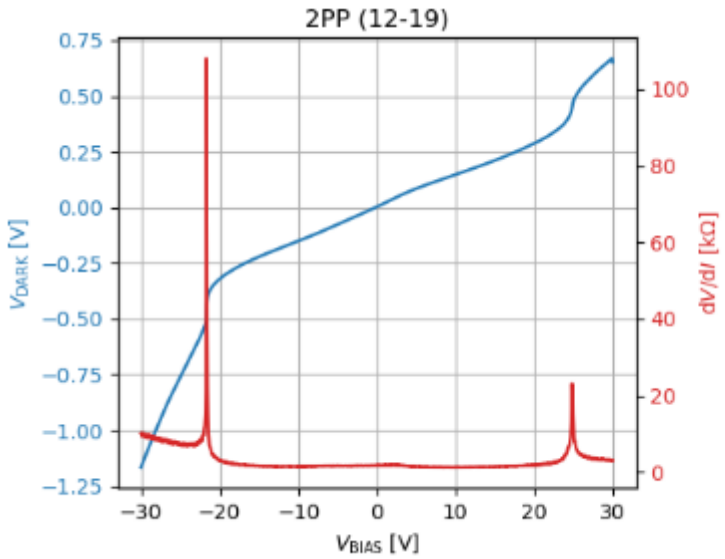
$$\boxed{V_{\text{PV}} \approx \frac{\mathrm{d}V_{\text{dark}}}{\mathrm{d}I} \cdot \delta I.} \tag{3}$$

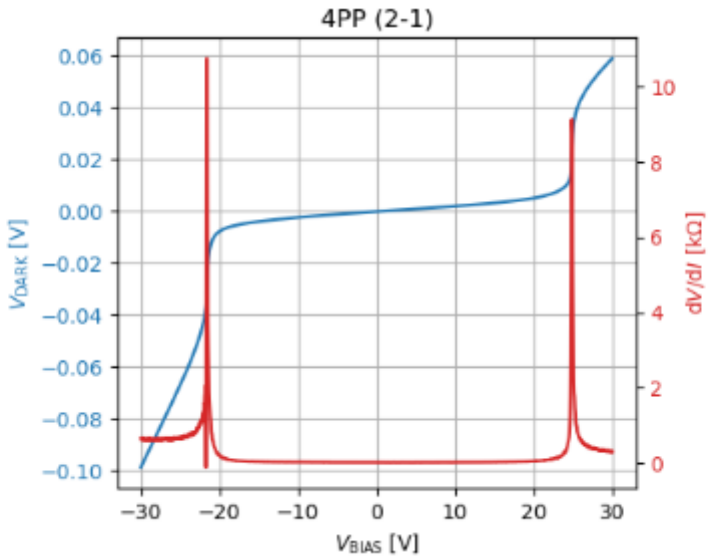
Current crowding / structural defect

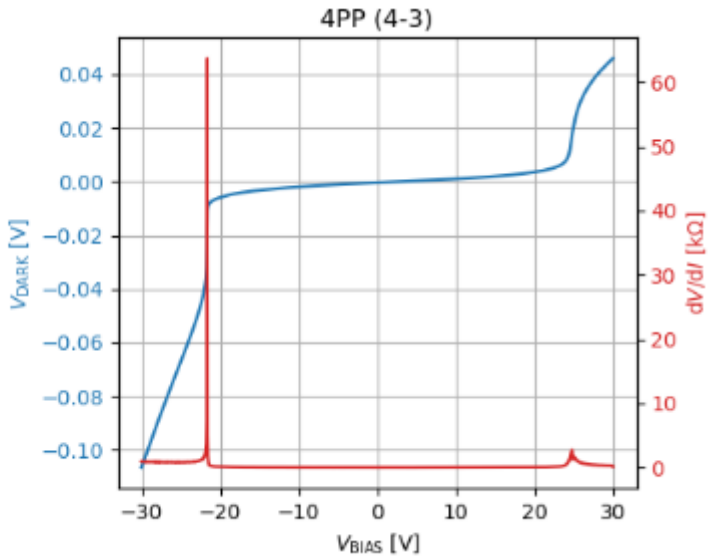


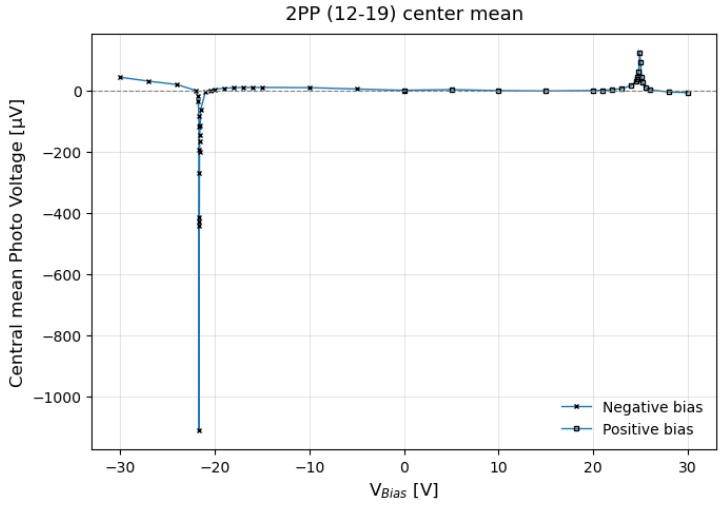


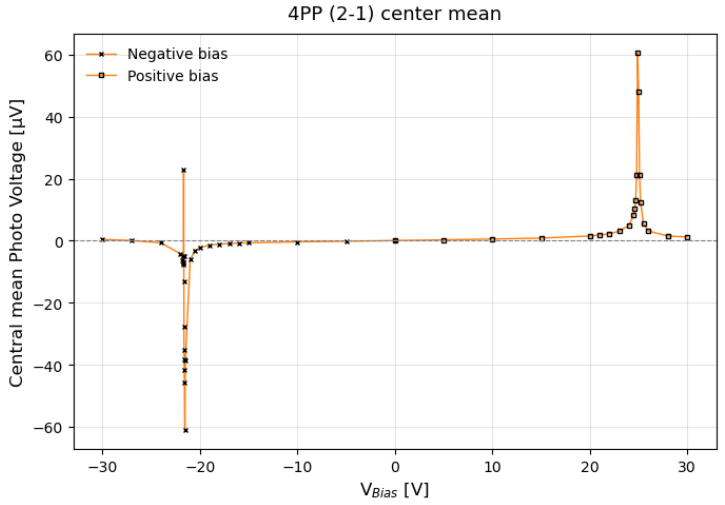
PV and dV/dI

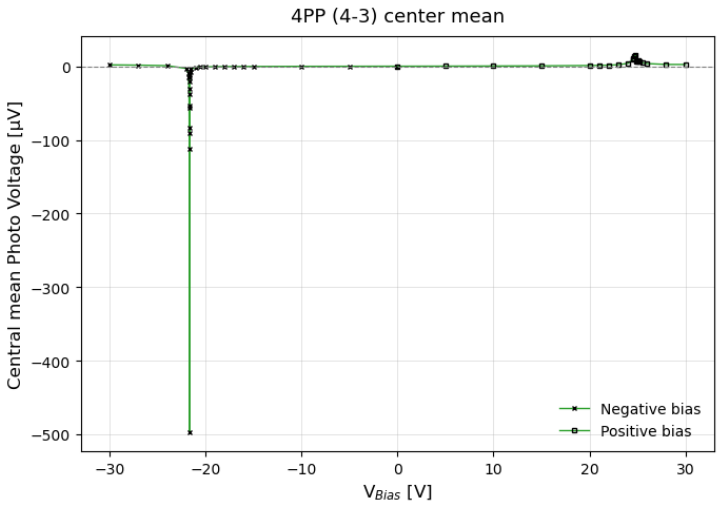

2PP (12-19)
V_{DARK} [V]
dV/dI [kΩ]
V_{BIAS} [V]


4PP (2-1)
V_{DARK} [V]
dV/dI [kΩ]
V_{BIAS} [V]


4PP (4-3)
V_{DARK} [V]
dV/dI [kΩ]
V_{BIAS} [V]

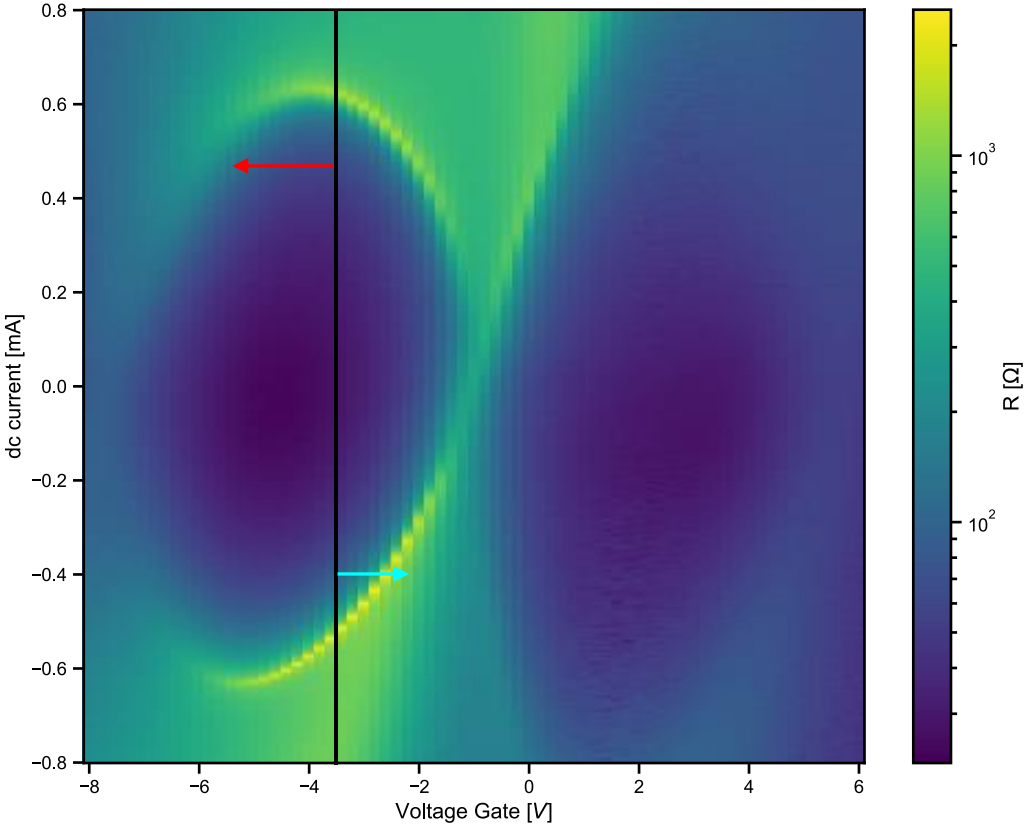

2PP (12-19) center mean
Central mean Photo Voltage [µV]
Negative bias
Positive bias
V_{Bias} [V]


4PP (2-1) center mean
Central mean Photo Voltage [µV]
Negative bias
Positive bias
V_{Bias} [V]


4PP (4-3) center mean
Central mean Photo Voltage [µV]
Negative bias
Positive bias
V_{Bias} [V]

Simulations :

- Use the transport measurement
- Given the sample being homogeneous, take self gating into account
- Each pixel has it's own R and dV/dI and V



- Initiation :
  - Given dc current input
  - Current through the sample is I_measured from the I_meas(I_sourced, n) map => Gives voltage drop on the sample
  - R for every pixel is taken as the first value : R_2PP.
  - V_SC at the contact is fixed : V = I_MEAS * R_2PP
  - Solve Laplace equation a first time : Get PHI(x,y) $\nabla \cdot [\sigma(x,y)\,\nabla\phi_{\text{unit}}(x,y)] = 0$
- Loop :
  - Compute the new n(x,y) (self-gating through phi) $n(x,y) = n_0 - C_g \cdot \phi(x,y) \,/\, e$
  - Associate the new R(x,y) thanks to the map (new R(I_measured, n(x,y)))
  - Compute the new potential solving Laplace equation with sigma(x,y)
  - It will have the right shape but needs rescaling : V_SC = I_MEAS * R_2PP is a limit condition that determines the rescaling needed
  - For stability : The new R(x,y) is 30% of the previous one and 70% of the one computed (stability)
- Converges ?
  - Convergence is checked via the maximum relative change in potential maps : taking the ratio between :
    - The max of the difference of the new and the old potential.
    - The max of the new potential
  - If this value is smaller than a given error (10e-5), algorithm stops